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CHOOSING THE BEST FORMAT PAIR SELECTION FOR SPGEMM ON KRONECKER MATRICES USING COST MODELS AND HARDWARE COUNTERS

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CHOOSING THE BEST FORMAT PAIR SELECTION FOR SPGEMM ON KRONECKER MATRICES USING COST MODELS AND HARDWARE COUNTERS

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Abstract

Sparse matrix-matrix multiplication (SpGEMM) is a core kernel in graph analytics, scientific computing and machine learning, and how fast it runs depends heavily on how the input matrices are stored. This thesis asks how the best pairing of storage formats changes with the size and structure of the input, and what the hardware is doing when it does. This thesis argues that, for sparse matrix-matrix multiplication using the kernels I chose and on the Kronecker graph data I tested on, choosing among storage formats is a simple decision. The dataflow of the format pair matters by orders of magnitude, and within the good pairs a fixed choice is within a few percent of the best. However, where the best pair does change follows quantities such as Number of Non-zeroes(nnz) or size, known before the multiply runs.

This thesis contributes the following: A benchmarking library that records timing and hardware counters (cache misses, branch mispredictions, page faults) around marked regions of code. Floating point and memory operation formulae for all sixteen pairings of four formats, which matched measured counts for fourteen of them, but operation counts turned out to be a poor predictor of runtime. A negative result in implementing a pipeline that picks a format for every block of a matrix based on the previous formulae and runs the multiply through a BLIS-style loop structure did not beat a single global format, because dispatch overhead dominated. A benchmarking sweep of every format pair over 19 different Kronecker matrices, which used my

benchmarking library to collect hardware counter data in addition to timing. This sweep demonstrated that inner-product dataflows are catastrophically slow, and that at low K a Gustavson dataflow using *CSRxCSR* is the best pairing, before giving way to outer product dataflows like *CSCxCSR* and *CSCxCOO*, with *CSCxCOO* winning out of the 2 when nnz/row is very low. And finally another partly negative, results of tiling on the same 19 matrices which found little benefit or in fact major performance loss. However, some matrices show minor promising trends where the relative cost of blocking fell as the matrix got larger. The thesis also includes an earlier FPGA design for streaming multi-format sparse matrix-vector multiplication, which motivated these questions.

These conclusions are limited to the specific kernel implementations I used and the Kronecker data sets I tested on.

Chapter 1

Introduction

Sparse matrix-matrix multiplication (SpGEMM), multiplies two matrices that are mostly zeros. It shows up in graph algorithms such as triangle counting and multi-source breadth-first search, in algebraic multigrid solvers, and in sparse machine learning workloads. Because most entries are zero, the matrices are not stored as full grids of numbers. They are stored in a *format* that records only the nonzeros and where they are, such as coordinate lists (COO), compressed rows (CSR) or compressed columns (CSC). The format decides which loop orderings the multiplication can use, how memory is accessed, and how much work is wasted on empty parts of the matrix.

The same product can therefore be computed in many ways, and they can differ in runtime performance by a significant degree. Two things make choosing the optimal combination difficult. There is no single best format, since it depends on the matrix. And for SpGEMM there are two input matrices, so the choice is a *pair* of formats, and the pair fixes the dataflow: row-wise (Gustavson), outer product, or inner product, see Chapter 2. Most of the existing work on format selection studies sparse matrix-vector multiplication on one matrix at a time. For SpGEMM, how the best pair changes with the size and structure of the input is less well documented (Section 3.6).

1.1 Central Question

This thesis asks one central question. For SpGEMM, how does the best pair of storage formats vary with the scale and structure of the matrix, and secondarily what hardware behaviour explains those differences? It began as a hardware question. My very earliest paper of this degree focused on an FPGA design (Chapter 7) which stored blocks of a matrix in different formats. While writing it I realized I came upon a number of questions which I hadn't answered but needed to. Questions like: which formats were worth implementing, how to decide, what to measure to decide, etc. The rest of the thesis tries to answer those questions.

1.2 Contributions

1. **Repetition Test Library.** I wrote a benchmarking library, the repetition test library (Chapter 4). It is capable of collecting a number of measurements that I thought would be important and explanatory.

2. **Analytical Formulae for Memory And Floating Point Operations.** I implemented kernels for all sixteen pairings of four formats, derived operation count formulae for them. The formulae were accurate to actual empirically observed counts on uniformly sparse matrices.

3. **Choose Best Format Pairing per-block for SpGEMM.** Used the above kernels and formulae to choose a format for each block of a matrix with an optimization solver (Chapter 5).

4. **Sparse BLIS.** Used the optimizer along with tiling and loop ordering in the spirit of the BLIS algorithm to choose the best format per pairing of blocks.

The results were quite poor performance-wise.

5. **Kronecker Sweeps.** I generated kernels for every format pair with the TACO compiler and swept them over Kronecker graphs at increasing scale, recording hardware counters (Chapter 6). Kronecker graphs can be made at any size, so at a fixed scale every graph has the same dimensions and only the structure differs, which lets size and structure be separated. This lead to the conclusions regarding the format-pair crossover points and their hardware and structural explanations.

6. **Kronecker Tiling Sweeps.** Finally, I swept over the same Kronecker graphs but this time with tiled formats. Tiling did not help at any sizes tested, but some matrices show a trend of the relative cost of tiling decreasing as matrix size increases.

1.3 Outline

Chapter 2 gives background on sparse formats, SpGEMM and its dataflows, TACO and Kronecker graphs. Chapter 3 reviews related work and the gaps this thesis tries to address. Chapter 4 describes the benchmarking library. Chapter 5 covers the operation-count formulae and the heterogeneous per-block pipeline. Chapter 6 presents the Kronecker sweeps with TACO-generated kernels. Chapter 7 describes the FPGA design. Chapter **??** discusses the results together and lists future work.

Chapter 2

Background

2.0.1 Sparse Matrix Storage Formats

The four formats considered in this work represent the standard vocabulary of sparse matrix storage. Compressed Sparse Row (CSR), see Figure 2.2, stores non-zeros row by row, with a row pointer array enabling efficient traversal of row entries. Compressed Sparse Column (CSC), see Figure 2.3, is the column-major analog to CSR; columns are stored contiguously, making it efficient for column-wise access patterns. Coordinate format (COO), see Figure 2.1, stores each non-zero as an explicit $(\text{row}, \text{col}, \text{value})$ triple, meaning more memory per entry and imposing a scan cost proportional to the total non-zero count during either row-wise or column-wise access. Dense format stores all elements explicitly in a flat row-major array, eliminating indexing overhead entirely and making any access pattern possible, making it optimal when most entries are non-zero, unsurprisingly. Each format encodes the same matrix but makes different trade-offs between storage overhead and the best access pattern. A standard reference for these formats in the context of sparse linear algebra compilers is TACO [1], who provide a unified description of format capabilities in terms of

per-dimension storage modes.

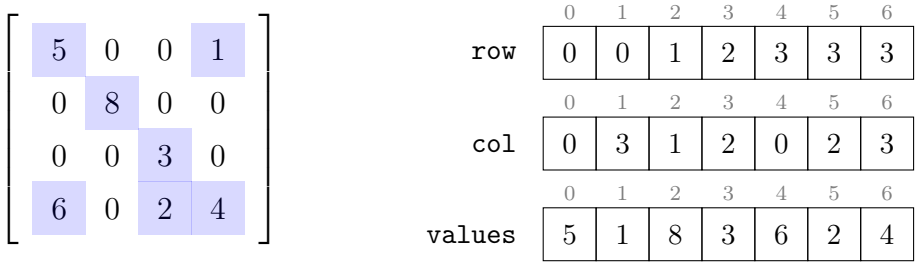


Figure 2.1: COO (coordinate) format. Each non-zero is stored as a (`row`, `col`, `value`) triple, so the k-th entry of the three arrays describes the k-th non-zero.

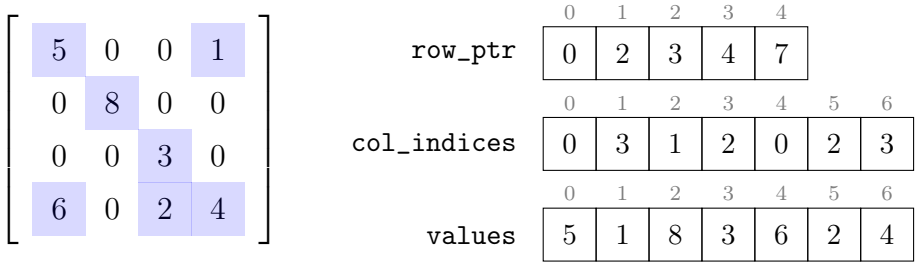


Figure 2.2: CSR (compressed sparse row) format. `values` and `col_indices` hold the non-zero values and their column indices, row by row. Row i occupies positions `row_ptr[i]` to `row_ptr[i+1]-1`; e.g. row 3 spans positions 4 to 6.

2.0.2 SpGEMM

SpGEMM computes $C = A \times B$ where A and B are sparse matrices. Unlike sparse matrix-vector multiplication (SpMV), SpGEMM involves two sparse operands whose non-zero structure interacts during multiplication: the output sparsity pattern is not known in advance and meaning many implementations compute the output symbolically before any actual numeric accumulation can begin. The operation appears as a core subroutine in algebraic multigrid solvers, triangle counting, multi-source breadth-first search, and high-dimensional similarity search, among many

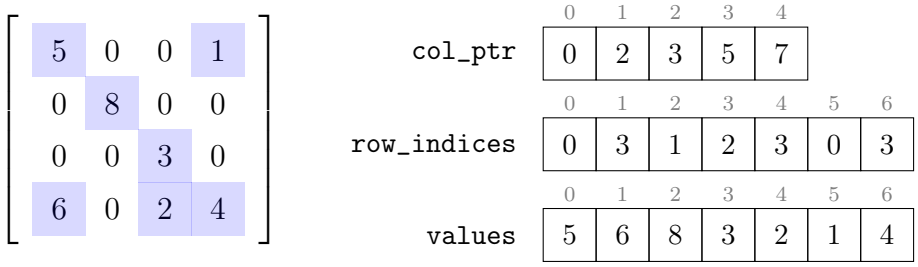


Figure 2.3: CSC (compressed sparse column) format. The non-zeros are stored column by column; `row_indices` holds their row indices. Column j occupies positions `col_ptr[j]` to `col_ptr[j+1]-1`; e.g. column 2 spans positions 3 to 4.

other applications [2]. Buluç and Gilbert [3] provide foundational parallel algorithms for SpGEMM. The most performance-relevant observation for this work is that SpGEMM is overwhelmingly memory-bound on current hardware. The irregular, format-dependent access patterns of sparse operands generate large numbers of short, hard to predict memory requests, keeping arithmetic units underutilized.

2.0.3 SpGEMM Dataflows

In SpGEMM, only products in which both a_{ik} and b_{kj} are non-zero contribute, I will call this quantity *matches*, since only values which 'match up' in both matrices will contribute non-zero output values. The three indexing variables i, j, k naturally give a triple loop nest, and which loop is innermost determines the *dataflow*, that is, the order in which non-zeros of the inputs are read and results are computed. Because a compressed matrix can be traversed efficiently only along its storage order, the choice of dataflow and the choice of formats are very tightly coupled. Table 2.1 summarizes the dataflows of interest to this thesis.

Inner product (i, j, k). Each output element is computed as the dot product of a row of A and a column of B, which is most conducive when one operand is stored

row-major and the other column-major (best represented as CSR×CSC). The two index lists must be *intersected*, and depending on density and structure, often most comparisons find no match, so non-zeros are unfortunately fetched and discarded. Each c_{ij} is finished in one pass, which gives good output reuse, but input reuse is poor and every possible output position must be examined even though the result is often sparse [4], [5].

Outer product (k, i, j). This dataflow is expressed as a sum of partial matrix products, $C = \sum_k A_{:,k} B_{k,:}$, pairing a column of A with the matching row of B (best represented as CSC×CSR). No index matching is required and every non-zero of A and B is read exactly once, however each term yields a partial result matrix, and all of them must at some point be reduced, giving poor output locality and intermediate storage that grows with the number of matches [5], [6], [7], [8].

Row-wise product / Gustavson (i, k, j). For each row of A, every non-zero a_{ik} selects row k of B, which is scaled and accumulated into a row buffer holding row i of C (best represented as CSR×CSR, the column-wise analog as CSC×CSC) [9]. It avoids index matching because the row pointer array gives direct access to row k of B, and it keeps the output row in the accumulator, and each element of A is touched only once, but reuse of rows of B across rows of A is irregular [10].

Accumulating partial products. Row-wise and outer-product algorithms must combine any partial products that land on the same output position. Common accumulators include the dense *sparse accumulator* (SPA) [11]; hash tables; and heaps or sorted merges. Of course, their costs and trade-offs differ [12]. Implementations also differ in whether the output structure is computed in a separate symbolic phase or allocated from an upper bound, which trades extra computation against extra mem-

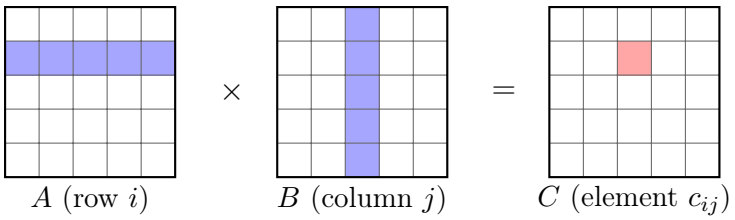


Figure 2.4: Inner-product dataflow. One output element c_{ij} is produced at a time by intersecting row i of A with column j of B and summing the matching products. Each output is complete when it is written, but most intersections are empty for sparse inputs, so effort is spent on index matching.

ory [12]. TACO defaults to a symbolic phase and many format combinations require specific scheduling commands to utilize any output accumulators. For the most part, this thesis ignores output accumulation. Most analysis later in the thesis will assume that accumulation can be done separately as a post processing step.

Relevance to format selection. Format pairs therefore largely determine the dataflow a compiler can reasonably generate, so the comparison of format combinations in Chapter 5 and Chapter 6 are, in part, comparisons of dataflows.

Dataflow	**Loop order**	*A* / *B*	**Input reuse**	**Output reuse**
Inner product	i, j, k	CSR / CSC	poor	good
Outer product	k, i, j	CSC / CSR	good	poor
Row-wise (Gustavson)	i, k, j	CSR / CSR	irregular	good
Column-wise	j, k, i	CSC / CSC	irregular	good

Table 2.1: SpGEMM dataflows, operand formats chosen by most naturally fitting to indexing order.

2.0.4 BLIS

The BLAS-like Library Instantiation Software (BLIS) framework [13] provides a portable infrastructure for implementing dense level-3 BLAS operations. Its key insight is that virtually all computation in dense matrix multiplication can be expressed

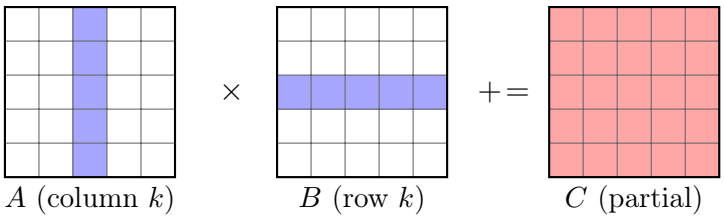


Figure 2.5: Outer-product dataflow. For each index k, column k of A is multiplied with row k of B, producing a partial result that touches many positions of C. Every input element is read once, but the partial results from different k must be merged.

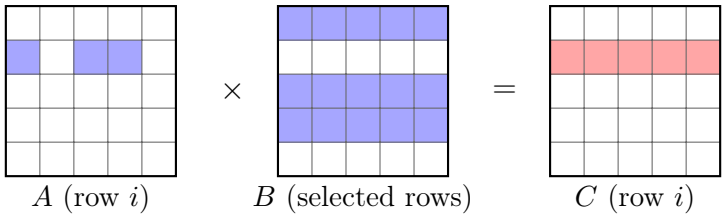


Figure 2.6: Row-wise (Gustavson) dataflow. Each non-zero a_{ik} in row i of A selects row k of B. The scaled rows are accumulated into row i of C. One output row is built at a time, so the accumulator for that row could potentially be kept in cache.

in terms of a small, hardware-specific micro-kernel operating on dense register-sized panels, surrounded by a fixed five-loop blocking structure parameterized by cache-level constants m_c, n_c, k_c and micro-kernel block sizes m_r and n_r. This separation allows the bulk of the framework to be written once in portable C while performance-critical sections are tuned per architecture. This thesis includes a chapter on an implementation which preserves this five-loop structure and blocking hierarchy, replacing only the innermost micro-kernel with a dispatch over heterogeneous sparse formats.

2.0.5 Roofline Model

The roofline model [14] provides an upper bound on achievable kernel performance as a function of arithmetic intensity (FLOPs per byte of memory traffic). For a given kernel and machine, the model draws a "roofline" as the minimum of the peak

compute throughput and the product of memory bandwidth and arithmetic intensity. A kernel is compute-bound if its arithmetic intensity exceeds the machine's compute-to-bandwidth ratio, and memory-bound otherwise. SpGEMM kernels consistently fall in the memory-bound regime due to low data reuse and irregular access patterns, see Chapter 5.

2.0.6 TACO

The Tensor Algebra Compiler (TACO) [15] automatically generates kernels for compound tensor algebra expressions over mixed dense and sparse tensors, including most format combinations relevant to this thesis. TACO provides a formalization of how iteration spaces and index expressions vary across storage formats, and its generated kernels served as the main reference for Chaper 6, as well as the starting point and correctness reference for Chapter 5's hand-instrumented implementations.

2.0.7 Kronecker Graphs

Kronecker graphs [16] are a family of self-similar synthetic networks generated by repeated Kronecker product of a small initiator matrix with itself. They were introduced as a model for real-world networks because their recursive structure naturally produces the heavy-tailed degree distributions and small diameters observed in real-world internet topology graphs and social networks. The datasets described in Chapter 6 are standard benchmark graphs which can represented as Kronecker graph instances, and which form the main datasets used for the experiments of this thesis. These were all generated using mfit [17]

Chapter 3

Literature Review

3.1 Outline

This chapter reviews prior work on the focus areas of this thesis. Section 3.2 covers how different sparse matrix formats are stored and those how storage formats are chosen. Section 3.3 covers analytical operation-count and performance models for sparse kernels. Section 3.4 discusses the inputs used to evaluate sparse kernels. Section 3.5 covers sparse linear algebra on hardware accelerators, relevant to Chapter 7. Finally, Section 3.6 discusses how this thesis hopes to address some of the gaps identified in the literature.

3.2 Sparse Formats and Format Selection

3.2.1 Formats beyond CSR, CSC, and COO

Many different storage formats have been proposed to exploit certain matrices or hardware. For hypersparse matrices, in which the number of non-zeros is very small

relative to the dimension, CSR and CSC can waste space on the pointer arrays that grow with their corresponding dimension. Buluc and Gilbert address this with their doubly compressed format (DCSC) that removes the empty-column pointers and uses space proportional to the number of non-zeros [7]. This is directly relevant to the Kronecker graphs studied in Chapter 5 and Chapter 6, where the number of non-zeros per row can be very small compared with the dimension.

Other formats target specific hardware. Nvidia's Bell and Garland study SpMV on throughput-oriented processors, specifically GPUs [18]. Their work proposes their hybrid (HYB) format, which builds off of ELLPACK, first proposed in [19]. In HYB, a regular part of the matrix is stored in ELLPACK format, which relies on consistent non-zeroes per row, and any exceptional elements are stored in COO. For wide SIMD units, Kreutzer et al. propose SELL-C-σ, a SIMD-friendly variant of sliced ELLPACK that combines ideas from GPU computing and vector processing, and show with fixed parameters that a single format can perform very well across a Intel Sandy Bridge CPU, an Intel Xeon Phi, and an Nvidia Tesla K20 [20]. Compressed sparse blocks (CSB) store a matrix as a grid of blocks so that both SpMV Ax and sparse transpose $A^T x$ can be computed efficiently in parallel [21].

3.2.2 Automated Format Selection

Because no single format is best for all matrices, one area of research focuses on automating this choice. The Sparsity framework and its successor OSKI tune SpMV with techniques such as register and cache blocking, symmetry, multiple vectors, variable-block and diagonal structure, and reordering, selecting parameters using empirical measurements together with properties of the input matrix [22], [23]. SMAT gives programmers a single CSR-based interface and uses a decision tree learned from

the features of 2,373 matrices in the University of Florida collection to choose among CSR, DIA, ELL, and COO at run time [24]. Sedaghati et al. show that no representation is uniformly best on GPUs and train a classifier on matrix features to select one [25]. Zhao et al. treat a matrix as an image and use a convolutional neural network for the same task [26]. Older investigations into automated tuning research seemed to focus on analytical or empirical methods, while newer research has focused on machine learning techniques.

Most of these selection methods choose a single global format for a whole matrix, and further most target SpMV. One question studied in this thesis, which pair of formats is best for a product of two sparse matrices, is not addressed by them. And indeed format selection is a harder question for SpGEMM in which 2 matrices must be considered. Like the learned selectors above, this thesis chooses by measurement as well as modeling, but it sweeps a controlled family of input kroneckers rather than a collection of unrelated matrices, see Section 3.4).

3.2.2.1 TACO.

Compiler work approaches the problem of format selection from the other direction. Chou et al. give a unified description of formats that lets a compiler generate kernels for arbitrary combinations [1], and TACO builds on this to generate tensor-algebra kernels for mixed sparse and dense operands [15]. TACO is the mechanism used to produce the kernels in Chapter 6 and for reference in Chapter 5. With this, format selection can be an experiment parameter to sweep across and identify crossovers. The rest of this section explains both how TACO represents a matrix and its code generation.

Matrices as coordinate hierarchies. TACO takes a tensor expression in index notation together with a format for each operand, and generates imperative code from them [15]. For a matrix product, the expression is $C_{ij} = \sum_k A_{ik}B_{kj}$. A format is described as a tree with one level per dimension, where each root-to-leaf path encodes the coordinates of one non-zero. Each level has a *level format* that says how its coordinates are stored. A *dense* level stores only the dimension size and implicitly contains every coordinate. A *compressed* level stores only the coordinates present, in a `crd` array, plus a `pos` array giving where each parent's segment begins. A *singleton* level stores one coordinate per parent [1], [15]. The 2018 extension also adds range, offset, and hashed levels, enough to express DIA, ELL, hash-map, and blocked formats [1], though this thesis will not examine these additional formats too closely. The formats in this thesis are then compositions in TACO's language. CSR is a dense row level over a compressed column level; CSC is the same with the dimension order swapped; COO is a compressed, non-unique row level over a singleton column level. Figure 3.1 shows one small matrix in each of these formats. Because a dense row level needs a `pos` entry per row whether or not the row is empty, part of a format's cost scales with the dimension, not the non-zero count, which is the overhead DCSR removes.

Format to Dataflow. The code generator reasons only about what each level can do. A dense level supports *locate*, finding a coordinate's position directly; a compressed level can only be walked in order [1]. TACO orders the loops with an *iteration graph*, so each operand is traversed top-down, and builds a *merge lattice* for each index. A product needs the *intersection* of its operands' coordinates. If neither operand supports locate, the generated code co-iterates over both sorted coordinate lists, like the merge step of merge sort. If one does, it iterates the other's non-zeros

Matrix M (4×4)

	5		1
7			
		3	8

COO

`row`	0	0	2	3	3
`col`	1	3	0	2	3
`vals`	5	1	7	3	8

CSR (dense → compressed)

`pos`	0	2	2	3	5
`crd`	1	3	0	2	3
`vals`	5	1	7	3	8

CSC (columns first)

`pos`	0	1	2	3	5
`crd`	2	0	3	0	3
`vals`	7	5	3	1	8

Figure 3.1: One matrix in three formats (indices from zero). In CSR, `pos[i]` to `pos[i+1]` delimits row i in `crd`; row 1 is empty, so `pos[1]`=`pos[2]`. CSC is the same over columns. COO repeats the row coordinate per non-zero and has no `pos`.

and looks each up, reducing the cost from $O(\text{nnz}(x) + \text{nnz}(y))$ to just $O(\text{nnz}(x))$ [1], [15].

Sparse × sparse products. In $A_{ij} = \sum_k B_{ik}C_{kj}$, the format pair therefore fixes how the sum over k is organized. With B and C both CSR, the order i, k, j works and the dense row level of C lets the code locate row k, giving the row-wise (Gustavson) dataflow. With B in CSC and C in CSR, both have k at the top, so k is the outermost loop, giving the outer-product dataflow. With B in CSR and C in CSC, k is innermost and two sparse lists are intersected per output element, giving the inner-product dataflow. Choosing formats thus largely chooses the dataflow, so the sweep over format pairs in Chapter 6 is also a sweep over algorithms.

Limits. Blocked formats are expressible with fixed-size dense levels, so block sizes must be uniform [1]. As well blocks of differing format are inexpressible. The published evaluation covers SpMV, sparse × dense multiplication, addition, and tensor kernels, single-threaded, on matrices up to about 1.3M × 1.3M [1], so sparse × sparse

products at the scales of Chapter 6 are beyond what those papers measure.

3.2.3 Heterogeneity within a matrix

A few approaches already store different parts of a matrix differently. HYB, as mentioned before, divides a matrix between two formats [18]. Vuduc and Moon split modify the traditional Block Compressed Sparse Row (BCSR) and propose Unaligned BSCR (UBSCR), in which a matrix is split into parts with different register block sizes to exploit variable block structure [27].

The closest prior work to the per-block pipeline of Chapter 5 that I am aware of is TileSpMV for GPUs, which partitions a matrix into tiles, implements seven SpMV kernels for tiles stored in different formats, and selects the format and algorithm for each tile based on its local sparsity structure [28]. This was extended to SpGEMM in TileSpGEMM, which adapts the tiling idea of dense matrix multiplication: it splits both matrices into fixed-size sparse tiles, so that work per tile is easier to balance and accumulate. It is evaluated on 142 square matrices against four GPU libraries (cuSPARSE, bhSPARSE, NSPARSE, and spECK) and is reported to be faster than the best of them, spECK, on 94 of the 142 [29]. TileSpGEMM is the closest SpGEMM analogue to the blocking of Chapter 5, but it targets GPUs, uses one sparse block format/representation rather than selecting among many formats, and is not concerned with choosing CPU-side format pairs.

Table 3.1 summarizes how these approaches compare.

3.3 Motivating Flop and Memory-Operation Models

For each pair of operand formats, Chapter 5 derives the number of floating point operations and memory operations the respective SpGEMM kernel performs. This

Work	Kernel	Granularity	Selection
Sparsity/OSKI [22]	SpMV	whole matrix	empirical profile plus heuristic
SMAT [24]	SpMV	whole matrix	learned model
Sedaghati et al. [25]	SpMV	whole matrix	learned classifier
Zhao et al. [26]	SpMV	whole matrix	convolutional network
TileSpMV [28]	SpMV	per tile	per-tile selection method
TileSpGEMM [29]	SpGEMM	per tile	none, single format

Table 3.1: Format-selection approaches compared by kernel, granularity, and selection method.

section motivates why such counts are a reasonable thing to model, what related models exist, and their limits.

3.3.1 Operation Counts

Sparse kernels often do little arithmetic per byte moved. Additionally, the work they do end up doing depends on the structure of the inputs, not just on their dimensions. The roofline model relates attainable performance to the number of operations performed per byte of memory traffic and to the hardware's compute and bandwidth ceilings [14]. It is the standard model for asking whether a kernel is limited by memory or by compute. Applying it to a given SpGEMM requires knowing how many operations and how many memory accesses the chosen algorithm and formats perform, see Table 5.1.

Operation counts also appear inside algorithm analyses. The sequential hypersparse kernel of Buluc and Gilbert has a running time that depends on the number of non-empty columns of one operand, the number of non-empty rows of the other, and the number of scalar multiplications. It is therefore independent of the matrix dimension, which is one reason the outer-product formulation suits hypersparse in-

puts [8], as we will see in Chapter 6. That bound is stated for one algorithm and one format, it does not compare formats.

3.3.2 Using operation counts for prediction

Two bodies of work suggest that quantities derived from operation counts predict which implementation is best. Nagasaka et al. find that the best SpGEMM algorithm on multicore and many-core processors varies with matrix size, sparsity, the compression factor of the product, and the operation being performed, and they distill their measurements into a recipe for choosing among hash-based, heap-based, and library implementations [12]. Looking into the field of custom accelerators, Flexagon argues that different workloads favor different dataflows depending on features of the kernel such as its dimensions, sparsity pattern, and sparsity degree, and builds hardware that runs each computation with the dataflow that best suits it [30].

Many of these lines of research choose between algorithms or dataflows. But as established in Chapter 2, format and dataflow are very tightly coupled.

3.3.3 Limits of operation counts

Counts describe what the algorithm asks the machine to do, not what the machine does. They do not capture cache behavior, branch misprediction, page faults and first-touch costs, etc. Hardware performance counters measure these effects directly, but they are often not exact. On eleven x86-64 implementations, events that should be deterministic, such as retired instructions, vary from run to run and overcount, with hardware interrupts and page faults among the sources [31]. Counts and counters are therefore complementary. We will find in Chapter 5 that the operation-count formulae predict runtime poorly, and Chapter 6 uses cache-miss, branch-misprediction, and

page-fault counters to hopefully examine and explain the discrepancy.

3.4 Benchmark Inputs for Sparse Kernels

3.4.1 Collections of real matrices

Evaluations of sparse kernels commonly draw on the SuiteSparse Matrix Collection [32], [33], which includes matrices from the SNAP network datasets [34], [35]. Studies typically report results per matrix across large sets. SMAT is trained on features of 2,373 matrices [24], and TileSpGEMM is evaluated on 142 square matrices, each requiring at least one billion floating-point operations for one SpGEMM [29]. Such evaluations are broad, but they do have a structural limitation: every matrix differs from every other in size, density, and structure at once. When a method wins on one matrix and loses on another, the data cannot easily say which of the three properties is responsible. Each matrix is a single point, and can not point us to a interpolatable trend.

3.4.2 Synthetic generators

Synthetic generators offer a bit more control. The R-MAT model recursively subdivides an adjacency matrix according to a small set of probabilities to produce graphs with power-law degree distributions [36]. Kronecker graph models generate self-similar graphs by repeated Kronecker products of a small initiator matrix and can be fitted to real networks [16], [17]. Because the number of repetitions is a free parameter, a single fitted model can be generated at many scales, so that structure is held approximately fixed while size varies.

3.5 Hardware Acceleration for Sparse Computations

3.5.1 SpGEMM accelerators and dataflows

Gustavson's row-wise algorithm is one classical formulation of sparse matrix multiplication [9] as it requires no intersection operation of non-zeroes. Recent accelerator work revolves around choosing among inner-product, outer-product, and row-wise dataflows. OuterSPACE pursues outer products [6]. SpArch observes that the outer product suffers from poor output locality because of the numerous partial product matrices it generates, and addresses this with on-chip merging, a condensed matrix representation that reduces the number of partial matrices, a Huffman-tree merge scheduler, and a row prefetcher [5]. MatRaptor adopts the row-wise product and introduces a storage format, C2SR, that lets parallel compute engines access sparse data in a streaming way [4]. Gamma builds on Gustavson's algorithm for GPU-like architectures [10]. Flexagon is a reconfigurable accelerator that supports all three dataflows [30]. Two themes recur: the dataflow determines which operand is reused and which output is accumulated, and accelerator designers have found it worthwhile to design the storage format together with the dataflow, as MatRaptor does. This parallels the format-pair question of this thesis, at the hardware level.

3.5.2 FPGA implementations

FPGA implementations of SpMV have been studied extensively because the kernel is dominated by irregular memory access. Fowers et al. propose a sparse matrix encoding that exposes parallelism across rows together with a banked vector buffer, and report competitive performance against CPUs and GPUs at lower power [37]. Hosseinabady and Nunez-Yanez describe a streaming dataflow engine for SpMV on embedded FP-

GAs built with high-level synthesis, combining loop pipelining, dataflow graphs, and data streaming to use most of the available memory bandwidth; they report about 3.25× the performance of embedded GPUs on small and medium matrices and about 8.9× better energy efficiency than the embedded CPU and GPU [38]. This is the design style used in Chapter 7. Work on irregular matrices addresses load imbalance [39], and CPU work selects among hybrid formats adaptively for SpMV [40].

3.6 Gaps Identified

Taken together, the literature reviewed suggests three points that this thesis tries to address.

1. **Evaluations are made at single points.** Collections of real matrices vary size, density, and structure together, so the dependence of the best choice on any one of them cannot be separated. Kronecker graphs fitted to real networks can be generated at many scales, and Chapter 6 sweeps them to show how the best format pair changes as size grows.

2. **Format selection is studied mostly for SpMV and mostly per matrix.** Selection of format *pairs* for SpGEMM appears less explored. Chapter 5 contributes flop and memory-operation formulae for sixteen format pairs, and Chapter 6 compares the pairs by measurement and with hardware counters.

3. **Heterogeneous formats within one matrix have been attempted for SpMV but their SpGEMM cost is less documented.** Chapter 5 presents a per-block selection pipeline for SpGEMM and reports that it unfortunately did not pay off along with an analysis of why.

Chapter 4

Repetition Test Library

This chapter discusses the library I created for benchmarking which was used in most experiments excepting any experiments utilizing FPGAs. Though not all functionality was present through all experiments using it. It was inspired and initially based architecturally on the one covered by Casey Muratori's Computer Enhance project [41]. It builds off of and requires my own base layer library of C code. It, like most of my C code, currently only supports Linux, but especially this particular library because it relies on the Linux 'Perf' [42] syscall API for collecting CPU hardware counters.

As one might assume from its title, it is primarily used to repeatedly test functions, collecting info about them such as timing, page faults, in addition to the aforementioned hardware counters for things such as branch and cache misses. In my own biased opinion, it provides this functionality in a straightforward and flexible way. It will continue to collect this info in a loop, collecting new minimums to try to eliminate any OS-scheduling or other noise. It allows programs to own their own control flow, while simply marking up important sections for benchmarking and data collection. It also allows users to provide additional info to calculate second order statistics such as

throughput by providing byte counts. Finally, it allows easy access to this data via options to output as CSV, containing both library generated data and user-provided data. It also pushes live info as it is collected to the terminal.

4.1 Motivation

The experiments in this thesis required a library/harness with a particular combination of capabilities which none of the options I considered provided all of.

Hardware counters for diagnosis. The experiments in Chapter 5 compare operation counts against runtime, and the gap between the two cannot be explained by operation counts alone nor do they necessarily even predict real performance. At this point the library did not collect hardware counter events, but explaining performance in reality requires cache misses, branch mispredictions, and other hardware events. Collecting these through source-level instrumentation, as I did with the counting macros in that chapter, perturbs the performance being measured, in addition to being tedious to insert. Hardware counters read via the Linux `perf` interface [42] avoid this, so a harness that exposed them directly was necessary.

Measurement of marked regions, not whole processes. Tools such as `perf stat` report counters for entire processes, which includes setup, input generation, verification, etc. I wanted measurements scoped to individual kernels, so the harness had to let the program and user mark the regions of interest.

Program-owned control flow. Many benchmarking frameworks invert control. Often they require the user to register a function and the framework or library decides when and how to call it. This can be awkward for kernels that need per-experiment

setup, such as matrix generation at a chosen density, or that we wish to be driven from a variable-sweep loop in the program, or that have special blocking transformations that need to be done on input matrices. I wanted the program to own its own control flow and simply mark the section(s) to be repeated and measured. An example program can be found in Listing A.1.

Simplicity. All the code for this thesis is written in C on top of a small base layer library of my own. I considered heavier weight libraries like PAPI [43], which would have potentially allowed me to collect even more cross-microarchitecture (TODO: better way to say 'cross-microarchitecture'?) hardware counter events such as flop/load/store instructions retired easier. However, I decided against it in the interest of keeping benchmarking complexity low. I wanted to save most of my mental complexity budget for dealing with the actual experiments. As well, its always more interesting to build your own thing than use a library.

4.2 Specific Measurements

This section defines the quantities reported in Chapters 5 and 6.

Time. Runtime is measured with the x86 timestamp counter (`RDTSC` instruction) and reported in timestamp "cycles." This counter advances at a fixed rate regardless of actual core frequency scaling, so these units are ticks of a constant-rate clock, not cycles of the core clock. Each kernel is repeated and the minimum observed time is kept, again to minimize scheduling noise, first touch page faults, and cold-cache effects. The minimum is therefore a best-case, warm-state estimate.

Hardware counters. Hardware-counted events are read through the Linux `perf` interface **perf**, scoped to the timed region. Two generalized events are used. *Cache misses* corresponds to `PERF_COUNT_HW_CACHE_MISSES`, which usually counts last-level cache misses, i.e., accesses that must go to main memory. *Branch mispredictions* corresponds to `PERF_COUNT_HW_BRANCH_MISSES`, the number of mispredicted branches.

Page faults. Memory returned by an allocator is often not immediately backed by physical pages. The first access to each page raises a *page fault*, which the operating system remedies by mapping an actual page. Page faults therefore measure first-touch cost: a kernel that writes freshly allocated memory incurs roughly one fault per page touched (typically 4 KiB, 2 MiB, or 1 GiB), and each fault costs kernel time that is included in the measured runtime. This matters for sparse formats because their working-set sizes can scale with both matrix dimension and the number of non-zeros. As well, TACO uses the C standard library allocator malloc and calloc which are not known to have the best behaviour regarding pre-faulting.

Allocations. In addition to counters, each run records the total bytes requested from the allocators malloc and calloc. This is *requested*, not actually mapped or touched memory, so it can exceed what page faults show if memory is allocated but never written, as is often the case with TACO generated code.

Chapter 5

Analytical Models and Sparse BLIS

5.1 Introduction

SpGEMM is a important kernel in graph analytics, scientific computing, and machine learning, and its performance can depend heavily on how the input matrices are stored. Dense, CSR, CSC, and COO each exploit different sparsity patterns. Usually a single format is chosen to compress an entire matrix. However, real-world matrices often have varying sparsity patterns across the whole matrix, e.g. dense diagonal bands surrounded by very sparse blocks, like those often seen in kronecker graphs. This chapter explores one avenue of exploiting this structured sparsity by partitioning each input matrix into smaller blocks and storing each block in whichever format would suit it best for the operation.

This chapter will investigate whether that idea paid off. The hypothesis was that assigning storage formats per block with a custom cost model and executing the multiply operation within a BLIS-inspired block loop structure would outperform a single global format. I went about this in this way:

1. I implemented SpGEMM kernels for all sixteen pairings of the four formats,

instrumented them to count floating point and memory operations, and derived analytical cost formulae for each.

2. I formulated per-block format selection as a mixed integer nonlinear program using those cost formulae, and built a sparse BLIS pipeline that executes the resulting heterogeneous assignment.

3. I evaluated the pipeline against uniform-format and global-compression baselines on synthetic banded matrices and finally on Kronecker graphs.

The results as you will see are mixed. The cost formulae predict operation counts quite accurately for 14/16 of the format combos, and some benchmarking reveals some crossover points between formats across varying density. Brining it all together in the heterogenous BLIS format did not succeed, however. There is too much overhead from both the dense blocking structure and from format dispatch. Even when the solver chose more 'optimal' formats it did not ever beat simple global compression. Operation counts also proved as a poor proxy for runtime. And finally, the solver does not scale beyond small matrices and proved intractable at sizes where blocking could potentially have benefits.

I believe that even where this experiment failed is still informative so I present all the results in full.

Attribution. This chapter is based on work conducted with Dr. Richard Veras and Moez Durrani. I implemented the kernels, instrumentation, solver pipeline, and experiments, and performed the analysis and wrote this chapter. Dr. Richard Veras proposed the initial research direction and provided supervision. Moez Durrani provided the kronecker datasets used in some of the experiments.

Listing 5.1: Instrumentation macros

```
#ifdef OBSERVE_FLOPS
#define FMADD(dst, a, b) dst += (a * b); \
  repetition_tester_count_flops(tester, 2)
#endif // OBSERVE_FLOPS
#ifdef OBSERVE_MEMOPS
#define LOAD(src) src; \
  repetition_tester_count_memops(tester, 1); \
  repetition_tester_count_bytes(tester, sizeof(src))
#define STORE(dst, src) dst = src; \
  repetition_tester_count_memops(tester, 1); \
  repetition_tester_count_bytes(tester, sizeof(dst))
#endif // OBSERVE_MEMOPS
```

5.2 Mechanism / Methodology

5.2.1 Instrumentation and Benchmarking

Each matrix multiplication kernel was initially derived using the TACO compiler as a reference, then rewritten by hand to incorporate a custom instrumentation layer. Each kernel is itself actually implemented as a C preprocessor macro, allowing the same implementation to be invoked both within the benchmarking framework and directly as a callable function in the sparse BLIS multiplication pipeline. The instrumentation layer is then built on top of this using a second set of macros, `LOAD`, `STORE`, and `FMADD`, which in their counting mode expand to the underlying operation plus a bit of bookkeeping to record memory and floating point operation activity. Each `FMADD` increments a floating point operation counter by two, while each `LOAD` and `STORE` increments both a memory operation counter and a byte counter. In their benchmarking mode the macros expand to only the base operations with no book-keeping overhead, allowing accurate runtime measurement. I observed the bookkeeping had a measurable and noticeable negative impact on performance.

Runtime performance is measured using a repetition testing framework inspired by Casey Muratori's Computer Enhance material [41]. For each kernel, the framework repeatedly executes the multiplication over a fixed time window and records the minimum observed cycle count, using the x86 hardware timestamp counter, _RDTSC. Taking the minimum rather than the mean reduces the effect of OS scheduling noise as well as cold cache effects.

By sweeping input matrices across a range of densities, both the counters and the runtime measurements can be collected. Of course one at a time to eliminate any noise from the instrumentation on performance. We can then obtain enough information for a direct comparison between predicted and observed operation counts as a function of sparsity. This sweep also provides the data required to plot a roofline model. The roofline performance bounds of peak floating point throughput and peak memory bandwidth were measured directly by running hand-written assembly loops designed to fully saturate the FMA and memory load ports of the target CPU. All results are visualized using a Python script built using Matplotlib [44].

The instrumentation macros actually ended up serving a dual purpose during development: beyond the data collection, their presence as visual markers in the kernel source made it straightforward to reason about which operations contributed to each term in the cost formula derivations described in the following subsection.

5.2.2 Formula Derivations

To enable format selection without runtime measurement, I derive analytical formulae for the floating point and memory operation counts of each kernel as a function of matrix dimensions and sparsity. For each kernel, the derivation follows a consistent procedure: we walk the loop nest from outermost to innermost, identify every `LOAD`,

Listing 5.2: Implementation macro example, backslashes ommitted for formatting, STATEMENT expands to a do while(0)

```
#define csr_x_csr_impl STATEMENT(
  for (usize left_row = 0; left_row < left.row_count; left_row++)
  {
    usize left_row_start = LOAD(left.row_pointers[left_row]);
    usize left_row_end = LOAD(left.row_pointers[left_row + 1]);

    for (usize i = left_row_start; i < left_row_end; i++)
    {
      usize left_col = LOAD(left.col_indices[i]);
      f64 left_value = LOAD(left.values[i]);

      usize right_row_start = LOAD(right.row_pointers[left_col]);
      usize right_row_end = LOAD(right.row_pointers[left_col + 1]);
      for (usize j = right_row_start; j < right_row_end; j++)
      {
        usize right_col = LOAD(right.col_indices[j]);
        f64 right_value = LOAD(right.values[j]);

        usize output_index = left_row * output.col_count + right_col;
        f64 current_value = LOAD(output.values[output_index]);

        f64 result_value = current_value;
        FMADD(result_value, left_value, right_value);

        STORE(output.values[output_index], result_value);
      }
    }
  }
)
```

`STORE`, and `FMADD` site, and multiply each by the number of times its enclosing loop body executes. The instrumentation macros serve as explicit markers during this process, making each counted operation visually identifiable in the source code. We also note that we assume a uniform sparsity.

In some cases the loop bounds are not immediately obvious. For instance a loop iterating from a sparse row pointer start to end appears to iterate over a single row, but summing across all rows reveals the total iterations equal the non-zero count of that matrix. Recognising these collapsed bounds where nested loops together amount to a single pass over all non-zeros was an important step in several of the derivations.

We'll adopt a more descriptive subscript notation in place of the conventional M, N, K.

- L_{RC}, L_{CC} — row and column count of the left matrix
- R_{CC} — column count of the right matrix (inner dimension is shared: $L_{CC} = R_{RC}$)
- L_{NZ}, R_{NZ} — non-zero counts of the left and right matrices
- matches $= L_{NZ} \cdot R_{NZ} / L_{CC}$ — expected number of multiply-accumulate operations under uniform sparsity with 2 sparse formats

Representative Derivations

I will present a few full derivations which I find representative of the whole process.

Dense $\times$ CSR The outer loop iterates over L_{RC} rows of the left matrix. Within each row, the middle loop iterates over all L_{CC} rows of the right CSR matrix, reading two row pointers per iteration, $2 \cdot L_{RC} \cdot L_{CC}$ reads in total. However, across all

iterations of this middle loop, the inner loop collectively visits every non-zero of the right matrix exactly once, so the total 2 inner iterations collapse to R_{NZ} per output row. This gives one left value load per non-zero per row, contributing another $L_{RC} \cdot L_{CC}$, and four reads/writes per matched non-zero the right column index, right value, output load, and output store giving $4 \cdot L_{RC} \cdot R_{NZ}$.

$$\text{flops} = 2 \cdot L_{RC} \cdot R_{NZ} \tag{5.1}$$

$$\text{memops} = 3 \cdot L_{RC} \cdot L_{CC} + 4 \cdot L_{RC} \cdot R_{NZ} \tag{5.2}$$

COO $\times$ COO The outer loop scans `left.row_indices` in full to group left entries into row segments, contributing L_{NZ} reads. Within each segment, the merge intersection advances `left.col_indices` through the segment, across all segments this totals another L_{NZ} reads. The right row index array resets to zero at the start of each left row segment and is fully scanned each time, costing $R_{NZ} \cdot L_S$ reads where L_S is the number of distinct left row segments. This is bounded above by both L_{RC}, the total number of rows, and L_{NZ}, since you cannot have more distinct rows than non-zeros, finally giving $L_S = \min(L_{RC}, L_{NZ})$. The min operation here helps to account for how this variable will change as density increases. At low densities L_{NZ} is very small and most non-zeroes exist on different rows, this flips as density increases and we approach a case where each row has at least one non-zero. Of course this holds true only for uniform sparsity. On a match, `left.values` is loaded once per matched left entry, giving L_{NZ} reads, and the right column index, right value, output load, and output store contribute $4 \cdot \text{matches}$. The expected match count assumes uniform sparsity, where each left non-zero at column k expects R_{NZ}/L_{CC} right non-zeros with row k, giving $\text{matches} = L_{NZ} \cdot R_{NZ}/L_{CC}$.

$$L_S = \min(L_{RC}, L_{NZ}) \tag{5.3}$$

$$\text{matches} = L_{NZ} \cdot R_{NZ}/L_{CC} \tag{5.4}$$

$$\text{flops} = 2 \cdot \text{matches} \tag{5.5}$$

$$\text{memops} = 3L_{NZ} + R_{NZ} \cdot L_S + 4 \cdot \text{matches} \tag{5.6}$$

Two format combinations, CSR×CSC and COO×CSC, present additional derivation difficulty compared to the rest. Both involve a merge intersection where both operands reset on different outer loop dimensions simultaneously: in CSR×CSC the left row entries reset per output column and the right column entries reset per output row, while in COO×CSC the same interaction occurs but compounded by the left row segment structure common in multiplies with COO. This seems to produce terms in the memory cost that are difficult, at least for us, to derive. The formulae given in Table 5.1 for these two cases capture the observed structure to the best of our abilities, though with considerably more error than the rest. All other fourteen combinations show strong agreement with observed counts across the full density range, as discussed in the Analysis section.

5.2.3 Format Selection Optimization

Given the cost formulae derived in the previous subsection, we can formulate the problem of selecting an optimal storage format for each micro-block as an integer optimization program. Since in matrix multiplication one block may actually be multiplied against many other blocks. The objective is to minimise the total estimated cost across all block multiplications required to compute the full matrix product

Left	Right	Flops	Memops
DNS	DNS	$2L_{RC} \cdot L_{CC}R_{CC}$	$2L_{RC}L_{CC}R_{CC} + L_{RC}R_{CC}$
DNS	CSR	$2L_{RC} \cdot R_{NZ}$	$3L_{RC}L_{CC} + 4L_{RC}R_{NZ}$
DNS	CSC	$2L_{RC} \cdot R_{NZ}$	$3L_{RC}R_{CC} + 3L_{RC}R_{NZ}$
DNS	COO	$2L_{RC} \cdot R_{NZ}$	$2R_{NZ} + L_{RC}R_{NZ} + 4L_{RC}R_{NZ}$
CSR	DNS	$2L_{NZ} \cdot R_{CC}$	$2L_{RC} + 2L_{NZ} + 3L_{NZ}R_{CC}$
CSR	CSR	$2 \cdot \text{matches}$	$2L_{RC} + 4L_{NZ} + 4 \cdot \text{matches}$
CSR	CSC	$2 \cdot \text{matches}$	$2L_{RC} + 4L_{RC}R_{CC} + L_{NZ}R_{CC} + L_{RC}R_{NZ} + 2 \cdot \text{matches}$
CSR	COO	$2 \cdot \text{matches}$	$2L_{RC} + L_{NZ} + L_{RC}R_{NZ} + 4 \cdot \text{matches}$
CSC	DNS	$2L_{NZ} \cdot R_{CC}$	$2L_{CC} + 2L_{NZ} + 3L_{NZ}R_{CC}$
CSC	CSR	$2 \cdot \text{matches}$	$4L_{CC} + 2L_{NZ} + 4 \cdot \text{matches}$
CSC	CSC	$2 \cdot \text{matches}$	$2R_{CC} + 4R_{NZ} + 4 \cdot \text{matches}$
CSC	COO	$2 \cdot \text{matches}$	$2R_{NZ} + 2R_S + 4 \cdot \text{matches}$
COO	DNS	$2L_{NZ} \cdot R_{CC}$	$3L_{NZ} + 3L_{NZ}R_{CC}$
COO	CSR	$2 \cdot \text{matches}$	$5L_{NZ} + 4 \cdot \text{matches}$
COO	CSC	$2 \cdot \text{matches}$	$L_{NZ} + 2R_{CC}L_S + L_{NZ}R_{CC} + R_{NZ}L_S + 2 \cdot \text{matches} + L_SR_{CC}$
COO	COO	$2 \cdot \text{matches}$	$3L_{NZ} + R_{NZ}L_S + 4 \cdot \text{matches}$

Table 5.1: Arithmetic and memory operation count formulae for all sixteen format combinations. matches $= L_{NZ} \cdot R_{NZ}/L_{CC}$, $L_S = \min(L_{RC}, L_{NZ})$, $R_S = \min(L_{CC}, R_{NZ})$.

Each input matrix A and B is partitioned into micro-blocks of fixed size $L_{RC} \times L_{CC}$ and $L_{CC} \times R_{CC}$ respectively. The dimensions can also be viewed as those usually found at the M_r and N_r size found in BLIS. The density of each block is computed by simply counting its non-zeros, and put into one of the density buckets. A cost table is then computed for all combinations of left format, right format, and density bucket pair using the above derived formulae, treating flops and memory operations as a combined cost weighted by an assumed memory access penalty.

For each block of A at position (i, p) and each block of B at position (p, j), we introduce binary indicator variables $a_{i,p,f}$ and $b_{p,j,f}$ denoting whether block (i, p) of A or block (p, j) of B is assigned format $f \in \{\text{dense, CSR, CSC, COO}\}$. A single-assignment constraint ensures each block is assigned exactly one format:

$$\sum_{f} a_{i,p,f} = 1 \quad \forall\, i, p \tag{5.7}$$

$$\sum_{f} b_{p,j,f} = 1 \quad \forall\, p, j \tag{5.8}$$

The objective minimises the total estimated cost over all block multiplications:

$$C = \min \sum_{i,j,p} \sum_{f_A, f_B} \text{cost}[f_A, f_B, d^A_{i,p}, d^B_{p,j}] \cdot a_{i,p,f_A} \cdot b_{p,j,f_B} \tag{5.9}$$

The product $a_{i,p,f_A} \cdot b_{p,j,f_B}$ makes this objective 'bilinear' in the decision variables, which constitutes a Mixed Integer Nonlinear Program (MINLP). The solver is implemented using Pyomo [45] with MindtPy [46] as the MINLP solver backend, using GLPK [47] for the MIP subproblems and IPOPT [48] for the NLP solver. Once solved, the optimal format assignments are written alongside the input matrices to a binary file consumed by the sparse BLIS multiplication program.

- $B_{row} = \texttt{MATRIX_SIZE}/L_{RC}$ — number of block rows in A/'Left'
- $B_{col} = \texttt{MATRIX_SIZE}/R_{CC}$ — number of block columns in B/'Right'
- $B_{inner} = \texttt{MATRIX_SIZE}/L_{CC}$ — number of blocks along the shared inner dimension

So far, the primary limitation of this approach is solver scalability. The number of decision variables grows as $O(B_{row} \cdot B_{inner} \cdot B_{col} \cdot |Formats|)$ where $|Formats| = 4$ is the number of formats, and the bilinear objective compounds this as problem size increases. In practice the solver becomes intractable for matrix sizes beyond approximately 512×512 with the micro-block sizes used in the experiments. One

direction may be increasing the micro-block size to alleviate the rapid growth of the number of decision variables.

5.2.4 Sparse with BLIS structure

The sparse BLIS implementation follows the hierarchical loop structure of BLIS [13], preserving its cache-level blocking parameters m_c, n_c, k_c and micro-kernel blocking parameters m_r, n_r, k_u as compile-time constants. Rather than operating on scalar elements or fixed-format compressed matrices, the innermost micro-kernel operates on heterogeneous sparse blocks — each of size $m_r \times k_u$ for the left matrix and $k_u \times n_r$ for the right — where each block may independently be stored in any of the four formats.

Prior to multiplication, each input matrix is packed into a `Multisparse_Matrix` structure by the `multi_sparsify` function, which partitions the dense input into blocks and converts each block to its assigned format according to the format assignment produced by the ILP solver, in the order consumed by the multiply. This packing is performed once and before the actual operation begins, as opposed to the standard BLIS practice where packing into the A_c and B_c panels happens during operation.

The multiplication itself is structured as the standard five-loop BLIS nest, but now iterating over block indices rather than element indices. At the innermost level, each pair of left and right blocks is dispatched to the appropriate kernel via a computed enum value encoding the format combination, covering all sixteen format pairings. The result of each micro-kernel is accumulated into a small dense temporary buffer of size $m_r \times n_r$ before being written back to the output matrix, very similarly to the original BLIS.

Listing 5.3: Multisparse Matrix struct

```
struct Matrix_Union
{
  Matrix_Format format;
  union
  {
    Dense_Matrix dense;
    CSR_Matrix csr;
    CSC_Matrix csc;
    COO_Matrix coo;
  };
};
struct Multisparse_Matrix
{
  // Overall.
  usize row_count;
  usize col_count;

  Matrix_Union *blocks;
  usize blocks_row_count;
  usize blocks_col_count;
};
```

Three configurations are benchmarked against each other: the solved optimal heterogeneous format assignment, a uniform block format baseline where all micro-blocks use the same user-specified format, and a global compression baseline where the entire matrix is compressed into a single format without any type of block structure. Correctness of the sparse BLIS output is verified by comparison against a reference multiplication within an epsilon.

5.2.5 Kronecker Graphs

Kronecker graphs were selected as an experimental dataset for two reasons. First, their recursive self-similar structure produces adjacency matrices with a hierarchical sparsity pattern, the same block density distribution repeats at every scale which we hypothesized might align naturally with the block-level format assignment of the approach, potentially allowing denser sub-blocks to be stored in dense or CSR format while sparser sub-blocks use COO. Second, the recursive structure of the cost model meant that format assignments computed at a smaller scale might be reusable at larger scales, potentially reducing solver cost. Experiments were conducted using the AS-Newman autonomous systems graph, generated as a Kronecker graph with varying scale parameter k, giving matrices of size $2^k \times 2^k$. We swept k from 7 to 15, running both the Sparse BLIS configuration and the global CSR×CSR baseline on each, with results reported in Section 4.3. In practice however, the self-similar sparsity of Kronecker graphs meant that all blocks at every scale fell into the same low density bin, rendering format selection trivial and providing no opportunity for heterogeneous assignment to reduce cost over a uniform sparse format. Due to solver scalability limitations discussed in Section 3.3, matrices larger than 256×256 were run using a constant block format assignment of CSC x CSR rather than the solver. This

Listing 5.4: Sparse BLIS Implementation

```
for (usize block_j_o = 0; block_j_o < block_n; block_j_o += BLOCK_J)
{
  for (usize block_p_o = 0; block_p_o < block_k; block_p_o += BLOCK_P)
  {
    for (usize block_i_o = 0; block_i_o < block_m; block_i_o += BLOCK_I)
    {
      for (usize block_j_i = 0; block_j_i < BLOCK_J; block_j_i += 1)
      {
        for (usize block_i_i = 0; block_i_i < BLOCK_I; block_i_i += 1)
        {
          Dense_Matrix temp =
          {
            .values = (f64[BLOCK_MR * BLOCK_NR]) {0},
            .row_count = BLOCK_MR,
            .col_count = BLOCK_NR,
          };
          for (usize block_p_i = 0; block_p_i < BLOCK_P; block_p_i += 1)
          {
            usize block_i = block_i_o + block_i_i;
            usize block_j = block_j_o + block_j_i;
            usize block_p = block_p_o + block_p_i;

            Matrix_Union left_block = left.blocks[block_i
                                    * left.blocks_col_count + block_p];
            Matrix_Union right_block = right.blocks[block_p
                                    * right.blocks_col_count + block_j];
            do_sparse_matmul(temp, left_block, right_block);
          }
          for (usize j_r = 0; j_r < BLOCK_NR; j_r += 1)
          {
            for (usize i_r = 0; i_r < BLOCK_MR; i_r += 1)
            {
              usize row = (block_i_o + block_i_i) * BLOCK_MR + i_r;
              usize col = (block_j_o + block_j_i) * BLOCK_NR + j_r;

              usize output_index = row * output->col_count + col;
              usize temp_index = i_r * temp.col_count + j_r;

              output->values[output_index] += temp.values[input_index];
            }
          }
        }
      }
    }
  }
}
```

was in practice inconsequential, because for all Kronecker graphs tested the solver assigned CSC x CSR uniformly anyway, as every block fell into the same low density bin and the cost table therefore selected the same format for every block regardless of position.

5.3 Analysis

5.3.1 Format Benchmarking and Instrumentation Results

For the graphs below, we swept both left and right matrices through a range of densities.

Figures 5.1 and 5.2 show observed and predicted FLOP and memory operation counts across the full density range for all sixteen format combinations, with both left and right matrices swept independently. The derived formulae show strong agreement with observed counts for fourteen of the sixteen combinations across the full density range, with relative error remaining below 5% for most format pairs at densities above 0.1, as shown in Figures 5.3 and 5.4.

The two exceptions, CSR×CSC and COO×CSC, exhibit higher error at low densities. Both involve a merge intersection where both operands reset simultaneously on different outer loop dimensions, producing memory cost terms that resist clean derivation under the uniform sparsity assumption. The formulae for these two cases capture the qualitative structure but with greater quantitative error than the remainder.

A notable property of the FLOP formulae is that they are exact, that is, floating point operation count is determined purely by the number of matched nonzeros and is independent of format, so all sparse×sparse combinations share the same FLOP formula. The memory operation formulae are where format choice has its effect,

encoding the differing access patterns of each combination.

Figure 5.5 shows the runtime performance of all sixteen format combinations across the density sweep. Several crossover points are visible as density increases: sparse formats such as COO and CSR are fastest at low densities where their ability to skip zeros outweighs their indexing overhead, while dense×dense becomes competitive and eventually dominant at higher densities where the cost of sparse bookkeeping exceeds the savings from skipping nonzeros.

Figure 5.6 shows the roofline model for all sixteen combinations. All kernels fall well below the memory bandwidth ceiling and far below the compute ceiling, confirming that these kernels are memory bound at all densities tested. The arithmetic intensity is low across the board, consistent with sparse formats that perform few FLOPs per byte loaded. No kernel approaches the compute roof, indicating that optimizing memory access patterns is more impactful than optimizing arithmetic throughput for this class of operation, unsurprisingly. A notable observation is that CSC×dense performs surprisingly close to dense×dense across the full density range, and was indeed selected by the solver for diagonal blocks in the banded diagonal experiments. This suggests CSC is an unusually cache-friendly format for the left matrix when the right matrix is dense, likely due to its column-major access pattern aligning well with the row-major accumulation into the output.

5.3.2 Format Selection Optimization Results

An important finding concerns the cost model used to drive the format selection. The analytical formulae derived in Section 3.2, while accurate predictors of operation counts, proved to be poor proxies for actual runtime when used directly as the solver objective. Format assignments produced using the formula-based cost model were

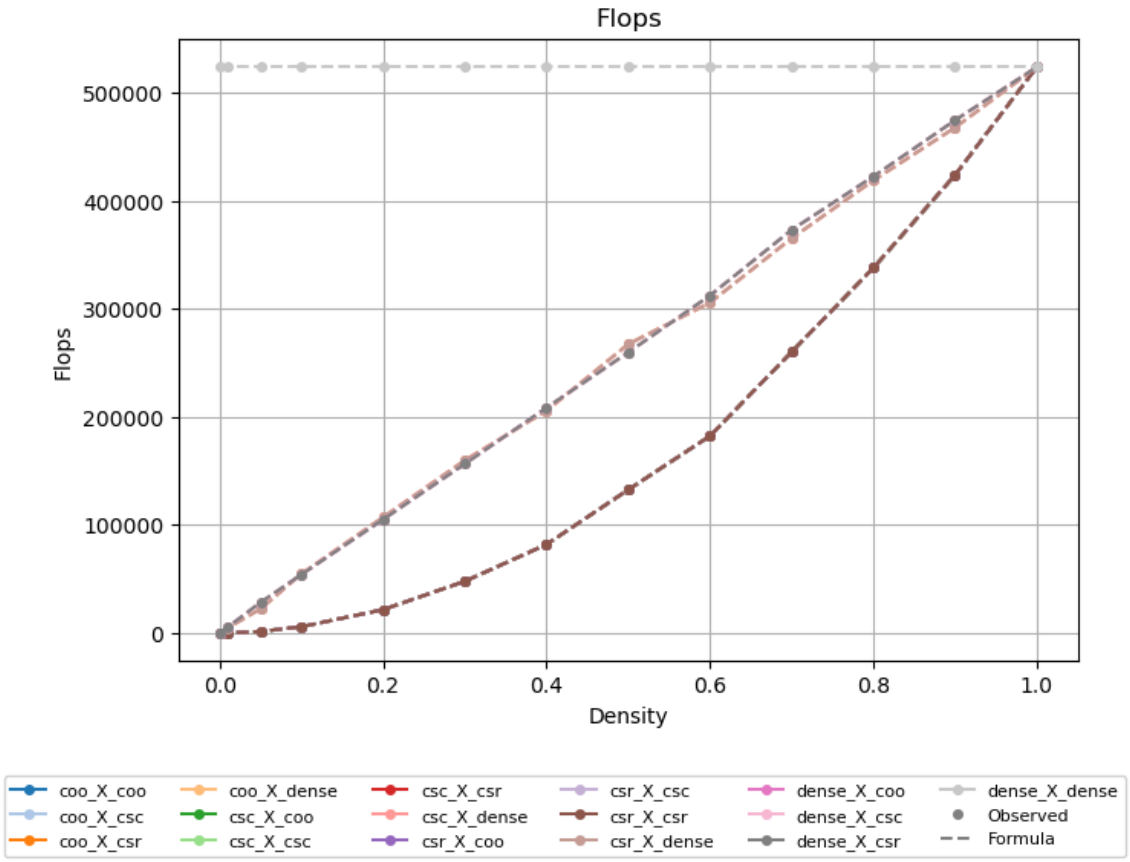


Figure 5.1: Observed and Formula FLOPs. Observed data points fall mostly on the dotted line plotted from formulas.

outperformed by assignments produced using empirically measured cycle counts from the density sweep as the cost function. This suggests that operation count alone is an insufficient proxy for runtime at this block size, likely because cache effects and branch misprediction dominate over raw arithmetic and memory operation counts for 32×32 blocks. In essence, the formulas mistakenly treat flops and memops as being comparable on their face, when they are not. The formulas could perhaps become more predictive at larger block sizes where compute cost dominates over fixed per-block overheads.

Figure 5.7 shows the performance of the three configurations on a synthetic banded

diagonal 256×256 matrix, the most favorable case for heterogeneous format selection as it exhibits clear density variation between diagonal and off-diagonal blocks. Even here, the solver-assigned configuration requires 408,725 cycles, compared to 239,189 cycles for global CSR x CSR, a 41% gap. The constant blocking configuration performs worst at 658,040 cycles, suggesting that solver assignment does at least select better formats than a naive uniform choice in this case, but neither blocking approach approaches the performance of the flat global baseline, unfortunately.

Figure 9 shows thread scaling. Performance does not improve meaningfully with additional threads and degrades at 8 threads, indicating the workload is too small and too irregular to benefit from parallelism at this scale.

Figure 10 shows the Kronecker graph sweep. Sparse BLIS scales poorly with matrix size while global CSR×CSR remains effectively flat, reflecting that block dispatch overhead grows with the number of blocks while the actual sparse compute remains minimal due to the uniformly low density of Kronecker graph matrices.

5.3.2.1 Blocking Sensitivity

Figure 5.8 shows performance across block sizes of 32, 64, and 128. Cycle count increases with block size, likely due to increased cache pressure from the growing dense accumulation buffer and greater work per block pair in the inner loop nest.

5.3.2.2 Thread Sensitivity

Figure 5.9 shows thread scaling. Performance does not improve meaningfully with additional threads and degrades at 8 threads, indicating the workload is too small and too irregular to benefit from parallelism at this scale.

5.3.3 Kronecker Graph Results

Figure 5.10 shows the Kronecker graph sweep. Sparse BLIS scales poorly with matrix size while global CSRxCSR remains effectively flat, reflecting that block dispatch overhead grows with the number of blocks while the actual sparse compute remains minimal due to the uniformly low density of Kronecker graph matrices.

As noted in Section 3.5, the solver assigned a uniform CSCxCSR format to all blocks for every Kronecker graph tested, collapsing the heterogeneous assignment to a constant blocking scheme identical to the baseline. This confirms that the density variation required to motivate mixed format selection is absent in these datasets.

Figure 5.11 presents a sweep over progressively larger block sizes for Sparse BLIS. While the conventional micro-block dimension is 32×32, this experiment extends into block sizes that can no longer be strictly classified as micro-blocks. Nevertheless, even at these larger granularities, the additional blocking overhead proves insufficient to overcome the inherent inefficiencies, at least for this dataset.

5.4 Summary

This chapter presented a study of SpGEMM across all sixteen pairings of four sparse matrix storage formats, deriving analytical formulas for floating-point and memory operation counts that show strong agreement with empirically measured values for fourteen of the sixteen combinations.

Building on these formulas, I implemented a BLIS-inspired [13] sparse matrix multiplication pipeline that assigns storage formats independently to each micro-block and dispatches to the corresponding kernel at runtime. Format selection is formulated as a MINLP and solved using Pyomo [45] with MindtPy [46]. Evaluation on banded

diagonal matrices and Kronecker graph datasets [16] shows that block dispatch overhead consistently dominates runtime, preventing the heterogeneous approach from matching a flat global CSR×CSR baseline despite producing better format choices than a naive uniform assignment.

Several directions could address these limitations. A more uniform blocking/adaptable format for the inner most microblocks could be explored. A lighter-weight greedy format selection heuristic, rather than a full MINLP solve, could reduce planning overhead. Finally, the approach could be better suited to architectures with less penalty for very branch-y inner loops.

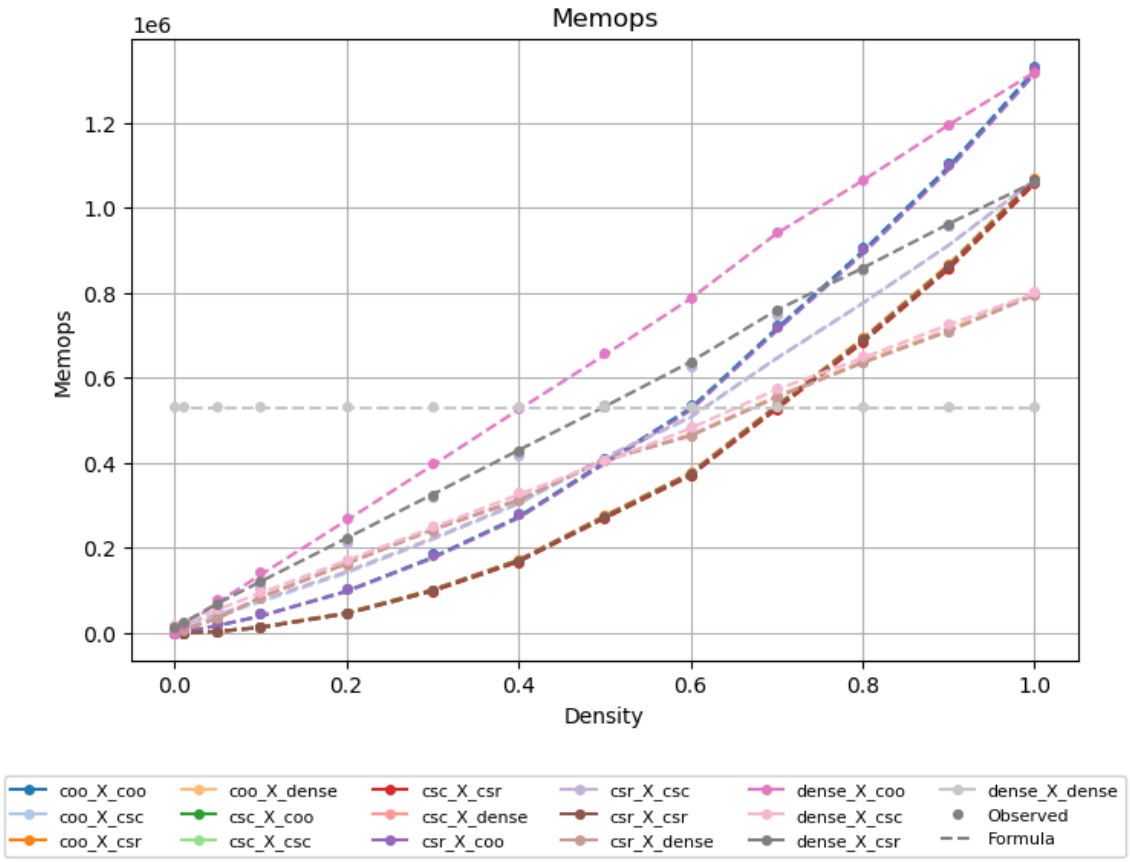


Figure 5.2: Observed and Formula memops. Observed data points fall mostly on the dotted line plotted from formulas.

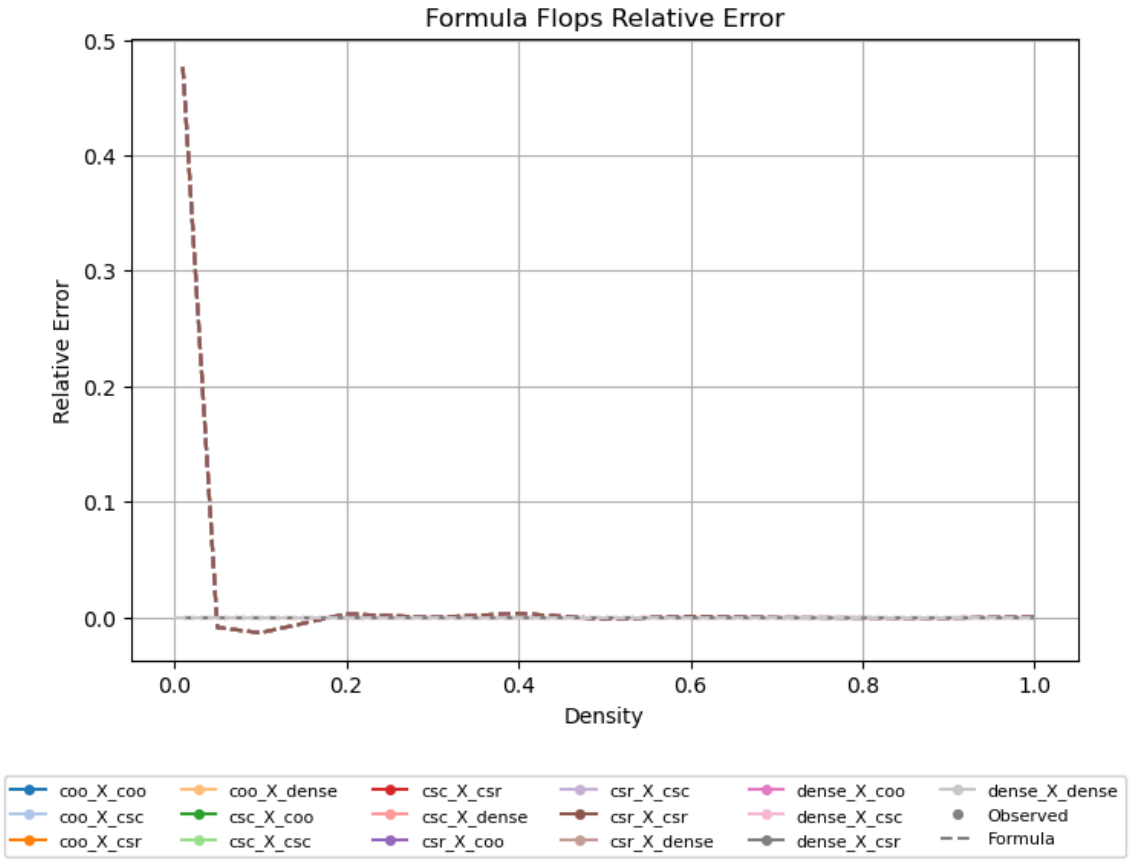


Figure 5.3: FLOPs: Relative error of Formula vs Observed. Very close fit by the formula, and flop count is determined entirely by compression combination.

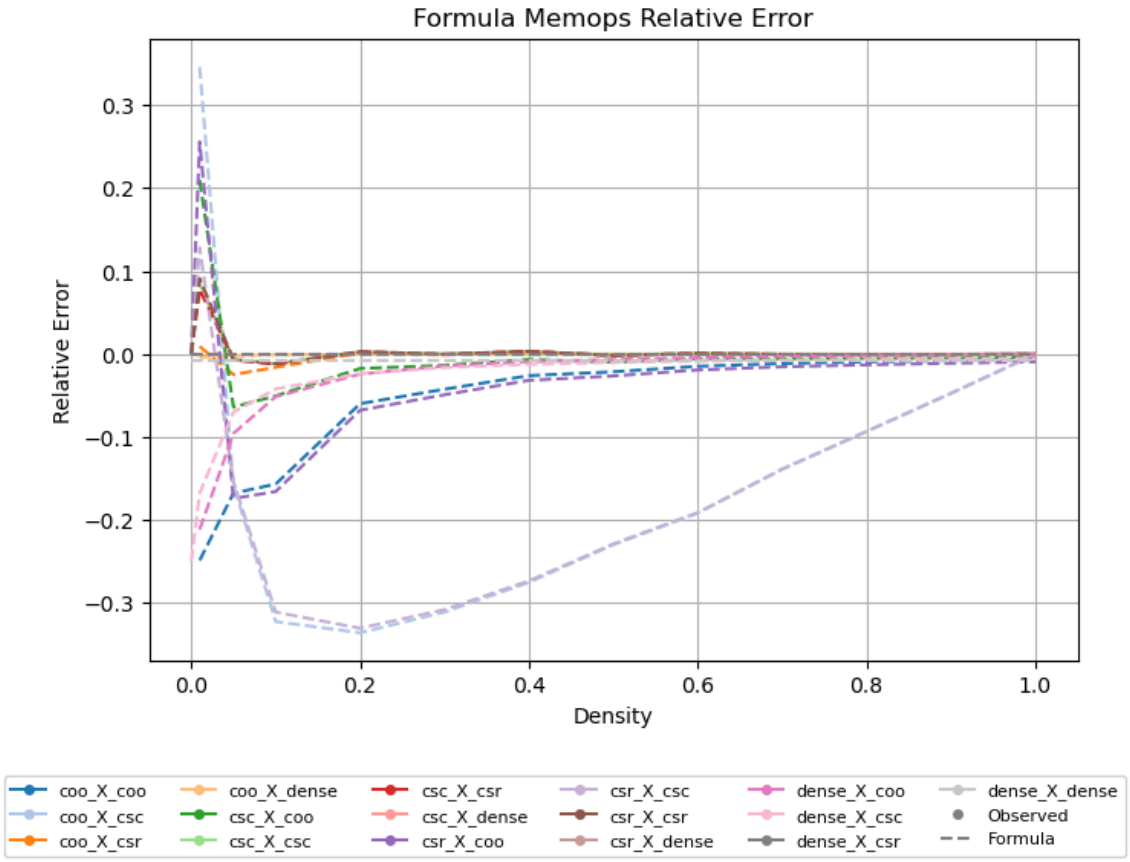


Figure 5.4: memops: Relative error of Formula vs Observed. Close fit by most formulas, variability due to the effect of variability of sparsity patterns.

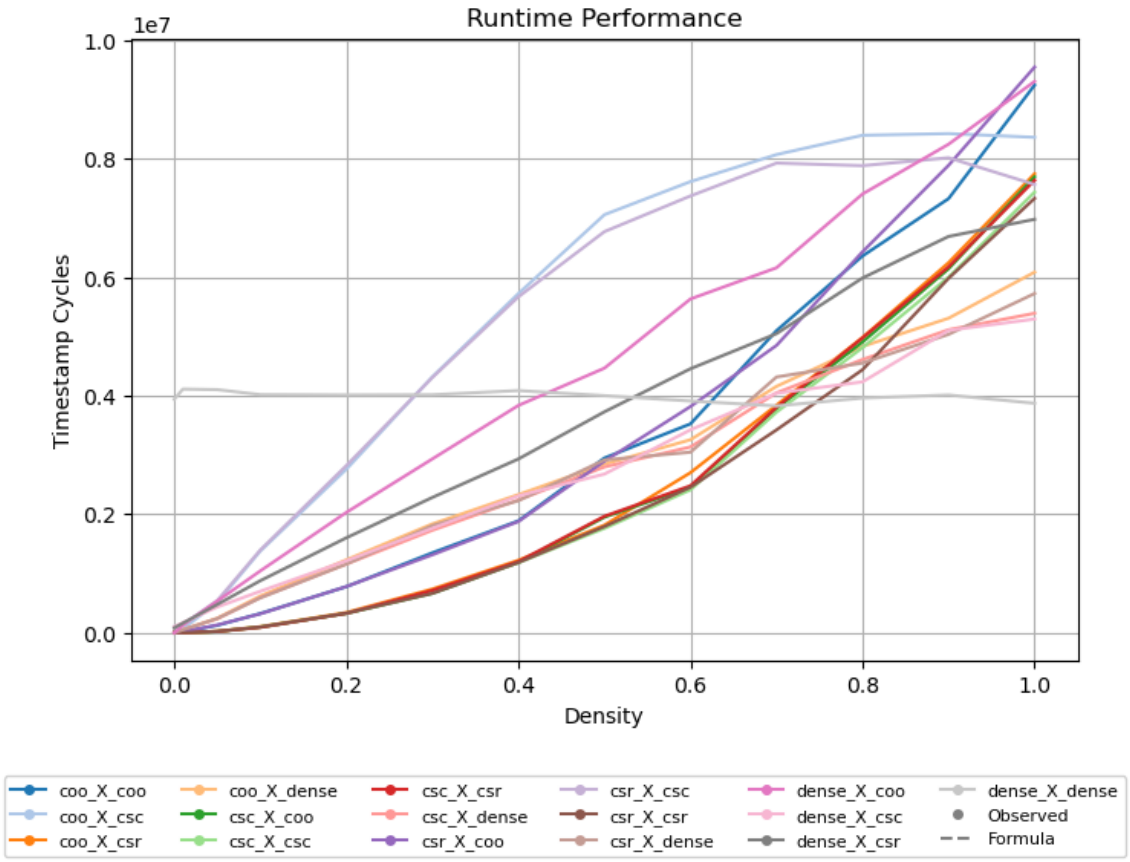


Figure 5.5: Runtime Performance: Timestamp Cycles. Crossover points at 0.7% density when performing a simple dense x dense is better.

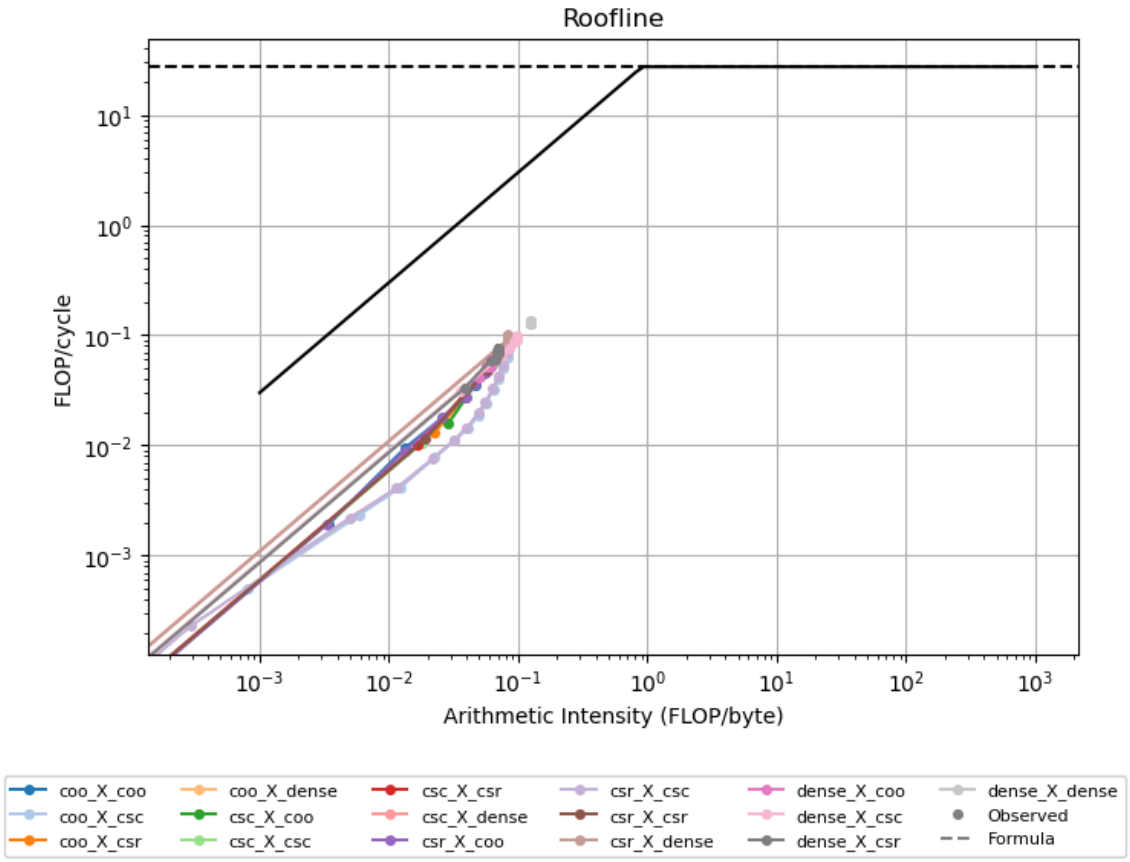


Figure 5.6: Roofline Model. Sparse formats are entirely memory bound.

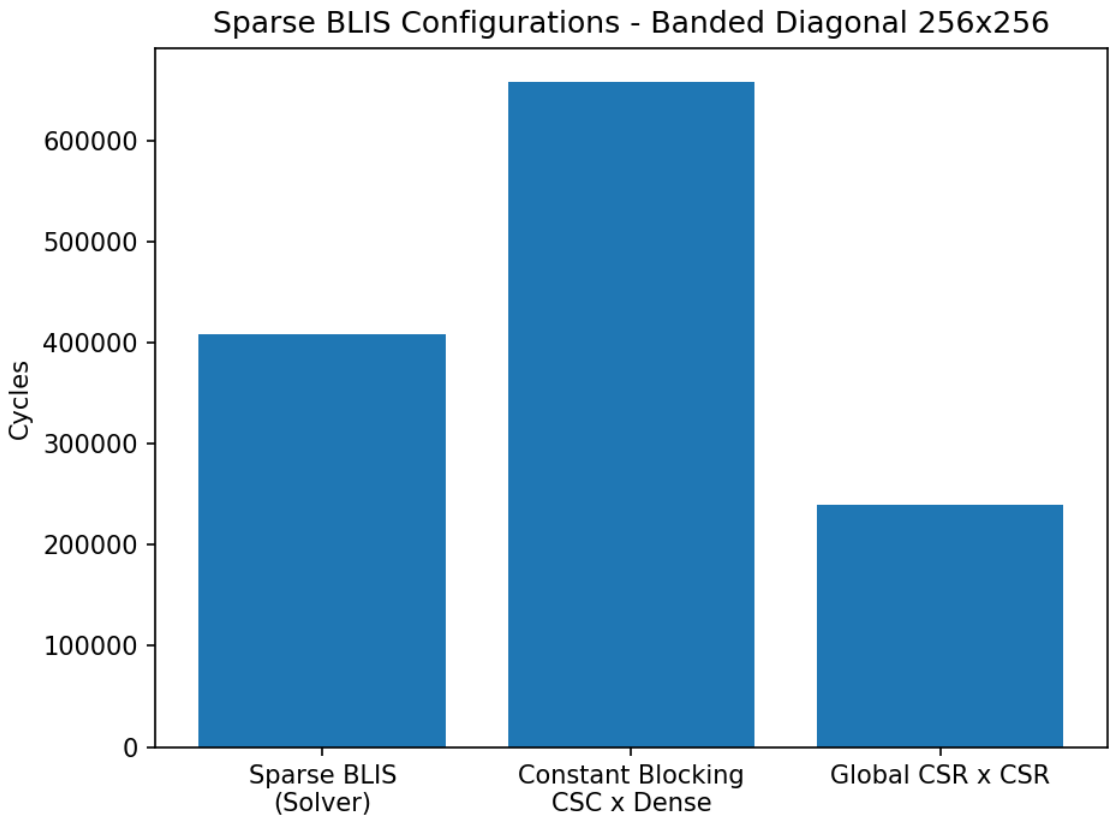


Figure 5.7: Diagonal Matrix Results. Poor performance compared to simple global compression, but optimal micro-format selection wins over naive format selection.

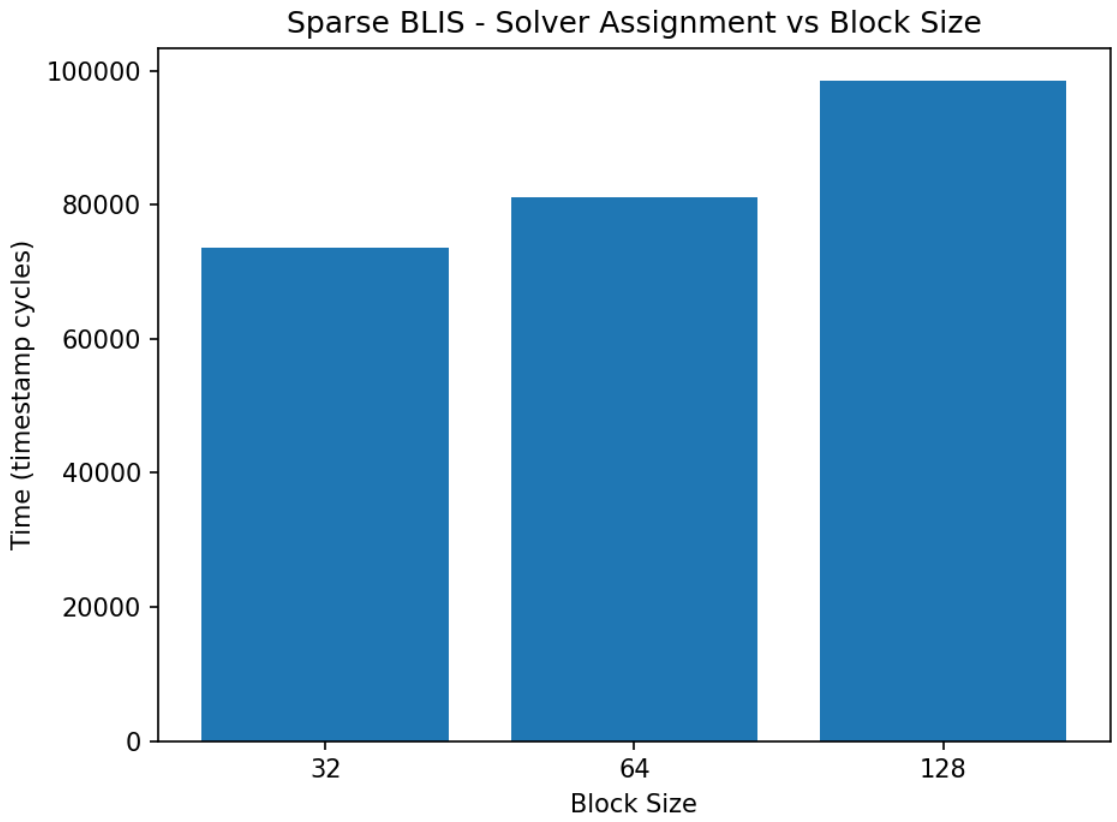


Figure 5.8: Block Size Performance Impact. Performance degrades as block sizes increase.

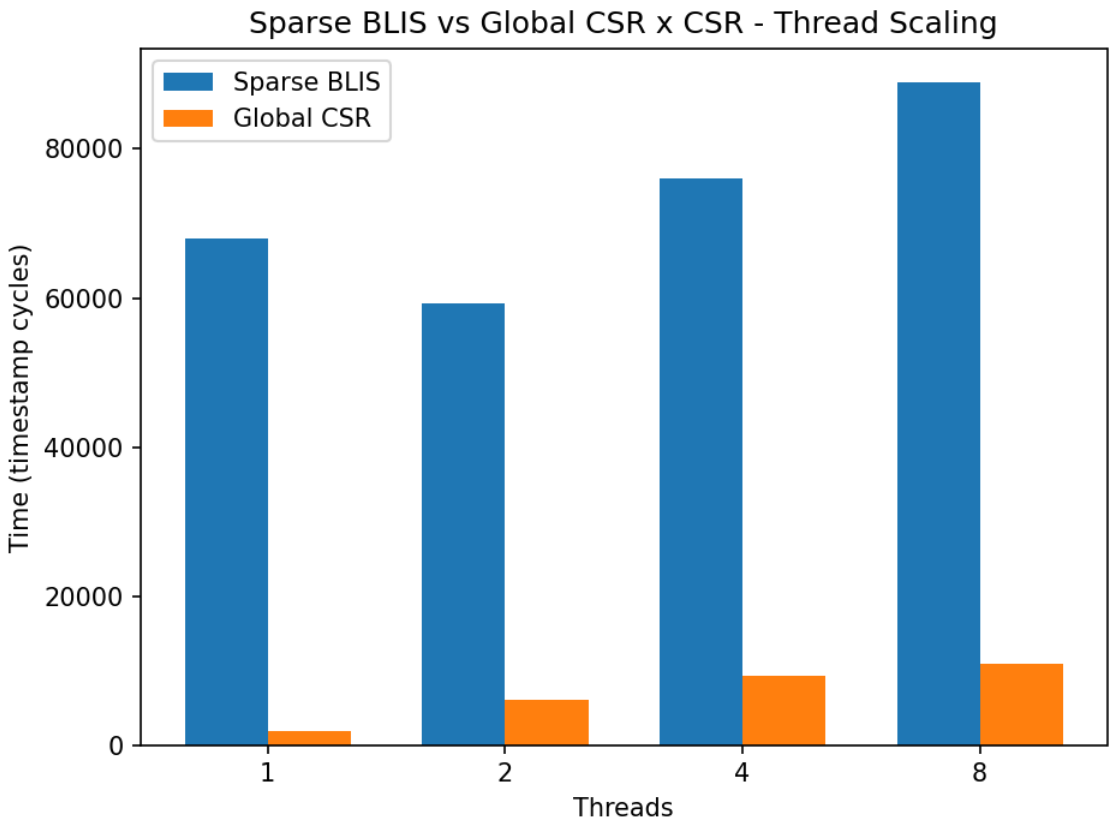


Figure 5.9: Thread Count Performance Impact. At matrix sizes that can be solved, threading provides minimal benefit beyond 2 threads.

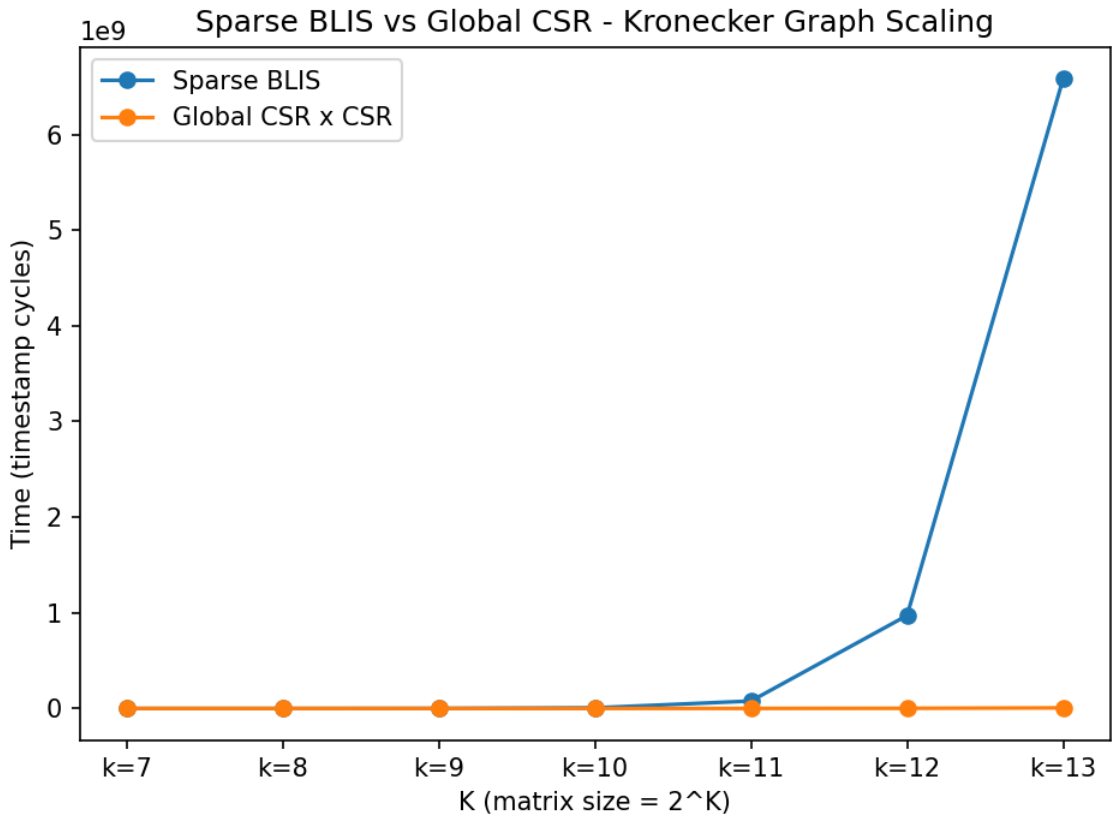


Figure 5.10: Kronecker Graph Sweep (AS-Newman). Sparse BLIS performs very poorly compared to a global compression strategy, this worsens as matrix size increases.

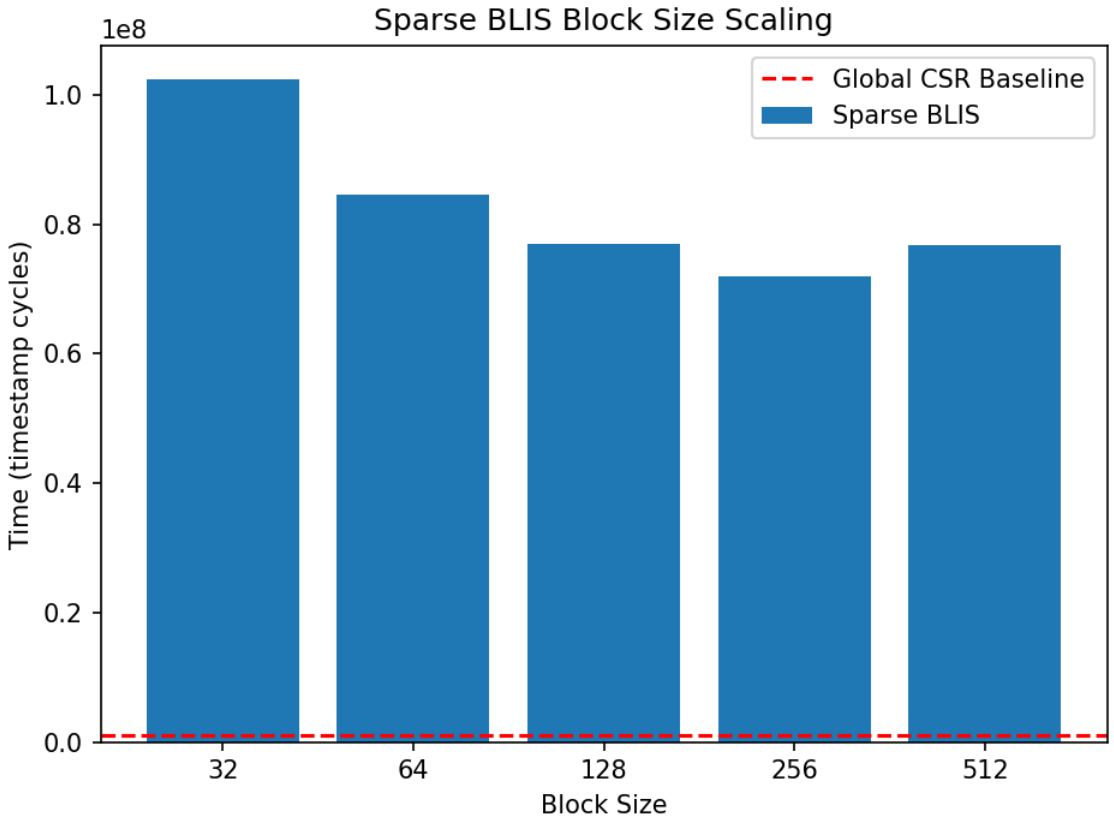


Figure 5.11: Kronecker Graph k=10 (AS-Newman) block size sweep. Block dispatch overhead generally decreases as block size increases, but not nearly enough to be comparable to simple global compression.

Chapter 6

Kronecker Sweep

6.1 Introduction

The preceding chapter left me with a few open questions. The analytical cost models of Chapter 5 predicted operation counts accurately, but operation counts proved a very poor proxy for runtime (perhaps predictably, a memory operation in one context is sometimes orders of magnitudes slower than a memory operation in another due to caching), and the heterogeneous per-block pipeline built on them did not outperform a single simple globally chosen format. Those experiments could show that the models failed to predict performance, but not exactly why. They also relied on a small set of synthetic inputs, so they said little about how the best format combination changes as a matrix grows or as its structure changes.

This chapter returns to the these unanswered with the upgraded reptest library and a bit more systematic experiment design. Rather than building a more elaborate selection mechanism first, I asked the more basic question I should have started with. For sparse×sparse matrix multiplication, how does the best pairing of storage formats vary with matrix structure and scale, and, importantly, what hardware behavior

explain these differences? This question informed the design choices I made below.

Structured inputs. Graph adjacency matrices are a common source of sparse matrix inputs, and Kronecker graphs [16] are a natural family to study. They are generated from a small 'initiator' matrix by repeated Kronecker products, so their structure is reproducible while still exhibiting structure commonly associated with real networks. In addition, the same graph family can be generated at many scales.

Sweeping scale instead of comparing individual graphs. Sparse linear algebra benchmarks are often reported as results on a collection of well-known, common matrices/graphs, typically as one bar per matrix. Such comparisons can be informative, but they confound size, density, and structure, and it can be hard to determine where or why performance behavior changes. Because Kronecker graphs can be generated at successive scales of k, each family can be swept across k, producing continuous curves in which crossover points between formats are visible directly.

Explaining performance with hardware counters. Chapter 4 describes the benchmarking library I built, which reads hardware counters through Linux `perf` sycalls. In the earlier experiments, only operation counts and cycle counts were available, and the gap between them could only be speculated about. In the new experiments, each reptest measurement also records cache misses, branch mispredictions, page faults so that differences in runtime between format pairings can be tied to specific observed behavior rather than inferred or guessed.

TACO-generated kernels. In Chapter 5, the kernels were rewritten by hand to embed instrumentation. In this chapter, kernels are generated by the TACO compiler [15], which derives code for any combination of formats from a single algebraic spec-

ification with 'scheduling' commands. This makes the comparison between formats more even-handed, since every pairing is produced by the same process. However, it also exposed practical limits of using TACO at scale. TACO generated code assumed 32-bit sizes in allocation, which fails for most kronecker matrices tested of roughly $2^{20} \times 2^{20}$ and above. I describe the changes made to support these sizes in Section 6.2.

Scope. I restricted my attention to sparse×sparse products, as it would be infeasible to densely materialize any of these matrices as we approach larger K's.

Blocking. I wanted to explore blocking one more time, as the implementation of a sparse BLIS in Chapter 5 left a lot to be desired. This time I used TACO's ability to define any number of additional dimensions to turn CSR × CSR and CSC × CSR into 4 dimensional matrices in which the outer 3 dimensions are dense. This, in effect, replicates the 'tiling' behavior we are after.

6.2 Taco Limitations

While conducting the experiments for this chapter I ran into and subsequently fixed a number of issues with TACO generated code. Most importantly, TACO has a bad habit of using int32_t types for calculating sizes to pass to malloc and calloc during the symbolic 'assemble' stage. This initially greatly limited the maximum Kronecker K I could test with, as these sizes were overflowing, becoming negative, and then being cast back into a size_t type when passed to malloc and thus requesting program crashing amounts of memory. This did however lead me to develop instrumentation to collect and check any allocation request from TACO code, which gave us another data point to analyze. The next issue I needed to address was that TACO can be quite inconsistent in how it sets up output accumulation. For some format combos

it defaults to a dense accumulator method like those found in Gustavson or outer product dataflows, while in others it does not and instead outputs as duplicated COO where the user is expected to accumulate themselves. Of course, these can be unified through careful use of the 'precompute' and 'reorder' schedules, but it was frustrating to learn that it is not consistent in whether that is required or not.

6.3 Experimental Setup

I, of course, made use of my Repetition Test library. All kernels were initially generated using TACO, and then fixed up as mentioned in Section 6.2. The program loads a specified Kronecker from disk into a COO format that TACO expects for their 'Pack' functions. It then sweeps over all of the kernels packing, assembling, and finally timing the compute function. All this is then saved to a CSV for use in python visualization/plotting scripts. All of this functionality is configurable via command line arguments.

As well, testing the blocked/tiled formats is exposed as a separate mode, as there is extra work to do in reorganizing the loaded Kronecker COO into a 4 dimensional COO for passing to the Pack functions.

The input matrices are the Kroneckers generated by mfit [17], fitted to the original real-world graphs presented in [16].

6.3.1 Kernels

Each kernel computes $C = A \times B$ with A and B stored in one of the formats CSR, CSC or COO and C is always stored in COO. Kernels, again, are generated by TACO. The pairs fall into the three dataflows as follows. It is important to note that any format pair which includes COO must do a scan for row or column-wise access even

if not requiring an inner-product like merge step.

- *Row-wise (Gustavson).* `CSR_x_CSR`, and `COO_x_CSR`. Rows of A index the outer loop and the matching rows of B are gathered.
- *Outer product.* `CSC_x_CSR` and `CSC_x_COO`. The index k is the outermost loop and each iteration forms the outer product of column k of A with row k of B.
- *Inner product.* `CSR_x_CSC`, `COO_x_CSC`, and `COO_x_CSR`. Each output entry is a dot product of a row of A with a column of B.

At $k \geq 16$ I ran only four pairs, `CSR_x_CSR`, `CSC_x_CSR`, `CSC_x_COO` and `COO_x_CSR`, because the inner-product like pairs were already orders of magnitude slower at smaller k.

Listing 6.1 shows the compute loops of the three kernels that matter most for the results. It is important to notice one thing for later, `CSC_x_CSR` iterates over all n values of the k index, including those for which column k of A or row k of B is empty, whereas `CSC_x_COO` walks only the stored entries of B.

Listing 6.1: Compute loops of the generated kernels, declarations omitted.

```
// CSR_x_CSR_COO: row-wise, gathers B's row for each stored A entry
for (int32_t i = 0; i < A1_dimension; i++)
  for (int32_t kA = A2_pos[i]; kA < A2_pos[i + 1]; kA++) {
    int32_t k = A2_crd[kA];
    for (int32_t jB = B2_pos[k]; jB < B2_pos[k + 1]; jB++)
      { C_vals[jC] = A_vals[kA] * B_vals[jB]; jC++; }
  }
```

```
// CSC_x_CSR_COO: outer product over every k
for (int32_t k = 0; k < B1_dimension; k++)
  for (int32_t iA = A2_pos[k]; iA < A2_pos[k + 1]; iA++)
    for (int32_t jB = B2_pos[k]; jB < B2_pos[k + 1]; jB++)
      { C_vals[jC] = A_vals[iA] * B_vals[jB]; jC++; }

// CSC_x_COO_COO: outer product over the stored k of B only
while (kB < pB1_end) {
  int32_t k = B1_crd[kB];
  int32_t B1_segend = kB + 1;
  while (B1_segend < pB1_end && B1_crd[B1_segend] == k) B1_segend++;
  for (int32_t iA = A2_pos[k]; iA < A2_pos[k + 1]; iA++)
    for (int32_t jB = kB; jB < B1_segend; jB++)
      { C_vals[jC] = A_vals[iA] * B_vals[jB]; jC++; }
  kB = B1_segend;
}
```

6.4 Results

6.4.1 The pair family matters most

Starting with the simplest and most predictable result: pair family matters by order of magnitude, but mostly in one direction. Inner-product forcing format combos are catastrophically slow, always. This is to be expected since, as mentioned previously, this requires a intersection step for each output value akin to the merge step in a merge sort, this is also reflected in the hardware counters, as evidenced by much higher branch mispredictions and cache misses compared to other format pairs. Just

avoiding the bad format combinations is one big win.

However, among those pairs which are not staggeringly bad, the best are often within a few percentage points of each other. Figure 6.4 shows that just using CSC×CSR always is withing 3% of the true best on average. The true best often follow outer-product or Gustavson formulations, though where these crossover will be more examined later.

The conclusion we can draw is that, on average, just avoiding terrible format combinations will do most of the work, and being very clever on which is the absolute best is not worth it.

6.4.2 Switch from CSR × CSR to CSC × CSR / CSC × COO

For relatively small input matrices `CSR_x_CSR`, is the fastest or with a few percent of the fastest for all graphs. However, at a few thousand nonzeroes almost every single graph switches over to `CSC_x_CSR`, see Figure 6.2. The moment of the switch is very similar across graphs whose structure is very different.

It is important to ask whether this is a structural effect or an artefact of how k can confound size and density. Within one graph the switch happens at one particular k, but that value of k is also a particular size, density and nnz. Plotting the winner against each variable separately, across graphs, shows that nonzeros aligns the switch best. Graphs of equal k switch at different times, and graphs of different k switch at much more similar nnz buckets.

Looking for hardware explanations, we can turn our attention to the branch misprediction counts, please see the extensive figures in the appendix for each individual graphs hardware counters, Chapter B. In `CSR_x_CSR` the inner loop's trip count is the length of a row of B selected by a column index $k =$ `A2_crd[kA]` that varies from one

iteration to the next, so the loop exit is hard to predict. In `CSC_x_CSR` the same trip count is reused for every entry of column k of A, so after the first iteration the predictor has most likely learned it. This also removes the random read of `B2_pos[k]`. This is not to say that I am certain this is the mechanism for the downstream performance crossover, but that it seems to have good explanatory power of those data I collected.

6.4.3 Outer Product Tie

Looking at Figure 6.1, we can see that once we cross over from `CSR_x_CSR` to the other formats, the difference between the 2 is often very small, and could be reasonably attributed to noise. However one outlier exists with ATP-GR-QC, where `CSC_x_COO` is up to 1.5x faster than `CSC_x_CSR`. To examine why I believe this happens we can look at the properties of the kernels and a few simple quantities of the graphs.

`CSC_x_CSR` and `CSC_x_COO` differ only in how B is traversed. `CSC_x_CSR` loops over all n values of the k index, including those with no work, at a fixed cost each. `CSC_x_COO` loops over the stored entries of B only, at a slightly higher cost per entry. If c_1 and c_2 are those two per-iteration costs, `CSC_x_COO` should win when $\text{nnz}/n < c_1/c_2$, that is, when there are few nonzeros per row.

The data agree (Table 6.1). With more than about 2.5 nonzeros per row the two kernels are within 1–3% of each other, which is inside the noise, so the "winner" there is arbitrary. Between one and 2.5 nonzeros per row `CSC_x_COO` is a few percent faster, and with fewer than one nonzero per row it is about 1.5 times faster.

Table 6.1: Median time of `CSC_x_COO` divided by `CSC_x_CSR`, by nonzeros per row.

nnz / row	median ratio	range	points
< 1 (ATP-GR-QC only)	0.664	0.657–0.705	9
1–2.5	0.957	0.880–1.083	50
> 2.5	1.011	0.894–1.115	99

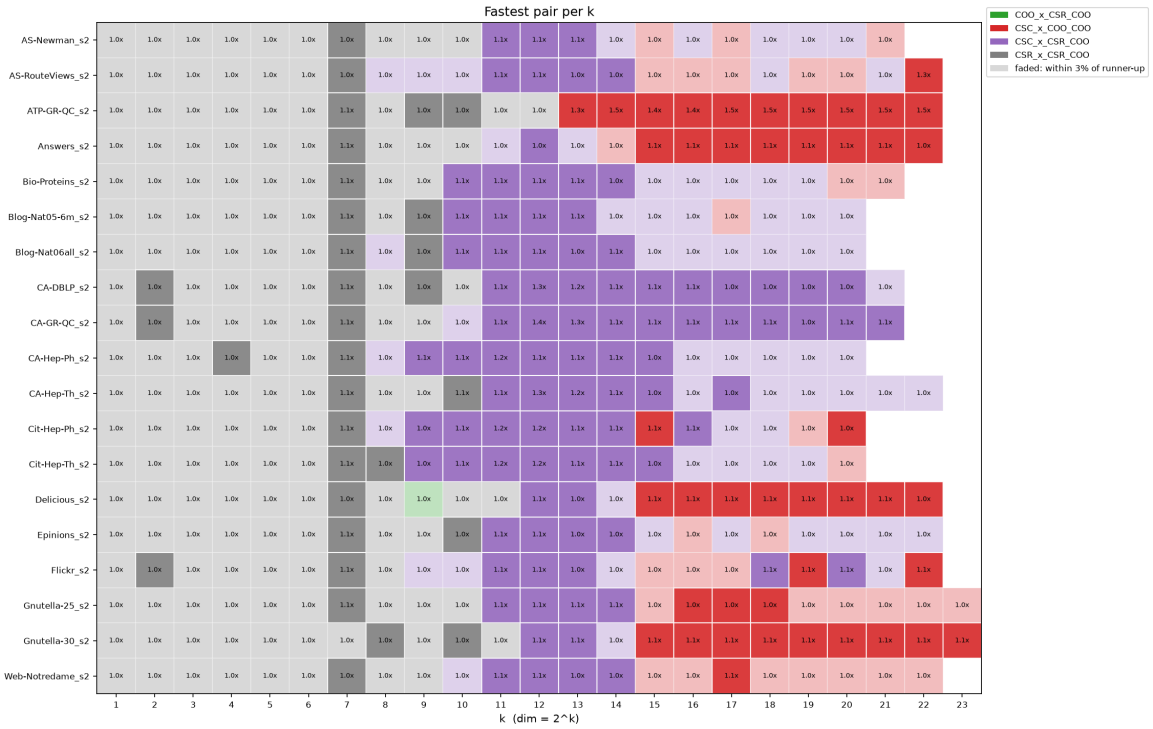


Figure 6.1: This figure shows the best performing format combo for each of the studied graphs at each K. Each square is also labeled with a speedup over runner up (e.g. 1.1x). There is a broad trend of the best being CSR × CSR at smaller K, then transitioning to CSC × CSR or CSC × COO at larger K. In most cases the difference between CSC × CSR beating CSC × COO or vice versa could potentially be contributed to noise as often they are runner ups of each other within 1.1x. 1 Outlier exists with ATP-GR-QC where CSC × COO beats CSC × CSR by an average 1.5x.

The sparsest case is ATP-GR-QC, which has fewer nonzeros than rows at every $k \geq 13$. ATP-GR-QC is the only graph below one nonzero per row, so the conclusion rests on just this one experiment, though it is the only outlier in this fashion.

6.4.4 Tiling does not help, at these sizes

Tiling is a standard way to improve locality in sparse kernels, so I also generated tiled versions of `CSR_x_CSR` and `CSC_x_CSR` and swept the block size at every k. The question I wanted answered was whether some block size beats the others, and whether the best size depends on k or on the graph. I swept block sizes by power of

2 grids of up to 32x32 blocks, whatever the inner size of those blocks would be.

The answer is that the best block size is the largest one tested, or in other words no tiling. At every k it is 2^k or within one or two powers of two of it, which means a single tile covering the whole matrix, and the same holds for both kernels and nearly all the graphs (Figure 6.5 and the per-graph heatmaps in Appendix C). The few graphs where the winner is a step or two smaller at $k \geq 17$ are differences of a few percent, reasonably inside noise.

This is a negative result for these kernels but it does not rule out tiling for other kernels, such as ones that accumulate might into a dense row or partial matrix, though Chapter 5 also found tiling to be net-negative. As well, a few matrices like AS-Newman, see Figure C.1, show a somewhat promising trend of the relative slowdown of blocking decreasing as the size of the matrix increases. On a sufficiently large matrix (which would probably required a distributed memory system), this trend might even cause some block sizes to beat no blocking at all, but that is just speculation.

6.5 Discussion and Summary

The data support three claims: the pair family is the dominant choice, a fixed pair among the best ones is pretty much good enough (at least to the degree I can't rule out measurement noise), and the cases where the best pair changes are predicted by nnz or nnz/n. A selector that uses only these two numbers and the known good family format pairs could, in theory, reproduce most of what an oracle would do.

This is not to say that that structure is irrelevant. At equal k the Kronecker graphs differ only in structure, and the winner does differ between them in some cells of Figure 6.1. But the claim is a bit narrower. Once nonzeros, nonzeros per row and output size are accounted for, the data finds no further effect that I could attribute

to structure, and the crossover points we looked at align with those quantities. Other structural properties should be examined in future to find if the crossover points fit more closely to them. As well, surprisingly, density (Figure 6.2 does not seem to have much explanatory power for crossovers.

This chapter presents every format pair on Kronecker graphs whose size is held fixed across graphs. The format pair can matter by orders of magnitude, though a single good pair is within a few percent of the best, and the remaining crossovers, namely row-wise to outer product at a few thousand nonzeros and `CSC_x_CSR` to `CSC_x_COO` below about 2.5 nonzeros per row can be predicted by quantities known before the computation runs.

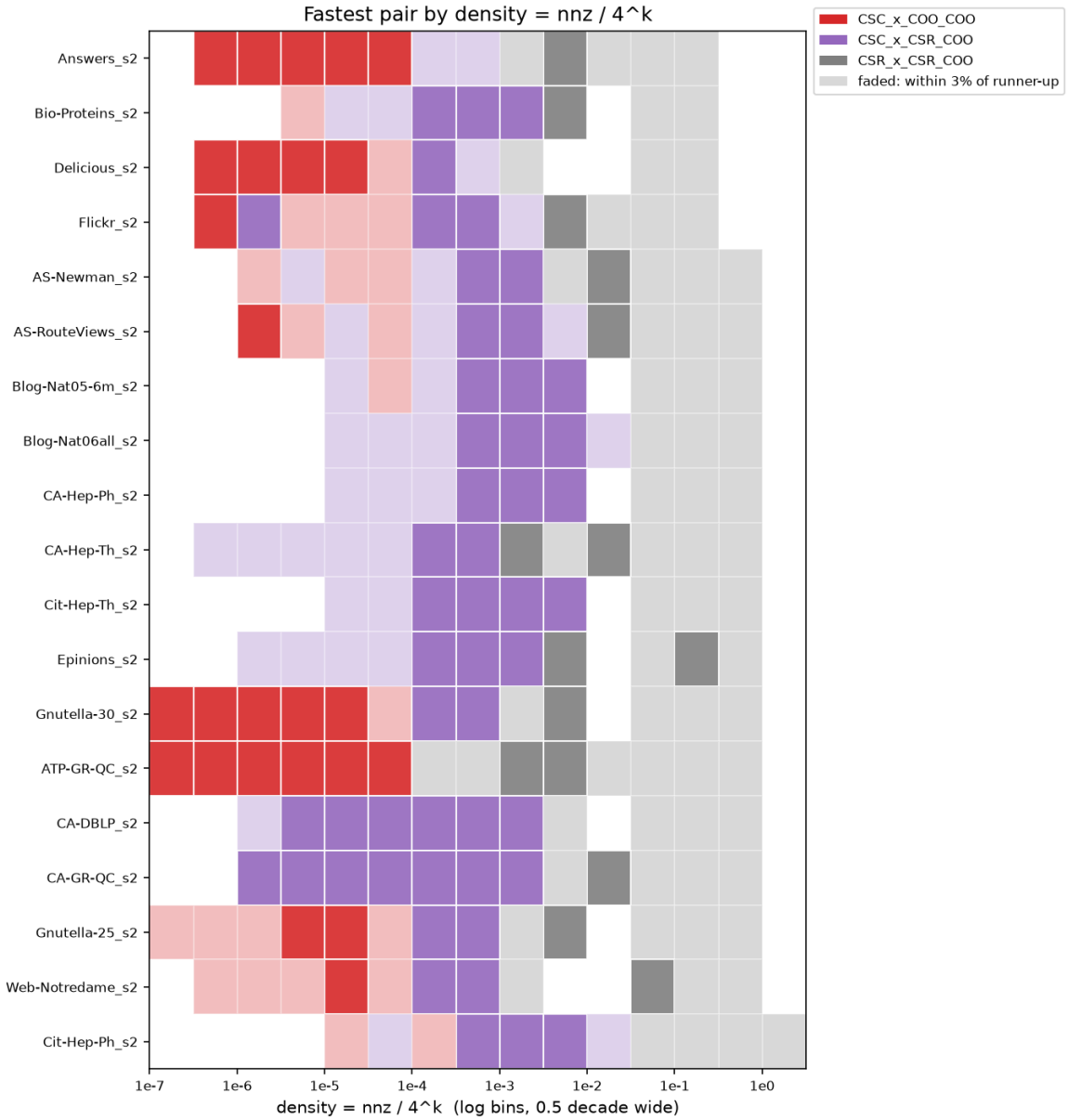


Figure 6.2: This figure shows the best performing format combo for each of the studied graphs at increasing density buckets. Each bucket collects densities in a range 0.5 decade wide. Again, there is a broad trend of the best being CSR × CSR at higher density, then transitioning to CSC × CSR or CSC × COO for sparser matrices. That this fits pretty well with K makes sense since Kroneckers become increasingly sparse as K increases.

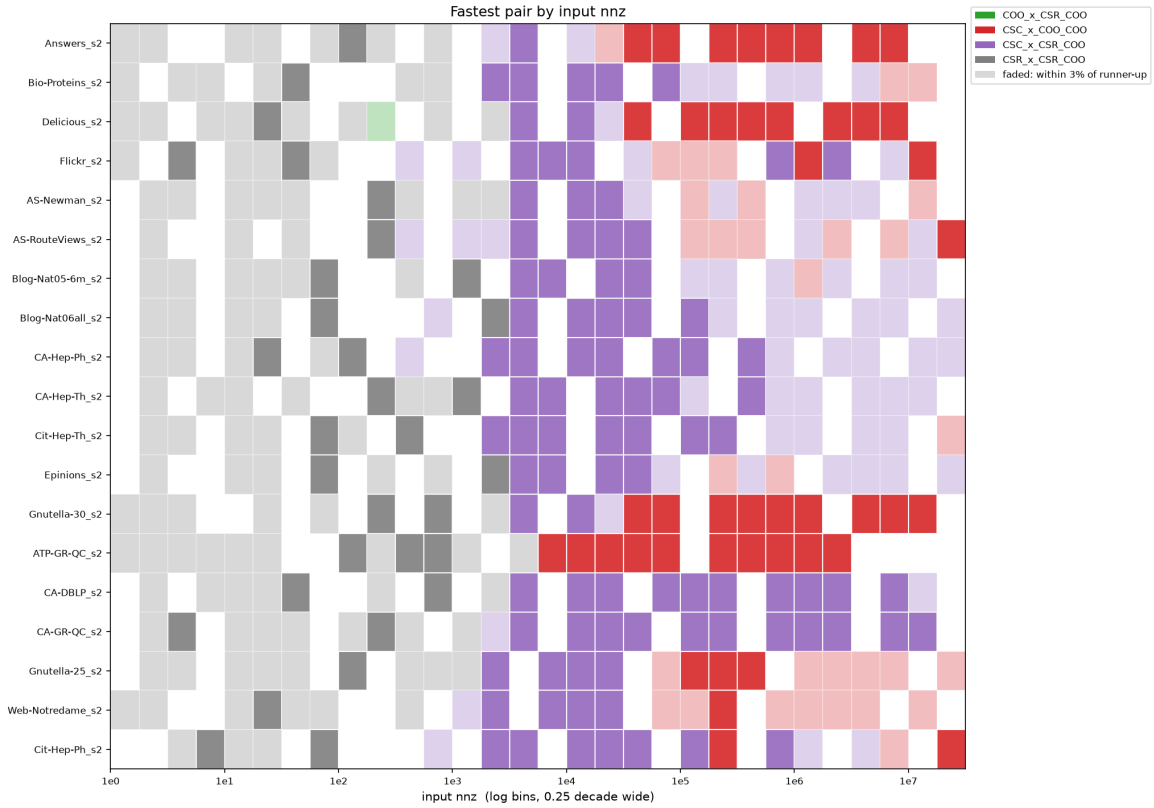


Figure 6.3: This figure shows the best performing format combo for each of the studied graphs at increasing nnz. Here the broad transition from CSC × CSR, to the other formats is much starker. Around 10^3 non-zeroes CSR × CSR stops being the best for nearly all datasets.

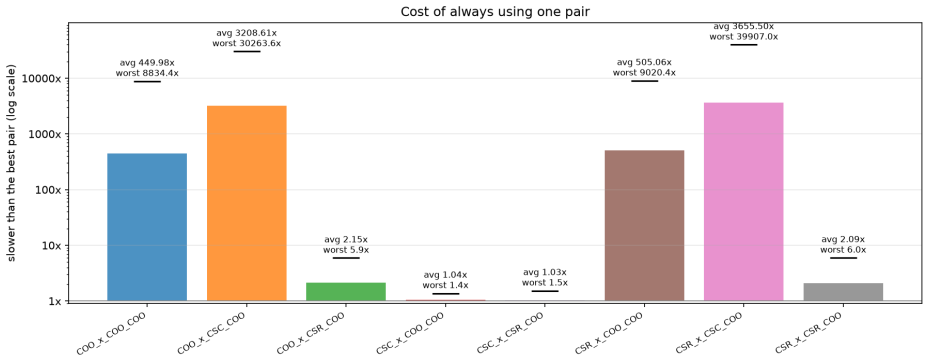


Figure 6.4: Cost of always using one pair, relative to the per-point best, all graphs together.

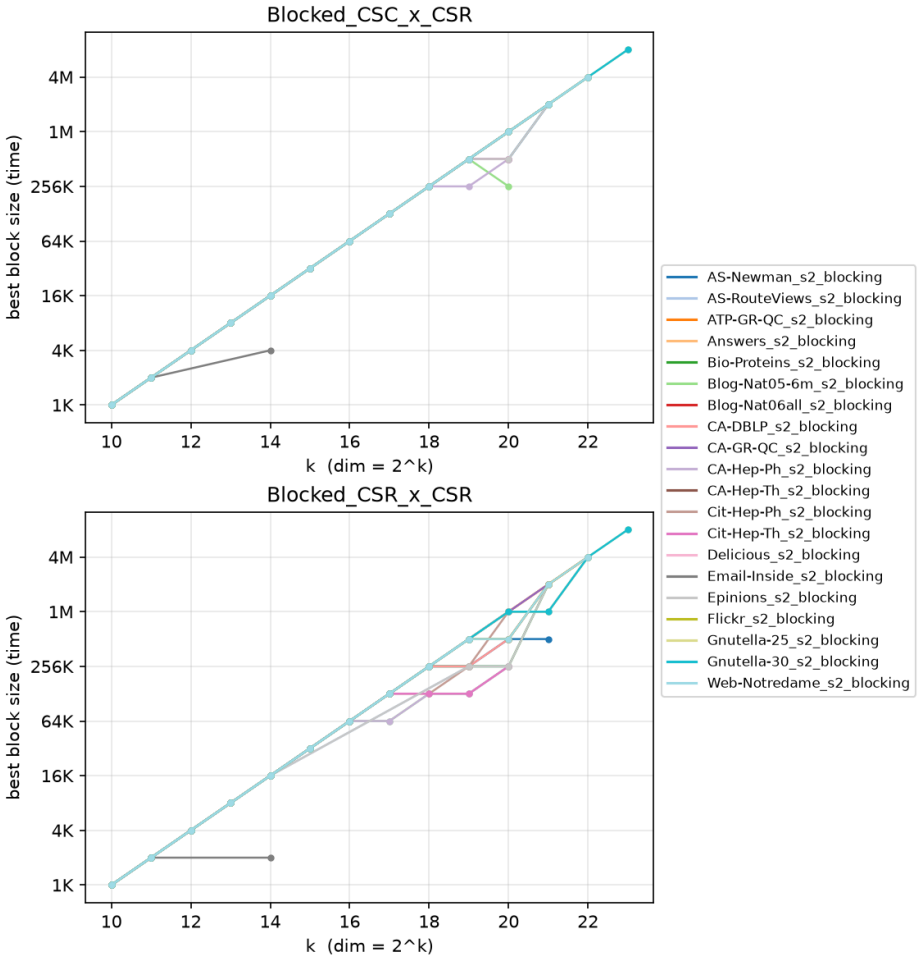


Figure 6.5: Size of block plotted against k. Almost always, the best tiling is a single tile at the matrix dimensions, or in other words, no tiling at all.

Chapter 7

Streaming SpMV for FPGA

7.1 Introduction

This chapter is based on a very early paper from my first graduate semester, way before my reptest library in Chapter 4 or the format exploration experiments of Chapter 5 and Chaper 6. It describes a streaming sparse-matrix vector multiplication kernel for an FPGA, storing blocks of that matrix in more than just one format. It allows both dense and coo formats as baseline blocks.

I include this chapter again not because of particularly good results, but because it was the impetus for the other questions driving this thesis. During the writing of the paper that forms the foundation of this chapter, I realized more questions needed to be answered before trying to commit anything to hardware. Questions such as what formats are even worth implementing kernels for? How do you even decide that? What measurements ought to decide that? Can this be generalized to other operations like SpGEMM? How does the actual input matrix and its shape inform these choices? The rest of this thesis hoped to answer these questions on CPU where exploring this problem space is much easier. This chapter should therefore serve more

as motivation for the rest of the thesis, and as a glimpse into possible future work now that more of these questions have been answered.

7.2 Theory

To take full advantage of the unique characteristics of highly irregular SpMV operations, this chapter present a new streaming multi-format structure. Where single-format matrices may succeed in general or uniform cases, there are others where the sparsity is not uniform. In these cases it may be beneficial to store parts of the larger matrix in the most efficient format. As well, because each value in a matrix for matrix-vector multiplication is read from only once, a streaming data structure may be particularly suited to the task. Furthermore, because of the heterogeneous logic involved in the different algorithms tailored to the different formats, current GPU hardware accelerators may not be sufficient, should any type of parallelism be pursued. Therefore, to address this unique challenge, this chapter proposes implementing this structure on FPGA. In summary, this chapter will discuss a multi-format matrix block stream, in which block sizes may be of arbitrary granularity and of arbitrary format to best exploit the data layout on a matrix by matrix basis.

7.3 Mechanism

7.3.1 Implementation

In this initial exploratory implementation, Figure 7.1, only two simple submatrix format options were chosen: Sparse coordinate (COO) and dense row-major / C-style. These were chosen for the simplicity of streaming. As well [39] found it to

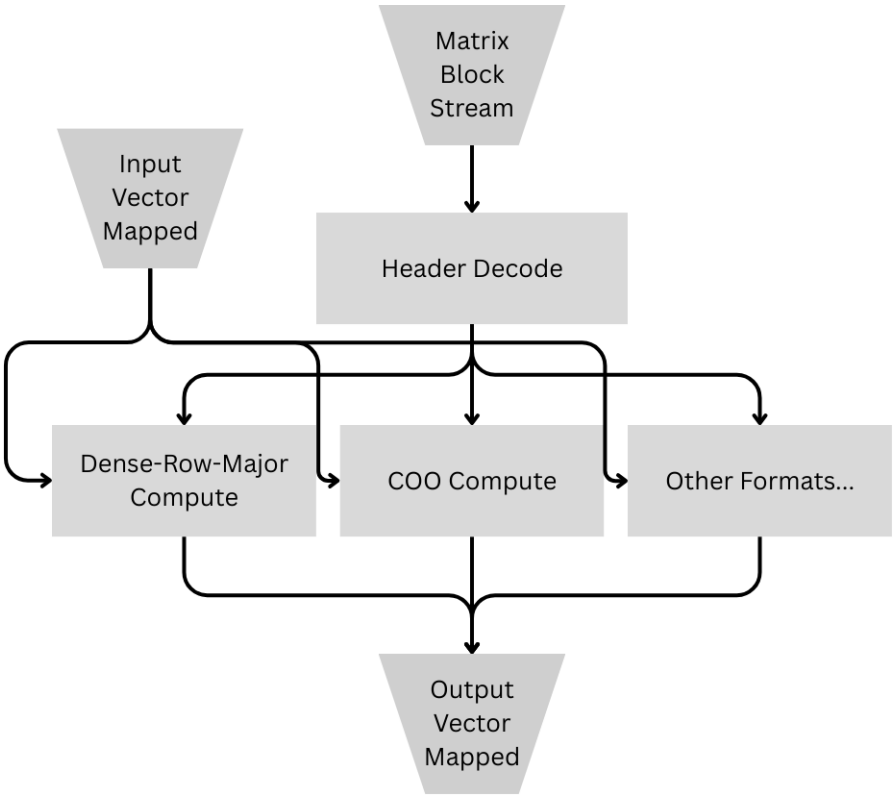


Figure 7.1: Current Sequential Implementation

be quite resistant to very irregular matrices, meaning a smaller variance. In matrix-vector multiplication, each element of the matrix must only be visited/read from once, making a stream a satisfactory data structure for storing and retrieving matrix elements as they are needed. Vitis also provides the hls::stream library for just this purpose. By default, this wraps around a FIFO port when placed on the top interface, so out of order accesses are not allowed [49]. This constraint is not an issue for the two formats chosen. The test bench takes matrices already in COO format sorted by row, then column and divides them into blocks. Each block is tested for a simple threshold of sparsity:

$$\text{Format} = \begin{cases} \text{Dense Row-Major}, & \text{if } 8mn \leq 24nnz, \\ \text{COO}, & \text{otherwise.} \end{cases} \tag{7.1}$$

Here, m and n denote the number of rows and columns in the block, respectively, and nnz is the number of non-zero elements in the block. The constants 8 and 24 represent the number of bytes required to store one element in dense and COO formats, respectively. This heuristic favors dense format when its storage cost is less than or equal to that of COO. Future work may explore more sophisticated format-selection rules.

After format-selection, a header is constructed for the data: Here, each header contains a bit of information about each following block, most importantly the storage format, and the block offsets into the respective input and output vectors. Matrix data is then written into the stream in the specified format, and in the order the values will be consumed. Values in this implementation are all doubles or 64 bit floats. For the COO format, first the number of non zeroes in the block as a uint64 is streamed, then values are streamed as triples of $(row, column, value)$, exactly the

Listing 7.1: Block header structure for mixed-format streaming.

```
struct Mix_Matrix_Header
{
    Matrix_Format format; // Enum: NONE = 0, DENSE = 1, COO = 2
    uint64_t vec_in_offset; // Input vector offset
    uint64_t vec_out_offset; // Output vector offset
    uint64_t vec_count; // Number of rows in the tile
    uint64_t data_size; // Stream offset to next tile
};
```

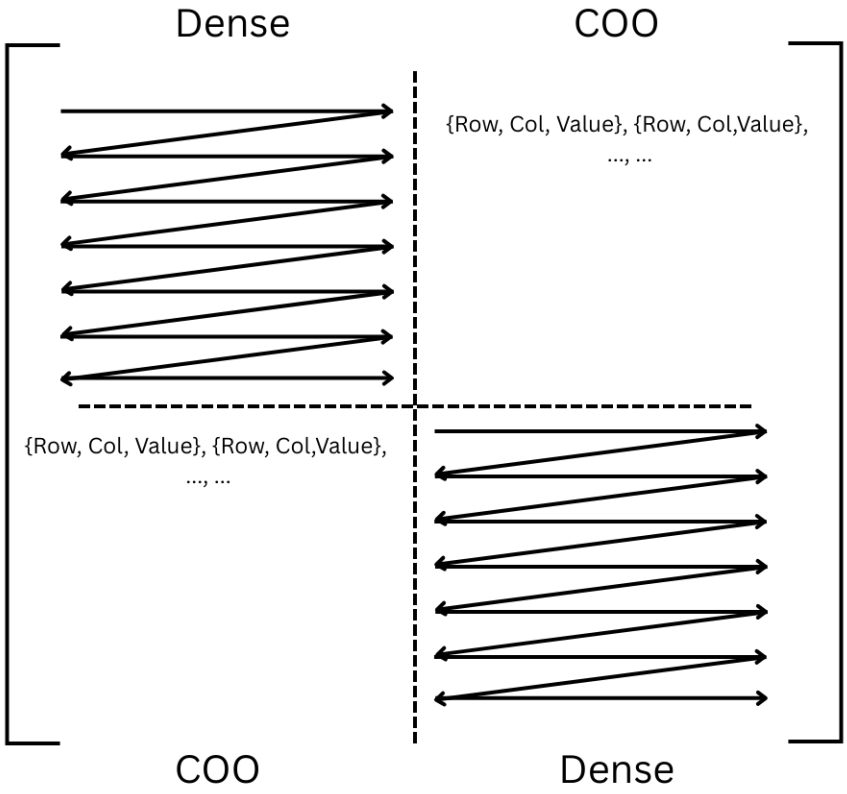


Figure 7.2: Conceptual View of Matrix Stream

order the COO compute function consumes them. And for Dense, first the number of rows and columns in the block are streamed first, again as uint64's, and then values in row major order. Conceptually, it might be thought of as Figure 7.2.

Table 7.1: Stream Memory Layout: fixed-size header followed by format-specific data.

Bytes	**Field**	**Description**
0–39	Header	Format ID, offsets, vector count
40–(40+N)	Dense Payload	$m \times n$ values
...		
X–(X+39)	Next Header	Same format as first header
...	COO Payload	$nnz \times (row, col, val)$ triples
Y–(Y+39)	NONE Header	Signifies the end of the stream

In reality the stream memory layout is demonstrated by Table 7.1. This stream is then consumed on FPGA, dispatching to the correct format compute function based on the header, Listing 7.2.

One can observe that this could be expanded to any number of potential formats, as long as they were conducive to streaming. The compute function/module for COO is given in Listing 7.3. In hindsight, choosing COO proved to be a little hazardous in the read-modify-write-cycle. Ensuring correctness required disabling pipelining for the COO_LOOP, greatly limiting potential throughput. This revealed a major limitation of some formats that have very irregular access like COO for the streaming, pipe-lined architecture of FPGAs. This as an important lesson guiding future work in exploring other formats. For now these are relatively naive implementations, but one could imagine using the given format's fastest implementation for each compute module, combining the most effective implementation with the most effective format for each blocks data layout. However, this first implementation is very sequential, having only one stream and writing potentially to the same rows in the memory mapped output vector. Both the input and output vectors are AXI Master memory

Listing 7.2: HLS Stream Processing on FPGA

```
void mix_matrix_mul_vector_top(Data_Stream *mix_stream,
                               f64 *vec_in, u64 vec_in_count,
                               f64 *vec_out, u64 vec_out_count)
{
  ...
  do
  {
    head = read_matrix_header(mix_stream);

    switch (head.format)
    {
      case Matrix_Format::NONE: { } break;
      case Matrix_Format::DENSE_ROW_MAJOR:
      {
        dense_row_major_mul_vec(mix_stream, head,
                                vec_in, vec_in_count,
                                vec_out, vec_out_count);
      } break;
      case Matrix_Format::COO:
      {
        coo_mul_vector(mix_stream, head,
                       vec_in, vec_in_count,
                       vec_out, vec_out_count);
      } break;
    }
  }
  while (head.format != Matrix_Format::NONE);
}
```

Listing 7.3: HLS Stream COO Block Compute

```
void coo_mul_vector(Data_Stream *mix_stream, Mix_Matrix_Header head,
                    f64 *vec_in, u64 vec_in_count,
                    f64 *vec_out, u64 vec_out_count)
{
  u64 non_zero_count = mix_stream->read();

  COO_LOOP:
  for (usize i = 0; i < non_zero_count; i++) {
    u64 coo_row = mix_stream->read();
    u64 coo_col = mix_stream->read();
    f64 coo_value = read_f64_from_stream(mix_stream);

    u64 vec_in_index = coo_col + head.vec_in_offset;
    u64 vec_out_index = coo_row + head.vec_out_offset;

    f64 current = vec_out[vec_out_index];
    f64 vector_in = vec_in[vec_in_index];

    vec_out[vec_out_index] = current + (coo_value * vector_in);
  }
}
```

mapped, since we are accessing these values multiple times. Nonetheless we can take advantage of Matrix-Vector multiplication's need to access each matrix value only once with the use of a streaming data structure. The tooling used for this implementation was AMD's Vitis HLS suite [49], and the board we used for our simulations was the Pynq-7000, specifically the xc7z020clg400-1.

7.4 Analysis

7.4.1 Preliminary Results

I created a simple script to generate sparse matrices in edge list format. For this simple exploratory implementation I focused on 2 cases a small 32x32 matrix for initial testing, and a larger matrix of 2048x2048, for a more rigorous test case. In all cases our simple heuristic for deciding format usage achieved a small data footprint, even including stream header overhead, see Table 7.2. However, the simplicity of the rule leaves much to be desired, and further work should be done, both in finding a more efficient rule perhaps even machine learning based, as Chen et al. have [40], as well as expanding the available formats. However, the implementation leaves much of the available hardware unused, as seen by Table 7.3. Ideally, I would like to achieve a much fuller saturation of the DSP units, as well as at least some usage of the fast BRAM. But, there is potential in the streaming structure [38], one could imagine having a much larger sparse matrix that one does not want to materialize all at once. One could instead only materialize by block, and at that moment also decide the format for streaming. This is especially useful as those blocks can then be dematerialized as they are consumed on FPGA, as stated before matrix-vector multiplication only requires one read for any one value of the matrix.

Table 7.2: Memory Usage for Matrix Size

Matrix Dimension	Memory Footprint (KB)
32x32	0.856
2048x2048	9827.224

Table 7.3: FPGA Synthesis Resource Usage

BRAM Usage	DSP Usage	LUT Usage	FF Usage
2%	6%	4%	10%

7.4.2 Future Work

Though there is a working implementation, I believe there is much more potential in this idea. So far, the implementation has not fully utilized the strengths of FPGA as a platform, namely parallelism and BRAM usage. Therefore I propose an improvement to the current design, seen in Figure 7.3. This design would include alongside the other decode and compute modules, a final reduction module. In this implementation, each block and compute module would receive a slice of BRAM to act as a personal subset of the output vector. In this way, each block could run completely in parallel once it received its data from the stream. A final reduce step would sum any overlapping subset output vectors and write back into the memory mapped true output vector. By allowing compute to be pipe-lined more effectively with reading from the stream, we would hopefully achieve much better memory and resource saturation. As mentioned before, slightly more regularly accessed sparse formats would aid in pipe-lining inner compute loops. In future, I would also like to explore a multiple stream implementation, allowing multiple blocks to be read in parallel.

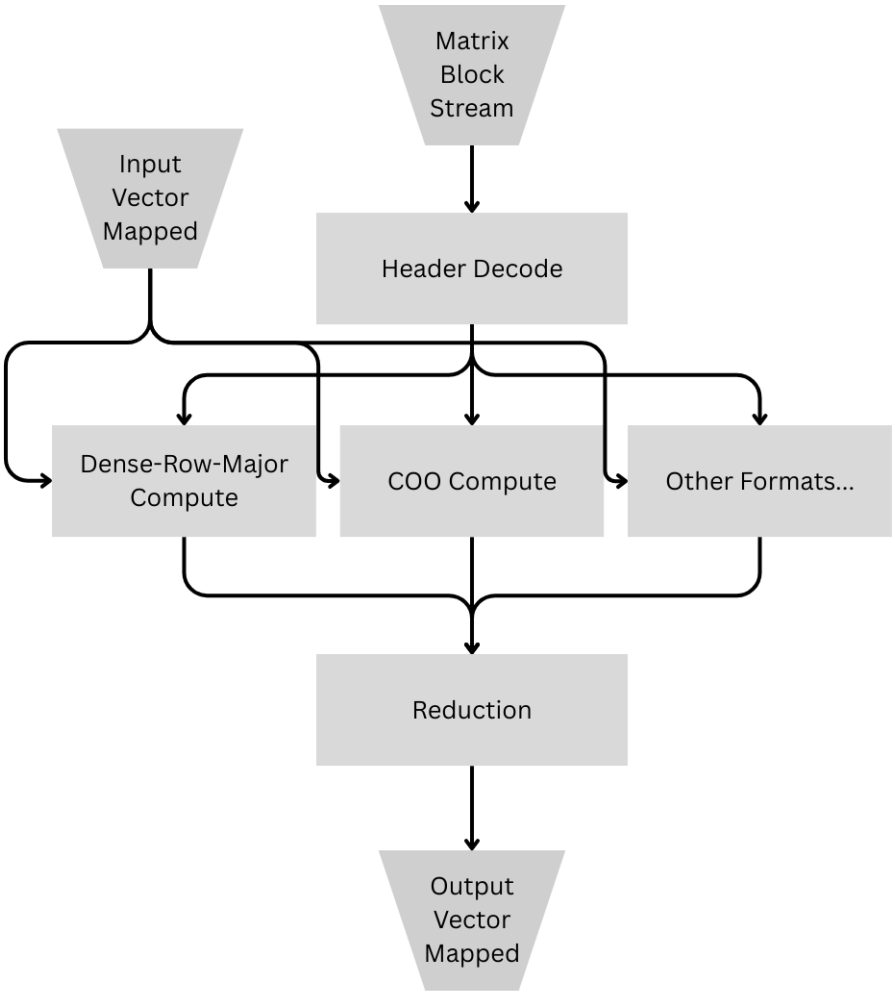


Figure 7.3: Future Parallel Implementation

7.5 Summary

In conclusion, we have motivated the need for new solutions to very irregular sparse matrices, as well as called attention to previous and older solutions. This chapter proposed a novel streaming structure that might combine the strengths of previous formats, tailoring them to the needed block granularity on an individual matrix basis. In addition, this chapter also describes an implementation of such structure on FPGA to enable future work in parallelizing the heterogeneous computation and unpacking logic of many different formats. Our baseline implementation demonstrated a working computation in which blocks of disparate types were consumed, as well as demonstrated that memory footprint was kept small even with streaming header overhead. Further, this chapter also discussed future work involving usage of BRAM and reduction functional modules for parallelizing computation in hopes of further saturating memory bandwidth.

Chapter 8

Discussion and Future Work

8.1 Putting everything together

I think this thesis is very much a tale of putting the cart before the horse, over and over again. It started with investigations into FPGA implementations (Chapter 7) before I really had a clue what would even be good on them. It then progressed into an investigation of cost models for choosing sparse sub-block formats, because I believed that this would yield good results. Of course, it did not in fact yield desirable results (Chapter 5). Though the analytical cost models were good at predicting actual empirical operation counts, those operation counts were actually a very poor proxy for real runtime performance. This lead me to ask the much more scoped and narrow questions of Chapter 6, now equipped with an upgraded reptest library that could measure real hardware events (Chapter 4).

On the Kronecker graphs I tested, the most important choice was just not choosing any format combinations that forced a bad dataflow. Avoiding inner-product type sparse intersections matters by orders of magnitude. But among the combos left, a single fixed combo is within a few percent of the best on average. Where the best

combination does change can be explained by simpler structural quantities about those particular matrices. Together this suggests a much simpler practical rule of beginning with any one good combination and switching when the crossover points (on whatever quantity that most line up with) point us to.

8.2 Limitations

- **Inputs.** All conclusions in Chapter 6 come from Kronecker graphs. They served their purpose in allowing structure to be held constant across sizes, but they are nonetheless all of the same synthetic family.

- **Kernels.** The TACO kernels I timed write one value per product and do not accumulate or merge, so they measure the multiply and store of partial products and not a complete SpGEMM. Algorithms that accumulate may behave differently, and may benefit from tiling in a way these did not.

- **Mechanisms for crossover are hypotheses.** The branch mispredictions explanation for the early switch is consistent with the counters and the generated code. The explanation for the hypersparse advantage of `CSC_x_COO` (it skips empty values of the contraction index) comes from reading the code and from the trend, as well as resting on a single graph, ATP-GR-QC. There may very well be other explanations that could be found with more data and experiments.

8.3 Future work

- **Other inputs.** Check whether the crossover thresholds hold on real matrix collections. I'd also like to look at structural properties beyond nonzero counts

such as shape. Though, I found no structural effect after accounting for nonzeros and nonzeros per row, but I tested only a few properties.

- **Try the simple selector.** As Chapter 5 demonstrates, a selector is only good if it actually produces performant computations. I'd like to actually test if the simple selector is any good.

- **Other kernels and algorithms.** Try kernels that actually accumulate. As well, revisit tiling there, since the absence of an accumulator is one possible reason tiling failed. As well more experiments into multithreading. The thread experiments in Chapter 5 found little benefit beyond two threads at the sizes that could be solved, so larger inputs are needed.

- **Hardware.** CPUs are not very good at the very branchy work required for variable per-block sparse format operations. Architectures with cheaper irregular control flow might suit it better. The FPGA design in Chapter 7 was a first investigation into this.

8.4 Closing

I think that most useful contributions from this thesis are really the measurments themselves. I set out to build fast implementations and realized I didn't really know enough to do that. And so this thesis hoped to answer the questions one should ask before building the fast implementation.

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Appendix A

Additional Listings

Listing A.1: Example usage of Repetition Test Library

```
// NOTE: Define the interface that all the functions we want to test conform
    to.
typedef void (*Test_Function)(u8 *buffer, usize buffer_size);
static
void read_buffer_linearly(u8 *buffer, usize buffer_size)
{
  ... // Omitted for brevity
}
static
void read_buffer_randomly(u8 *buffer, usize buffer_size)
{
  ... // Omitted for brevity
}

// NOTE: Each entry is just going to be the function and
```

```
// then a little string to name it.
typedef struct Function_Entry Function_Entry;
struct Function_Entry
{
  const char *name;
  Test_Function function;
};
// NOTE: Ok now we can just make an array of all the functions we want to
    test.
Function_Entry entries[] =
{
  {"linearly", read_buffer_linearly},
  {"randomly", read_buffer_randomly},
};

int main(int argc, char **argv)
{
  Arena arena = arena_make(.reserve_size = GB(64));
  u64 cpu_timer_frequency = estimate_cpu_timer_freq();

  // NOTE: How many seconds to keep trying for a new min, will reset once
  // we find a new minimum.
  u32 seconds_to_try_for_min = 1;

  // NOTE: User may provide custom fields.
  Repetition_Series *series = repetition_series_make(60 * STATIC_COUNT(
      entries), ((const char *[]){"buffer_size", "function"}));
```

```
for (usize buffer_size = MB(1); buffer_size <= MB(4); buffer_size *= 4)
{
  for (usize test_func_idx = 0; test_func_idx < STATIC_COUNT(entries);
      test_func_idx++)
  {
    // NOTE: Allocating and deallocating every time to make it fair page-
        fault wise.
    u8 *buffer = os_allocate(buffer_size, OS_ALLOCATION_COMMIT);

    Function_Entry *entry = &entries[test_func_idx];

    Repetition_Tester tester = repetition_series_new_tester(series,
        buffer_size,
                                                            cpu_timer_frequency,
                                                            seconds_to_try_for_min
                                                                ,
                                                            "---␣%s␣@␣%lu␣MB␣---
                                                                ", entry->name,
                                                            buffer_size / MB(1))
                                                                ;

    // NOTE: User may set fields of current test.
    repetition_series_set_field(series, "buffer_size", "%lu", buffer_size);
    repetition_series_set_field(series, "function", entry->name);

    // NOTE: The main loop, it will continue testing until the minimum time
```

```
            has not decreased in user-provided time-frame.
        while (repetition_series_is_testing(series, &tester))
        {
          repetition_tester_begin_time(&tester);
          entry->function(buffer, buffer_size);
          repetition_tester_close_time(&tester);

          repetition_tester_count_bytes(&tester, buffer_size);
        }

        os_deallocate(buffer, buffer_size);
      }
    }

    // NOTE: Self-explanatory
    repetition_series_save_csv(series, "out.csv");
}
```

Appendix B

Format Comparison Figures

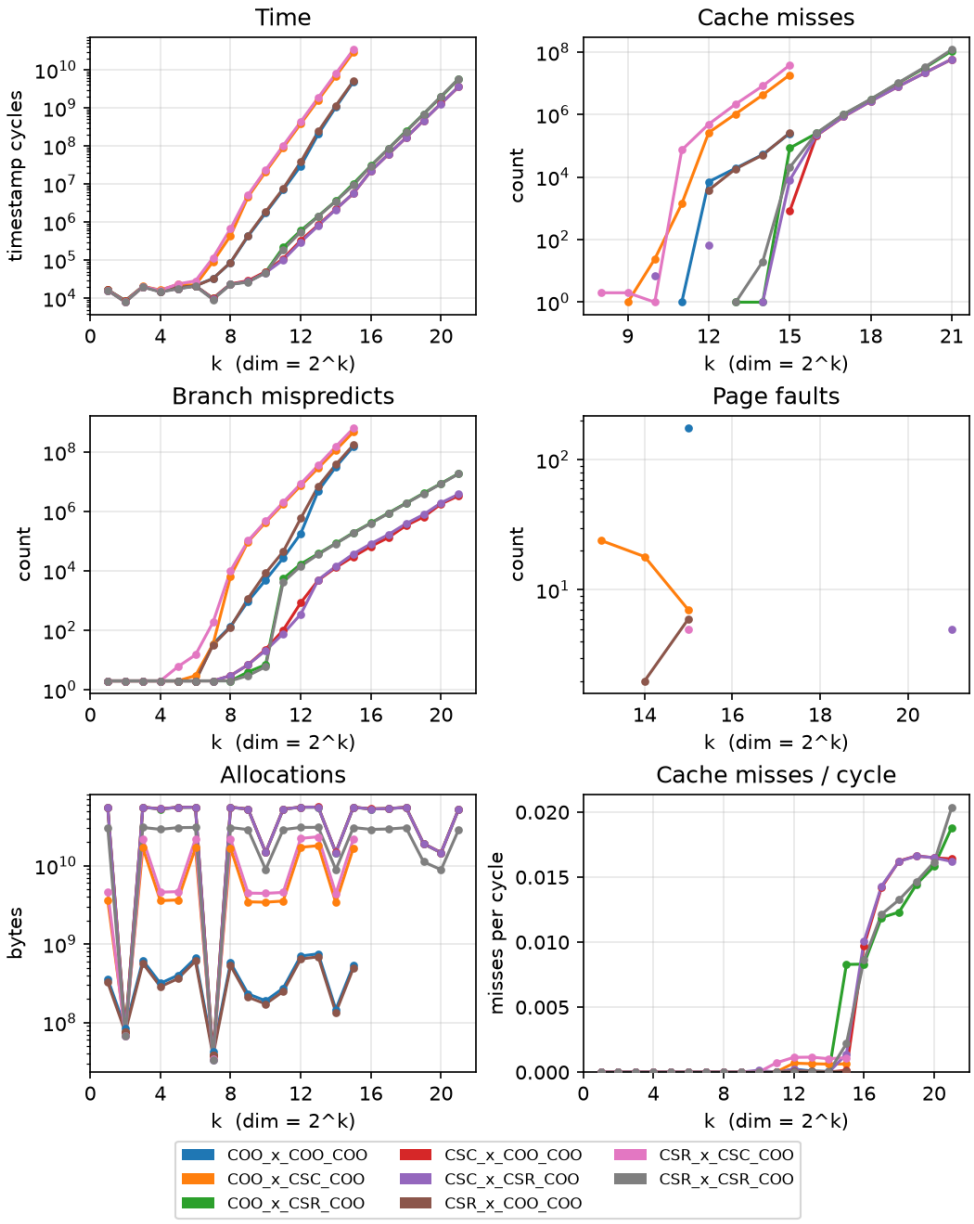


Figure B.1: AS-Newman Sample 2 comparison.

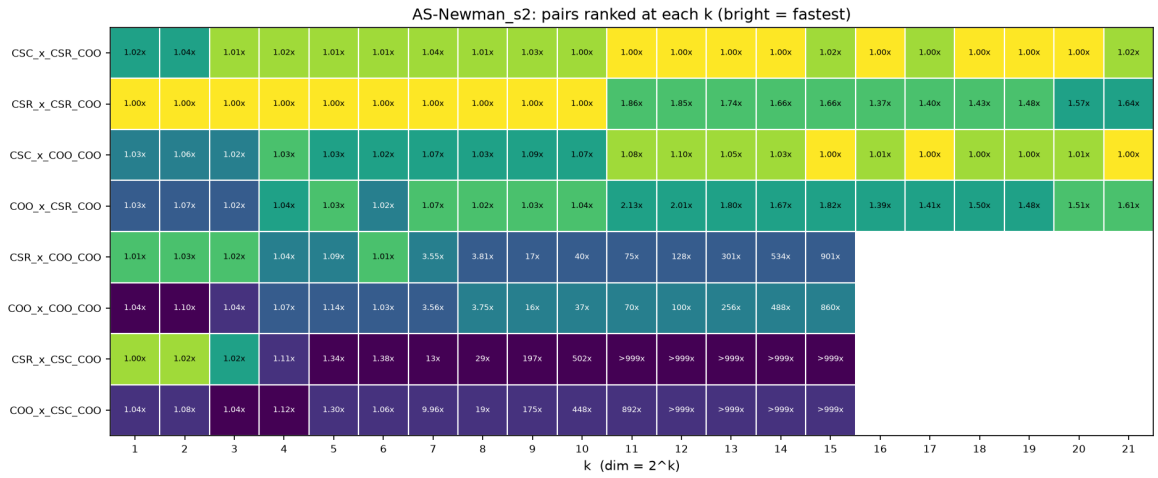


Figure B.2: AS-Newman Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

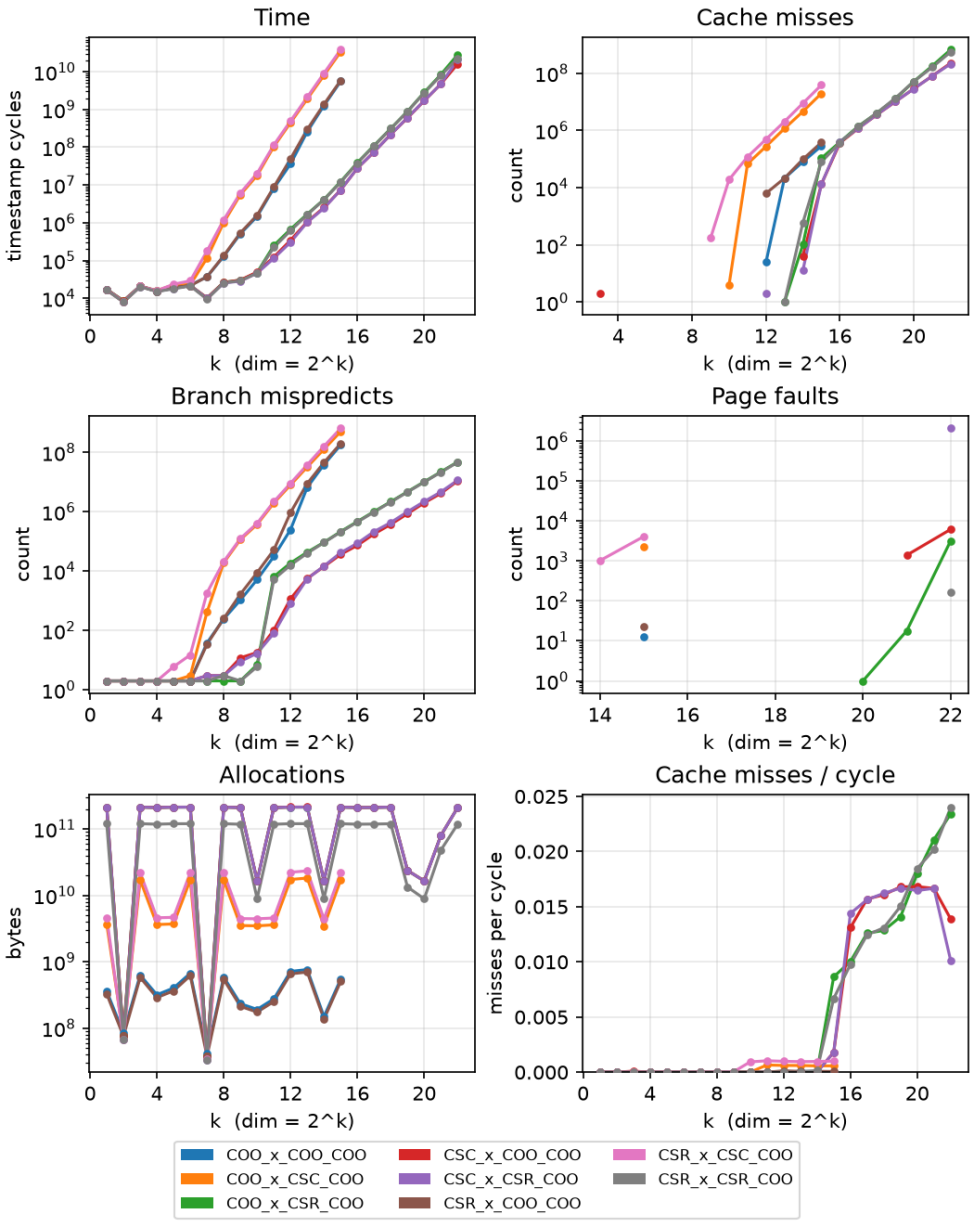


Figure B.3: AS-RouteViews Sample 2 comparison.

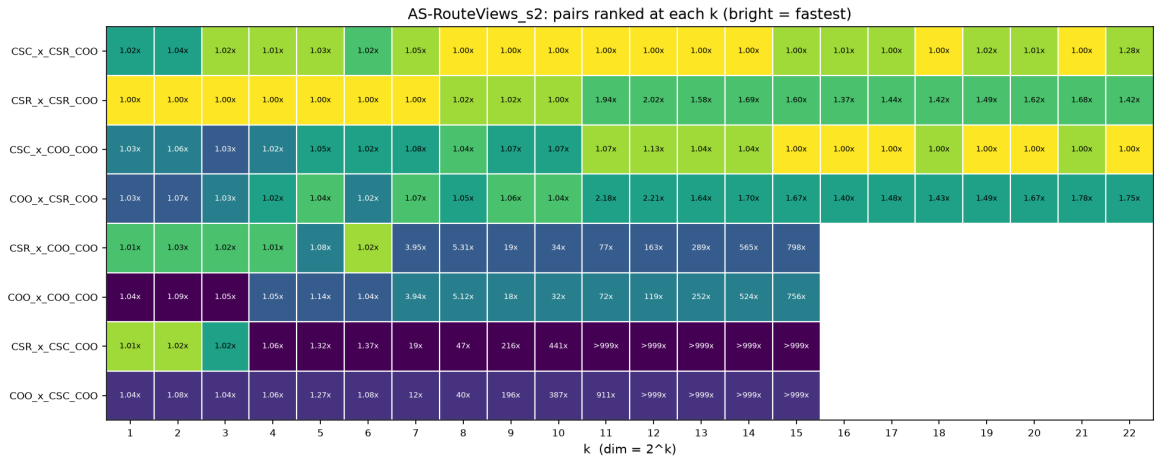


Figure B.4: AS-RouteViews Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

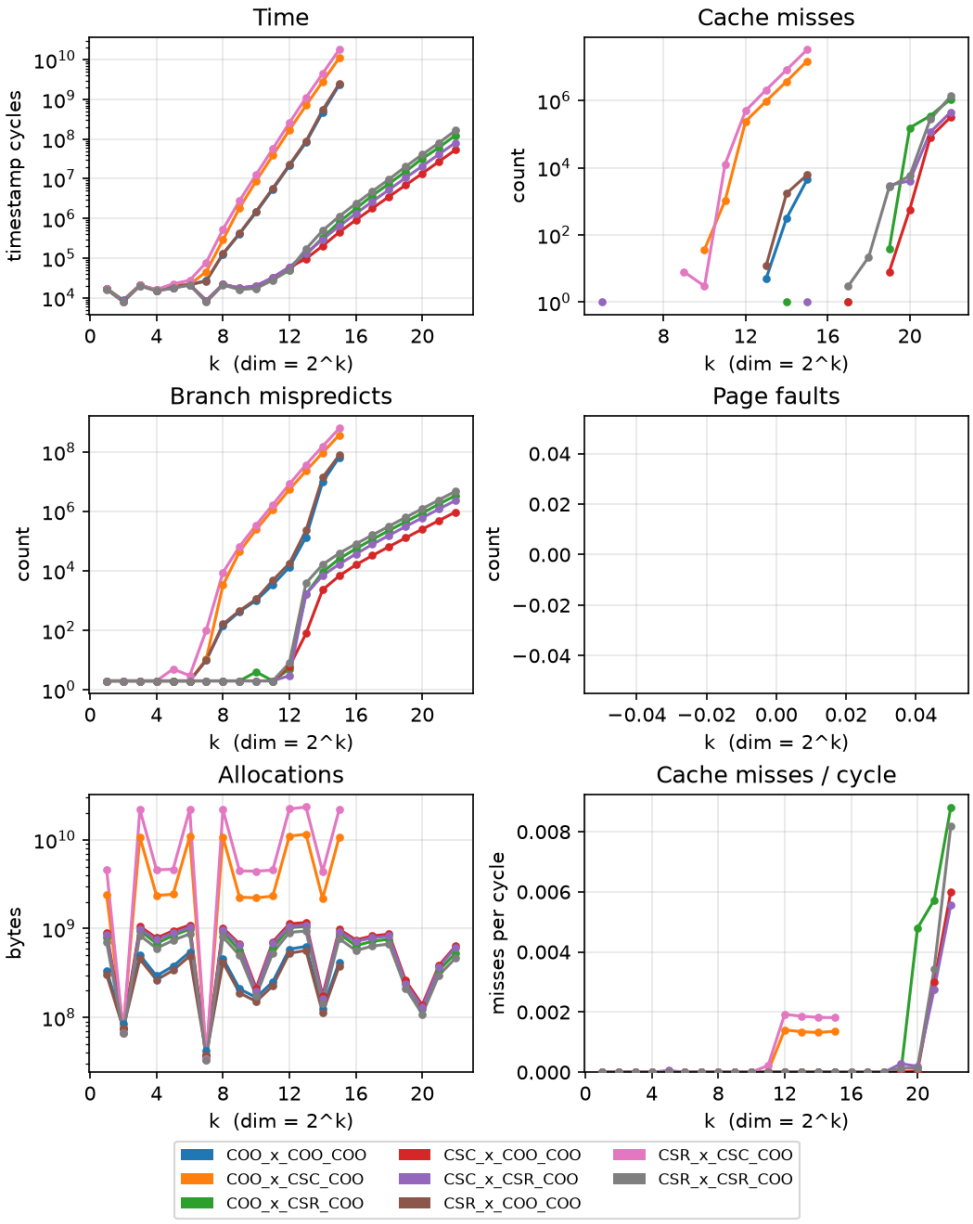


Figure B.5: ATP-GR-QC Sample 2 comparison.

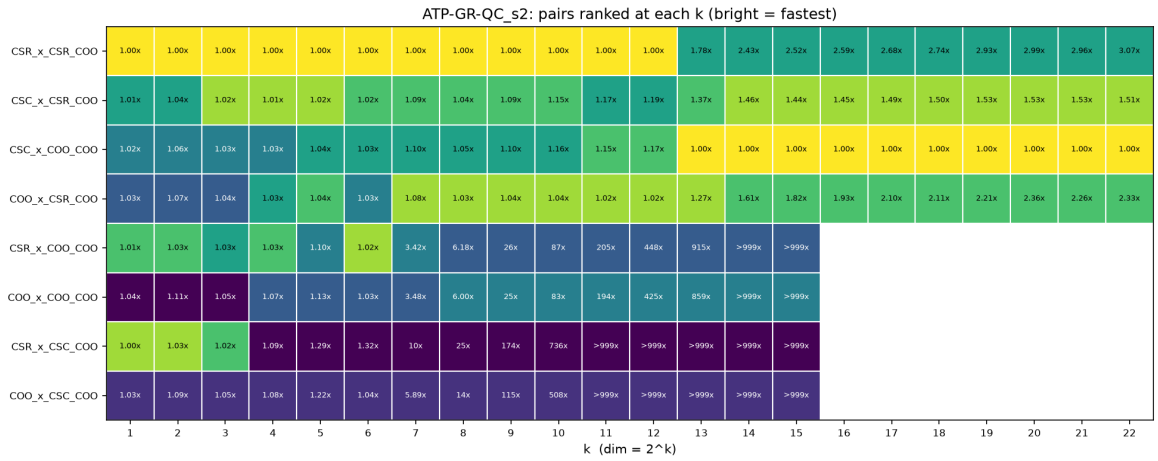


Figure B.6: ATP-GR-QC Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

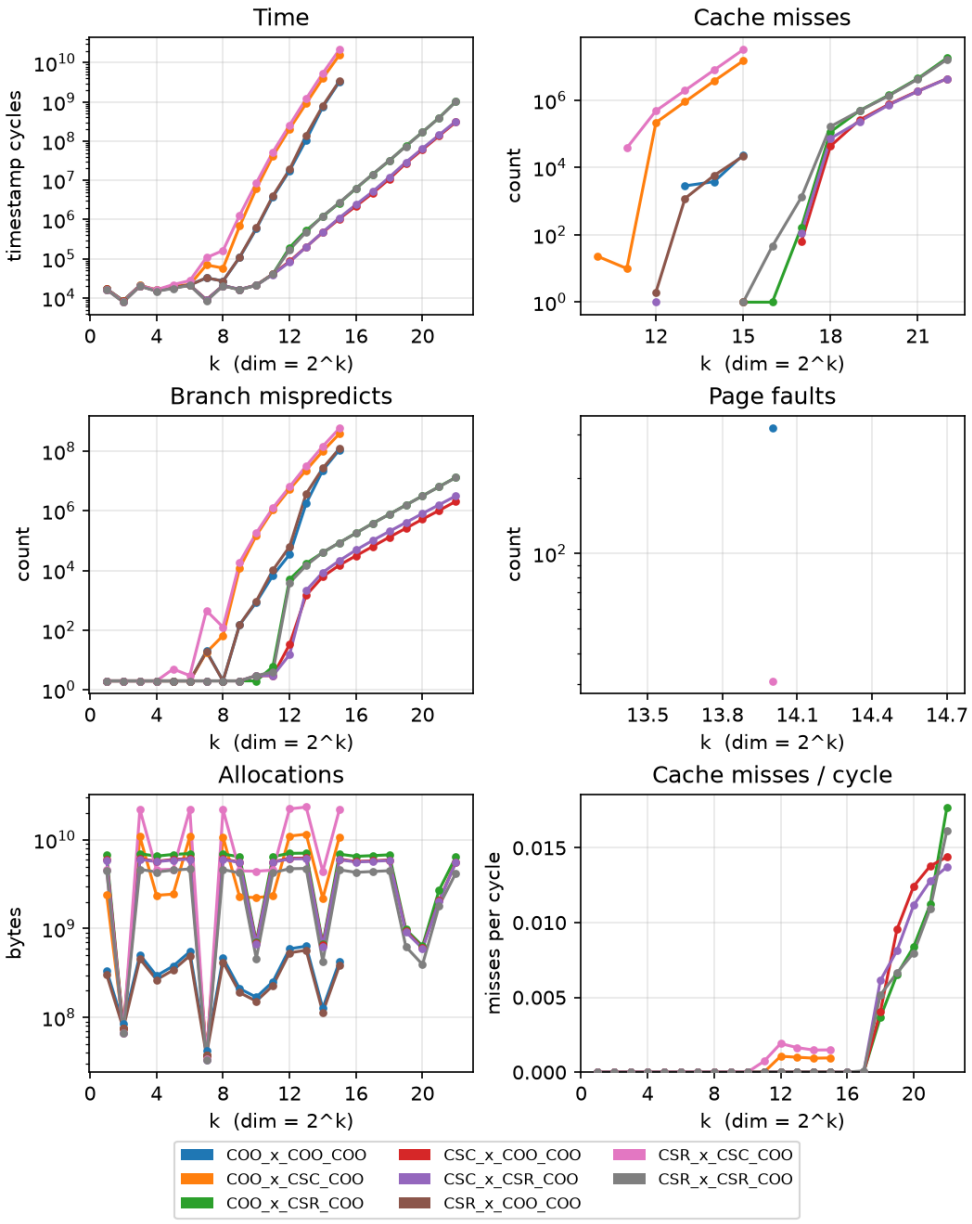


Figure B.7: Answers Sample 2 comparison.

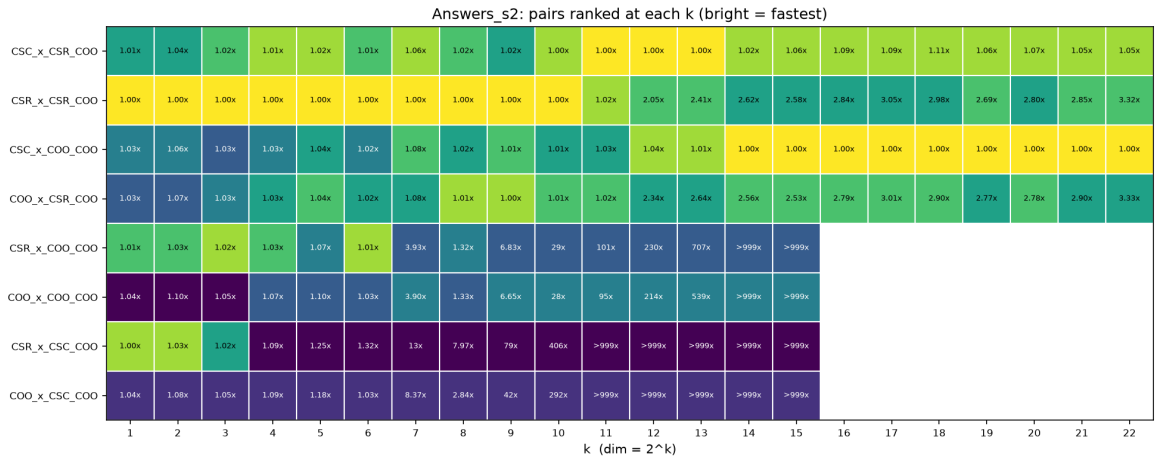


Figure B.8: Answers Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

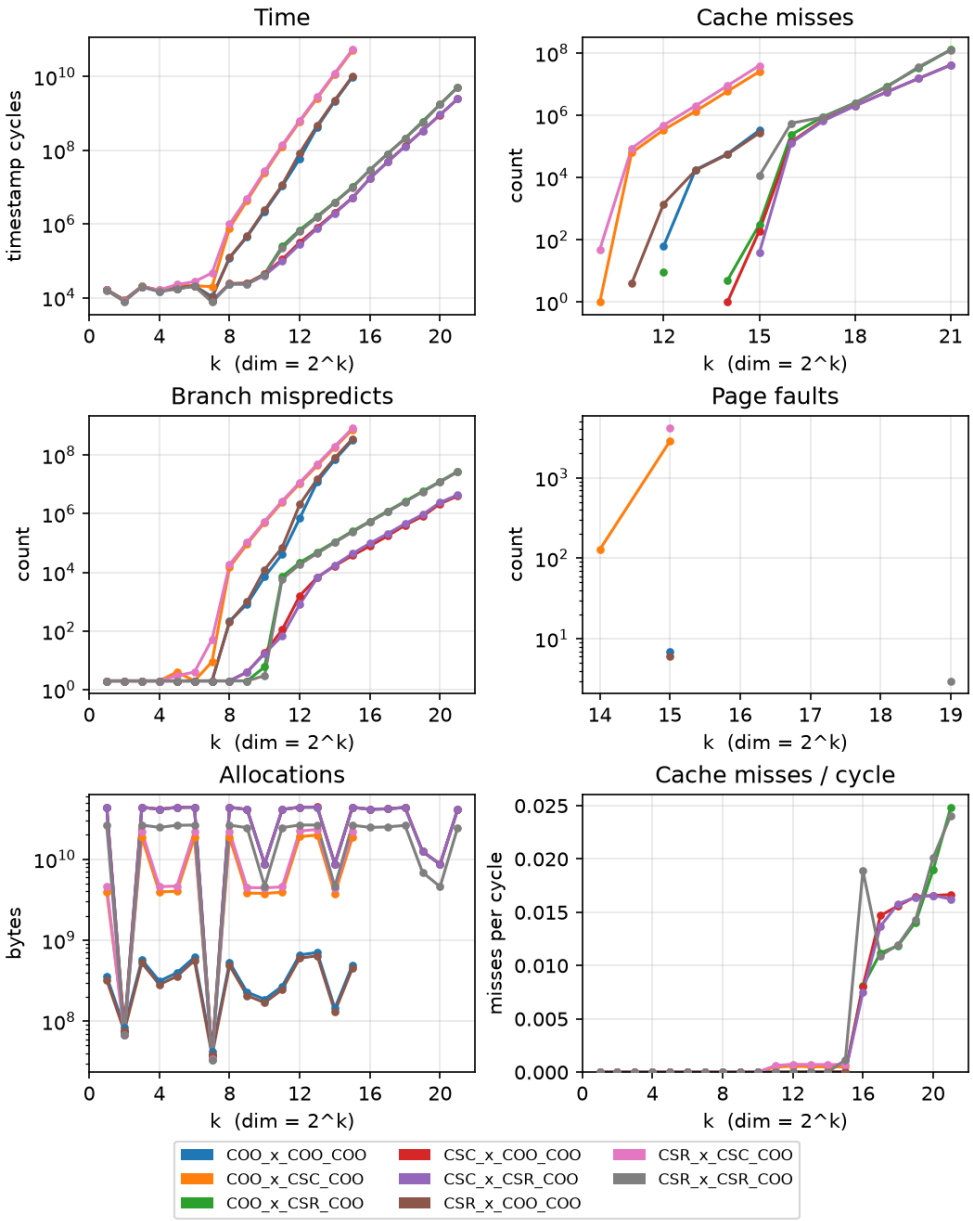


Figure B.9: Bio-Proteins Sample 2 comparison.

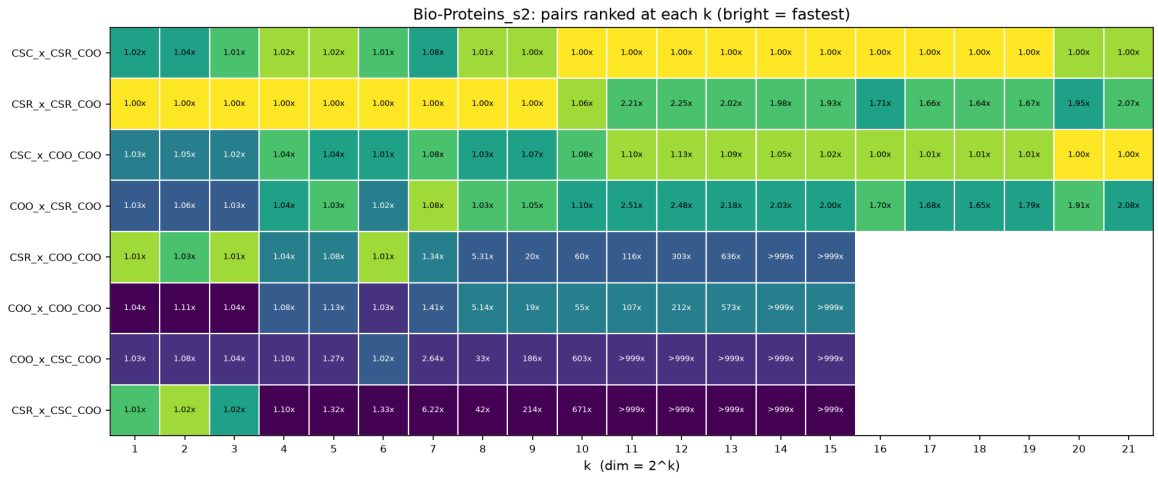


Figure B.10: Bio-Proteins Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

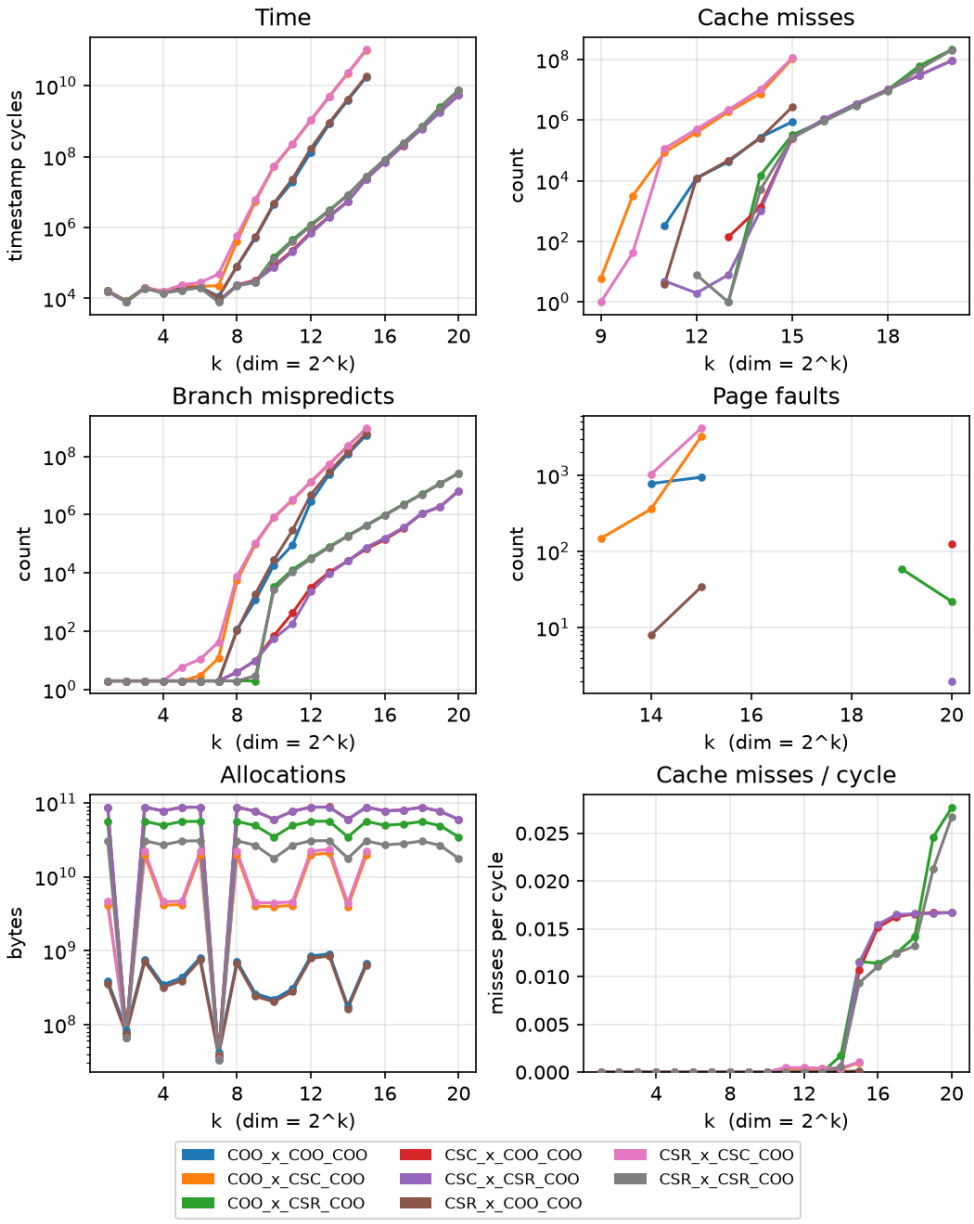


Figure B.11: Blog-Nat05-6m Sample 2 comparison.

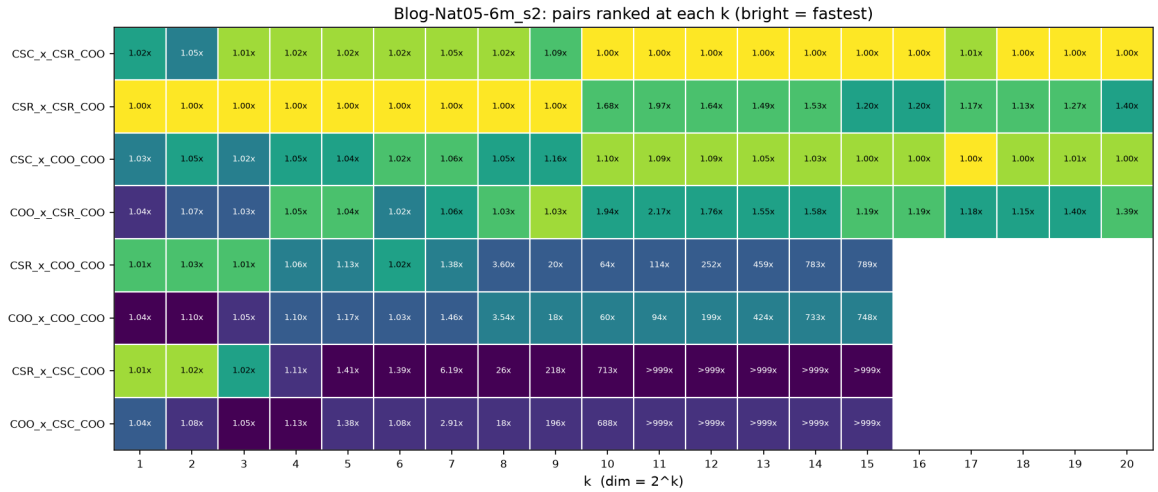


Figure B.12: Blog-Nat05-6m Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

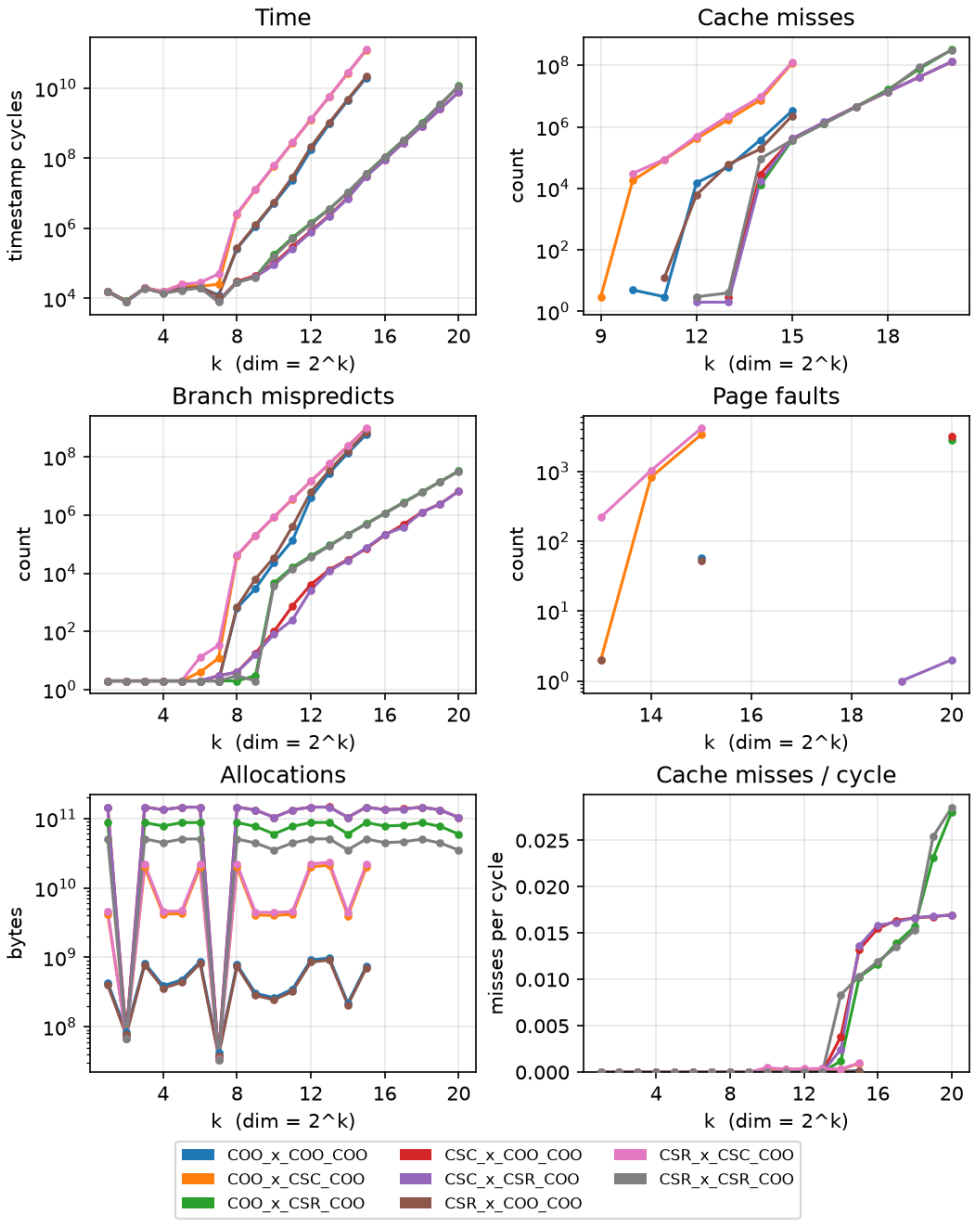


Figure B.13: Blog-Nat06all Sample 2 comparison.

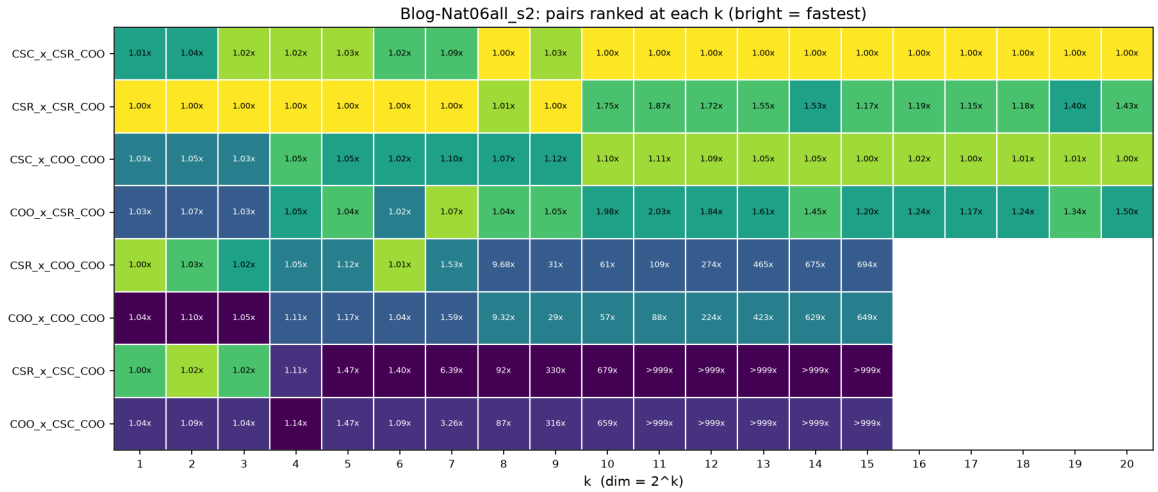


Figure B.14: Blog-Nat06all Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

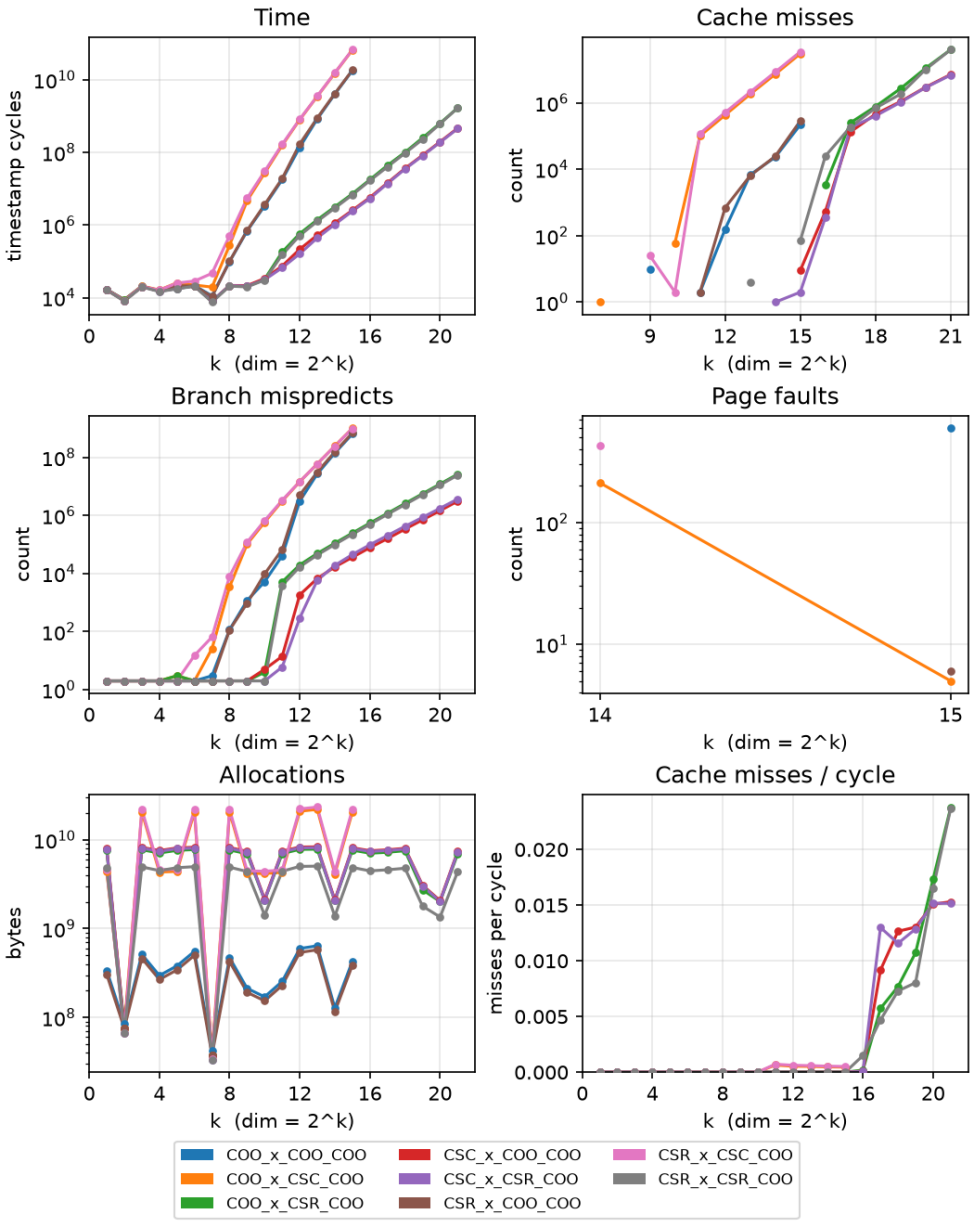


Figure B.15: CA-DBLP Sample 2 comparison.

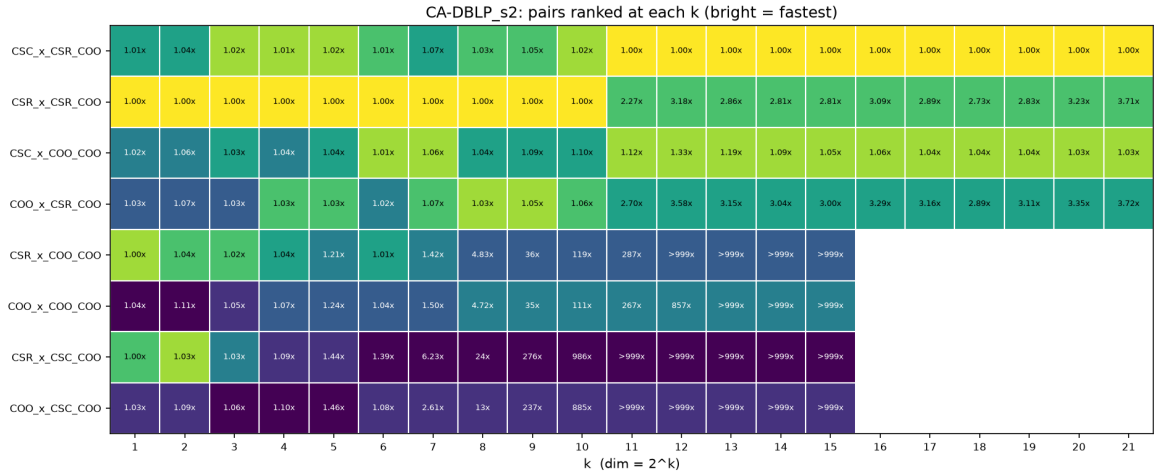


Figure B.16: CA-DBLP Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

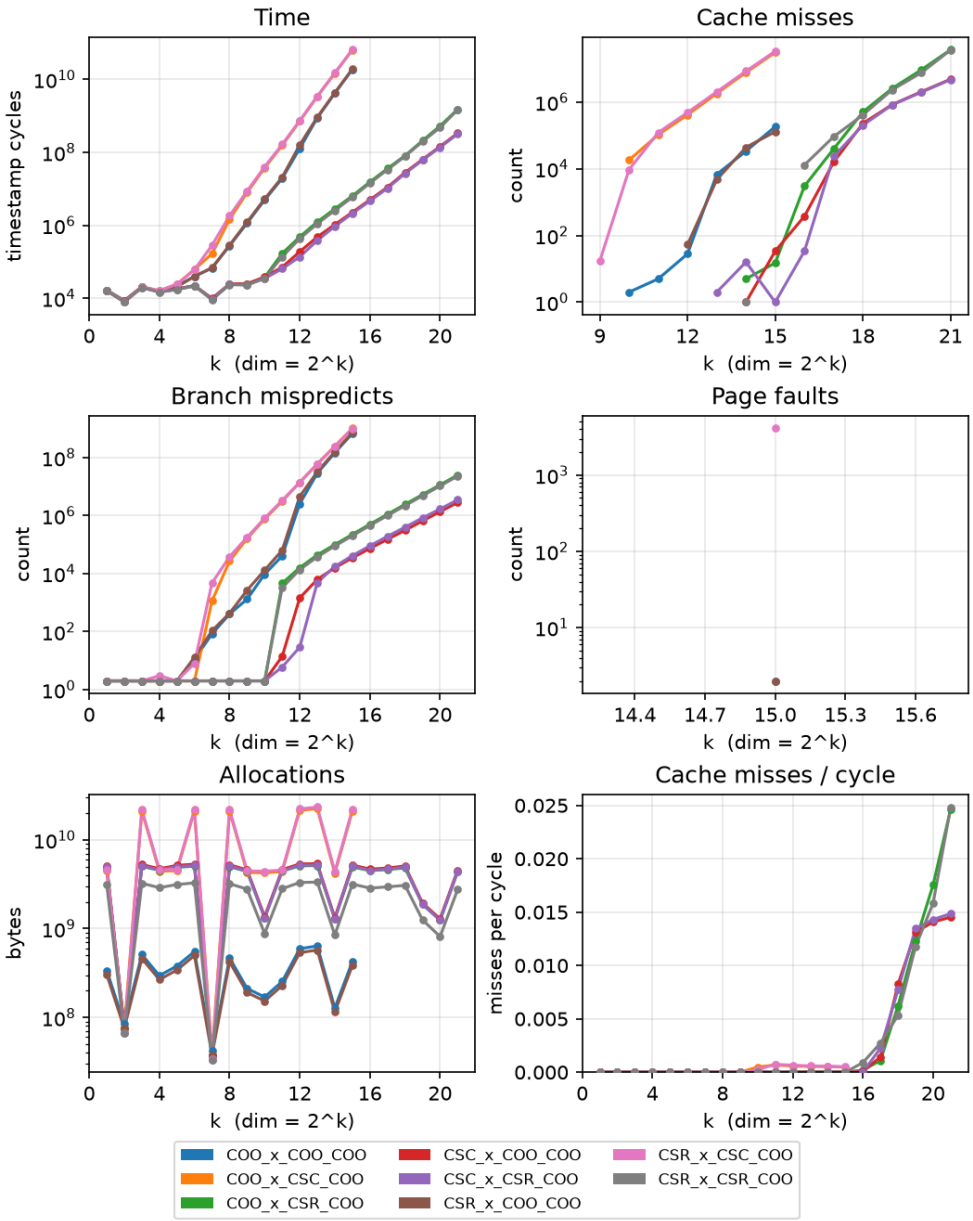


Figure B.17: CA-GR-QC Sample 2 comparison.

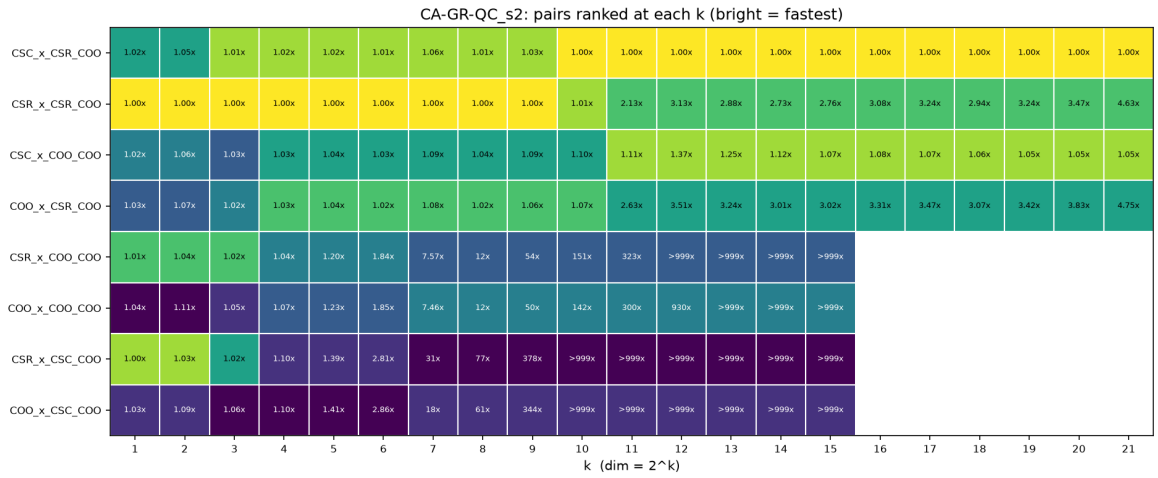


Figure B.18: CA-GR-QC Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

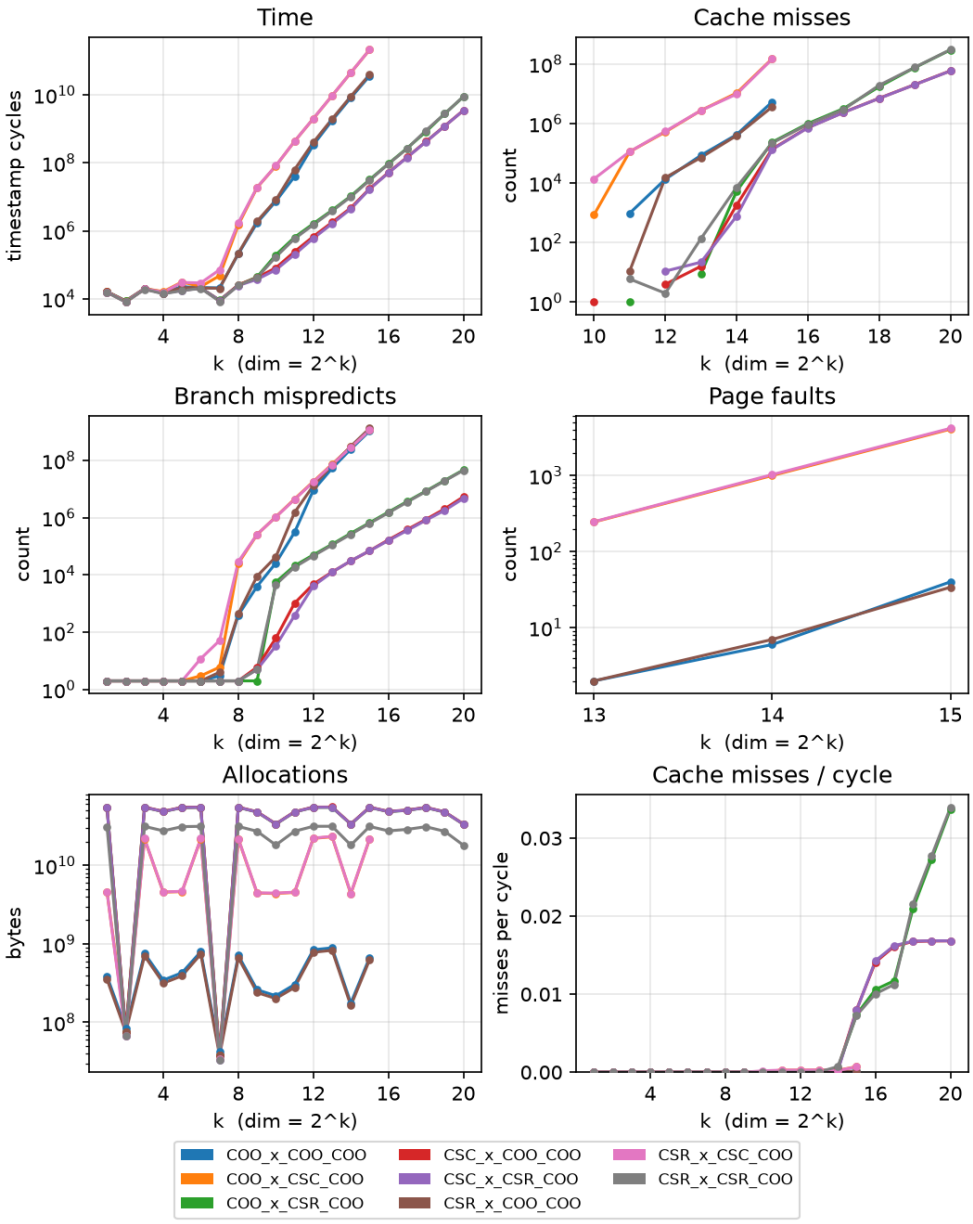


Figure B.19: CA-Hep-Ph Sample 2 comparison.

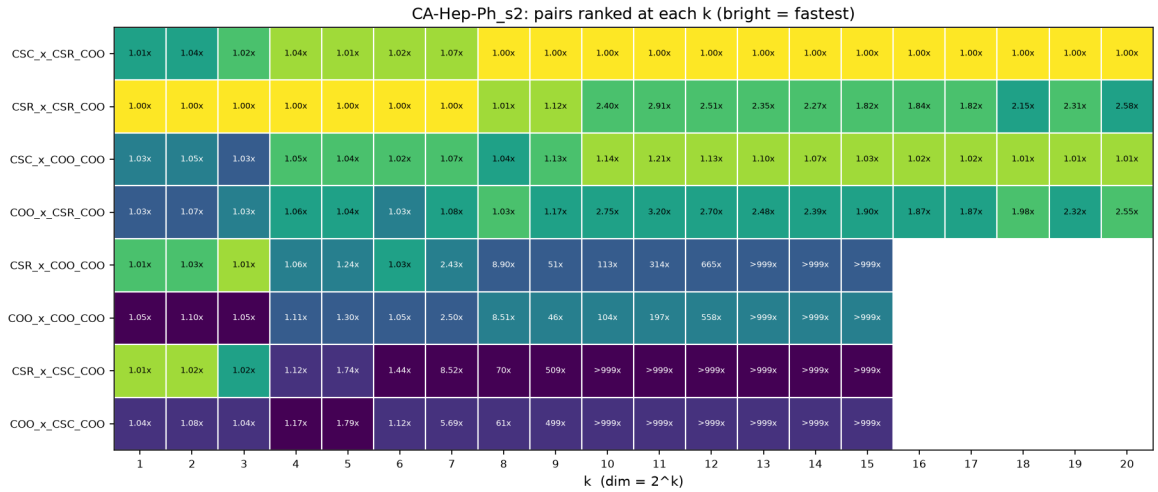


Figure B.20: CA-Hep-Ph Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

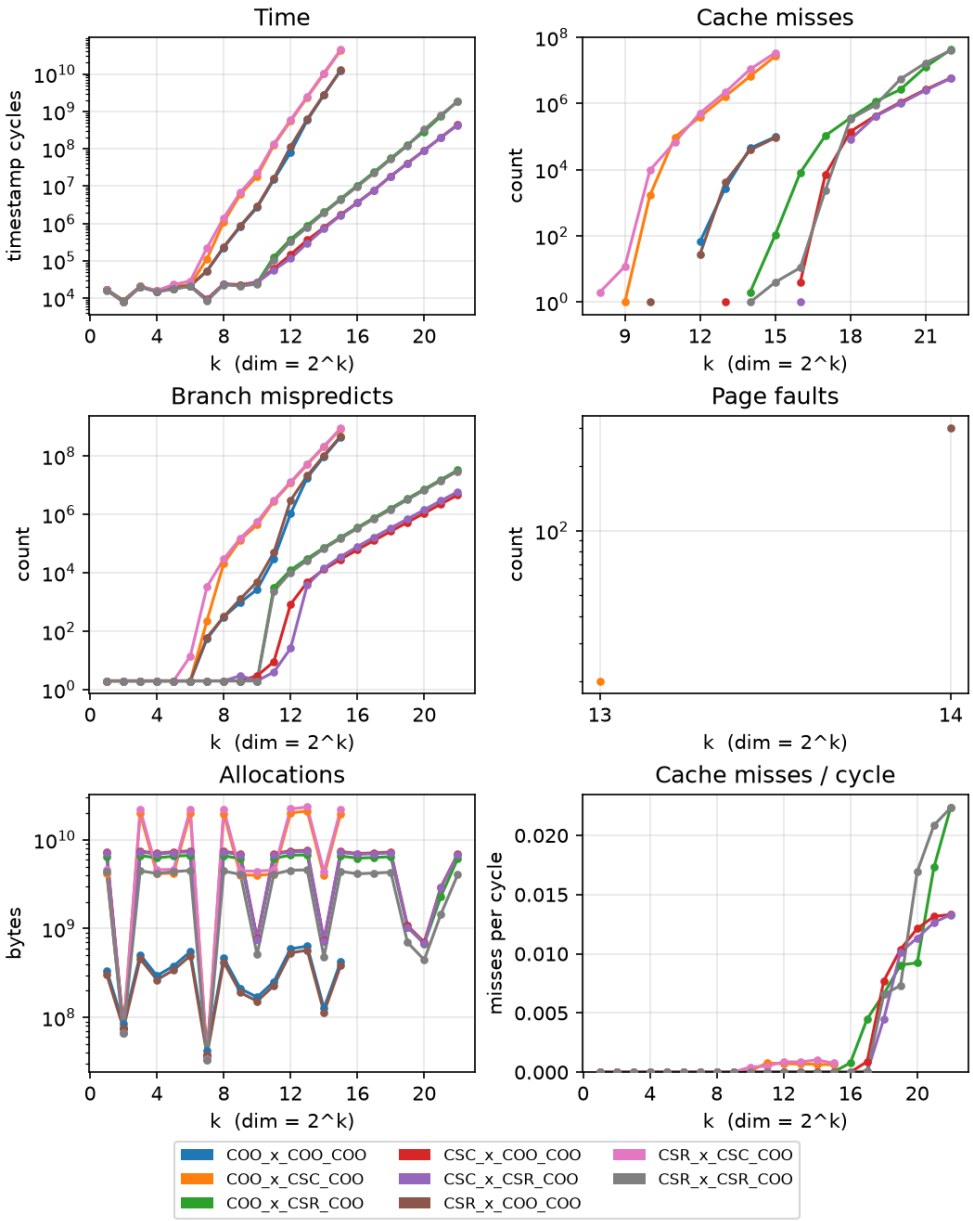


Figure B.21: CA-Hep-Th Sample 2 comparison.

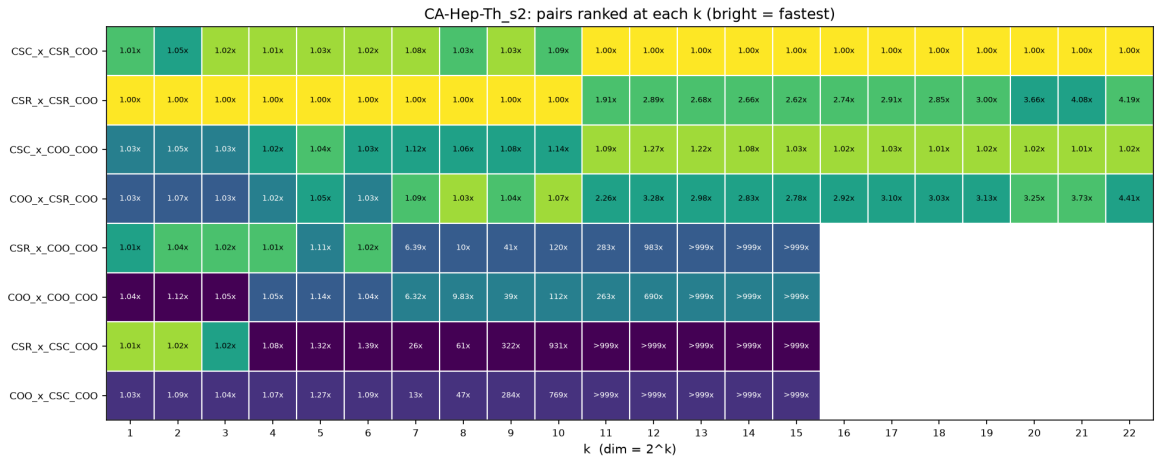


Figure B.22: CA-Hep-Th Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

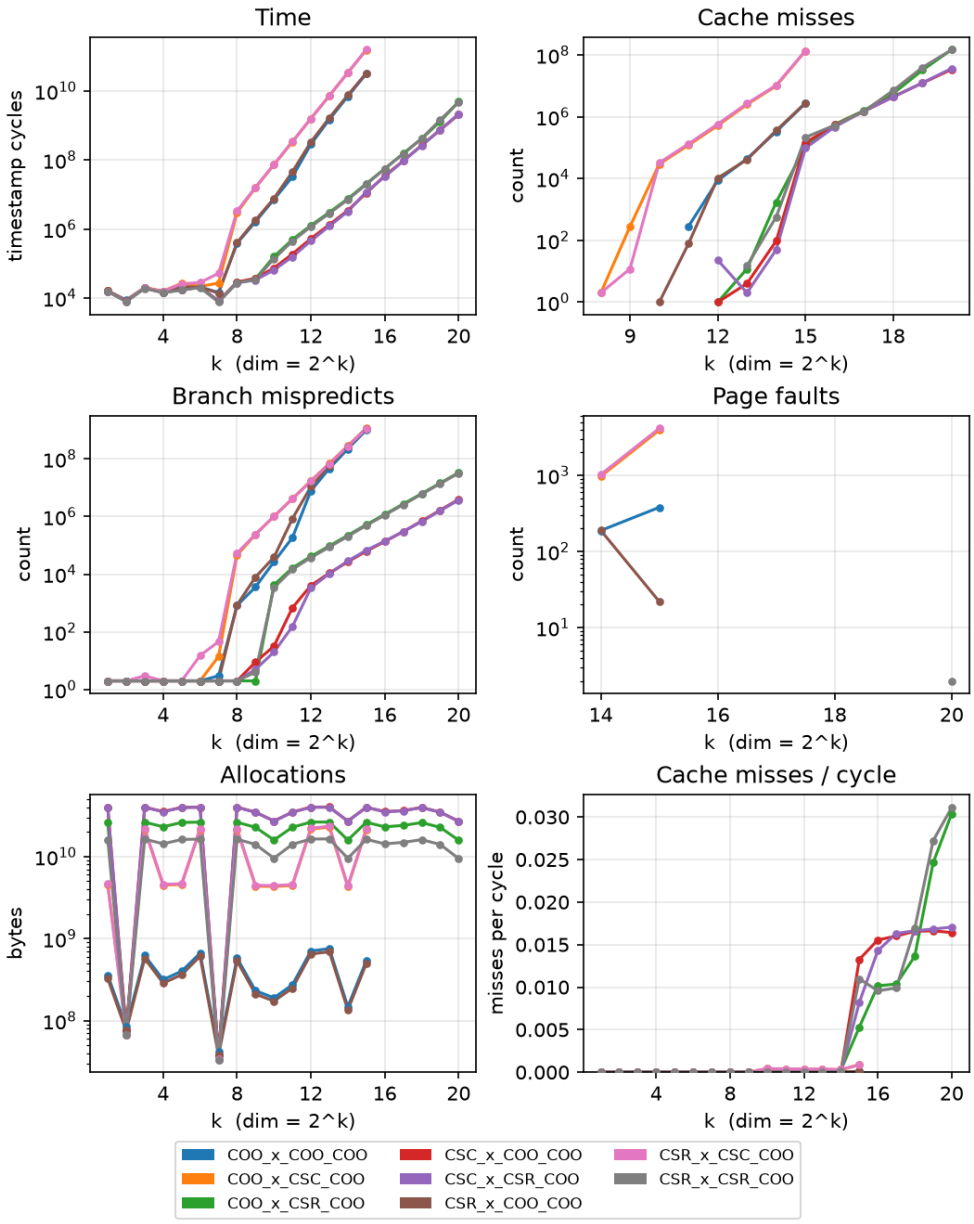


Figure B.23: Cit-Hep-Ph Sample 2 comparison.

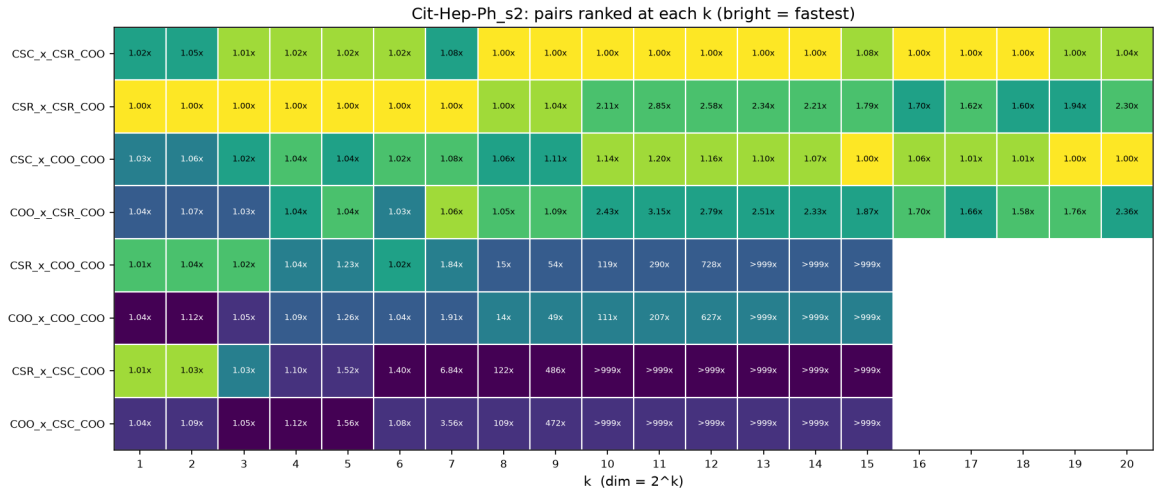


Figure B.24: Cit-Hep-Ph Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

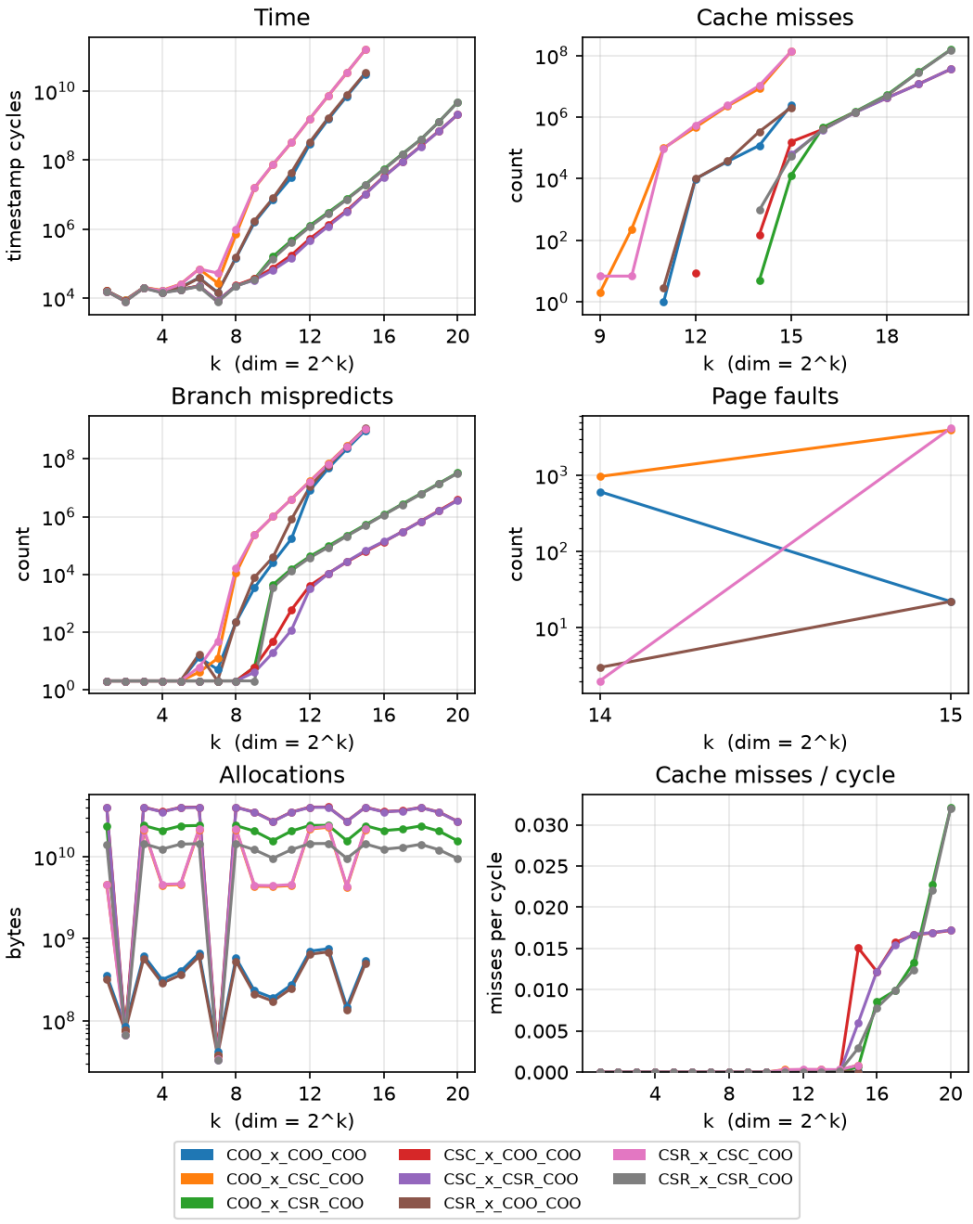


Figure B.25: Cit-Hep-Th Sample 2 comparison.

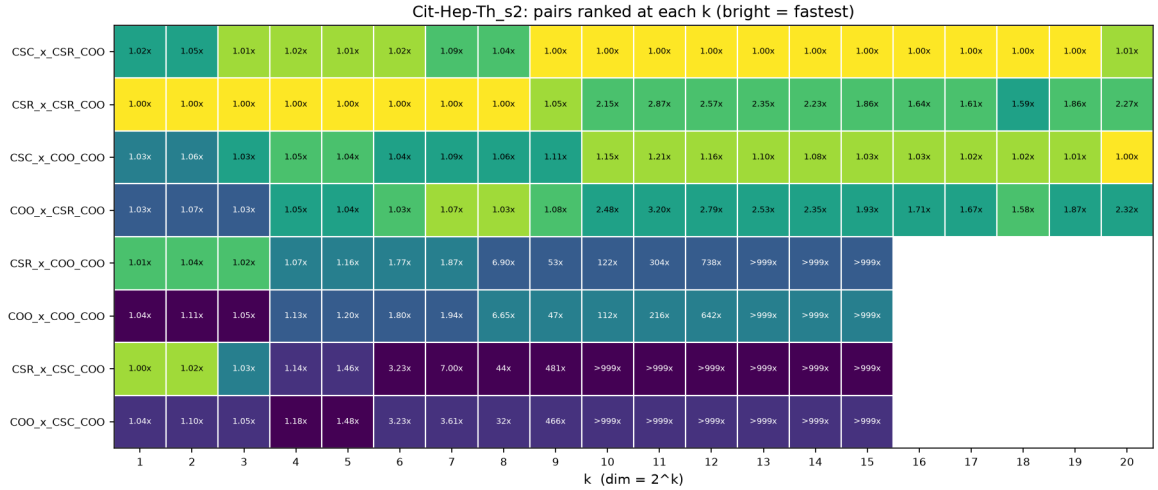


Figure B.26: Cit-Hep-Th Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

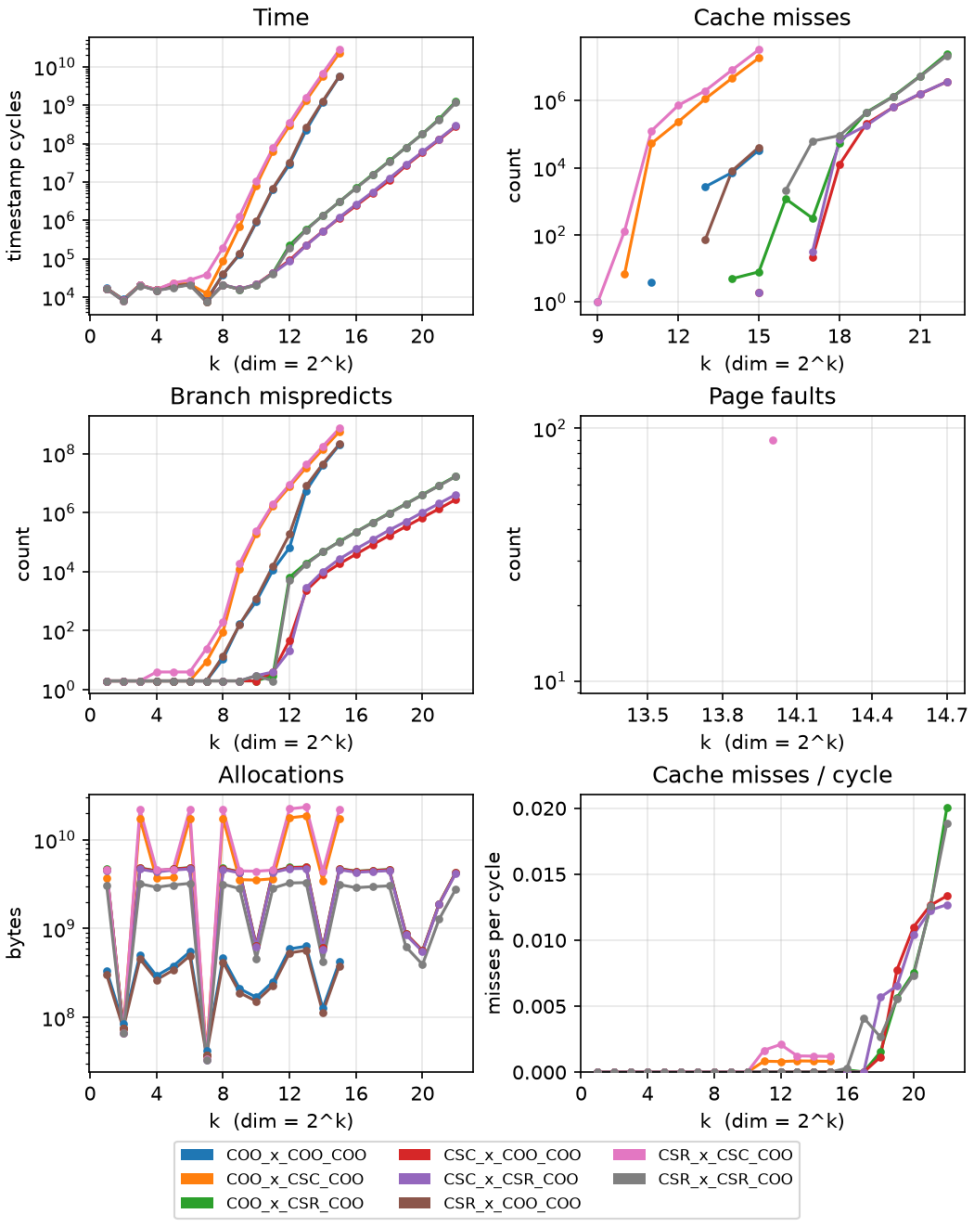


Figure B.27: Delicious Sample 2 comparison.

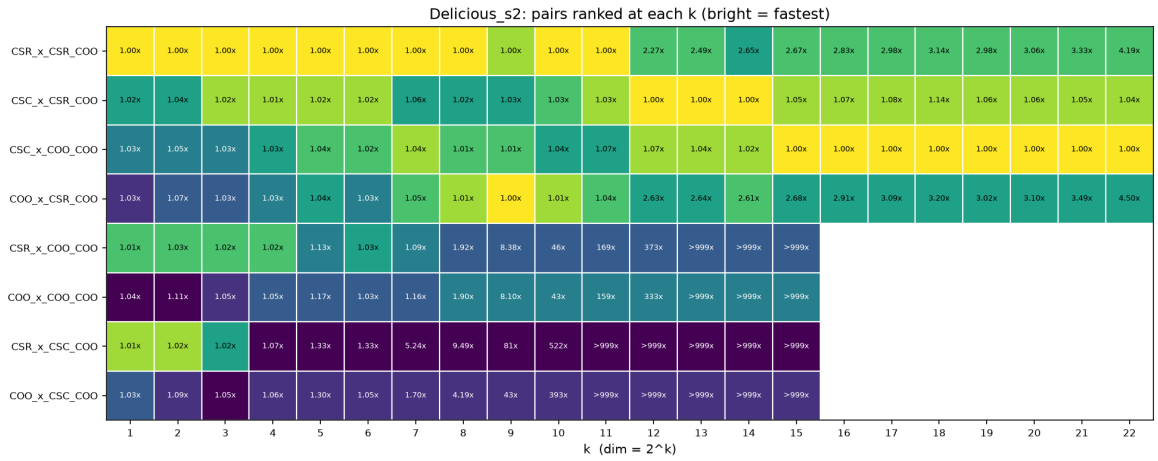


Figure B.28: Delicious Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

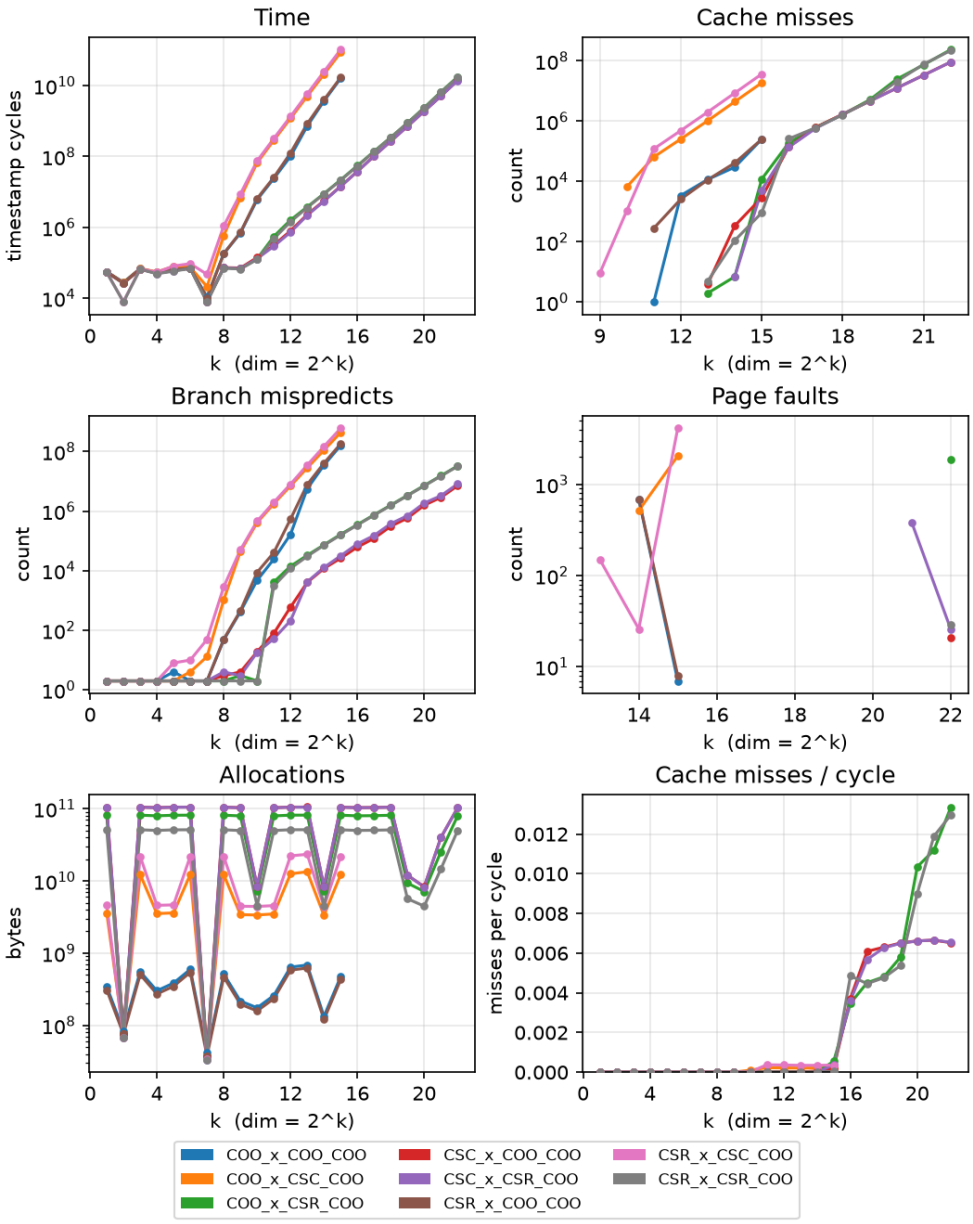


Figure B.29: Epinions Sample 2 comparison.

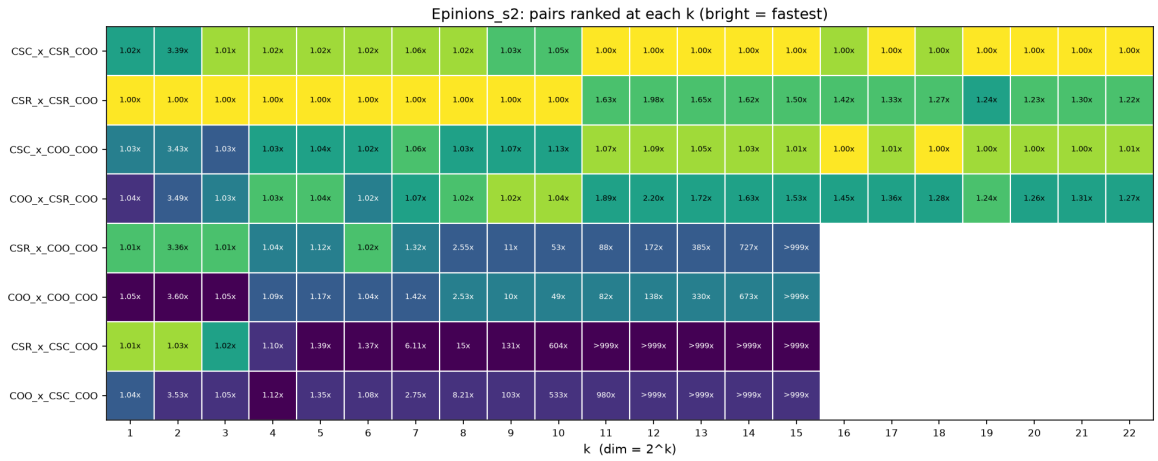


Figure B.30: Epinions Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

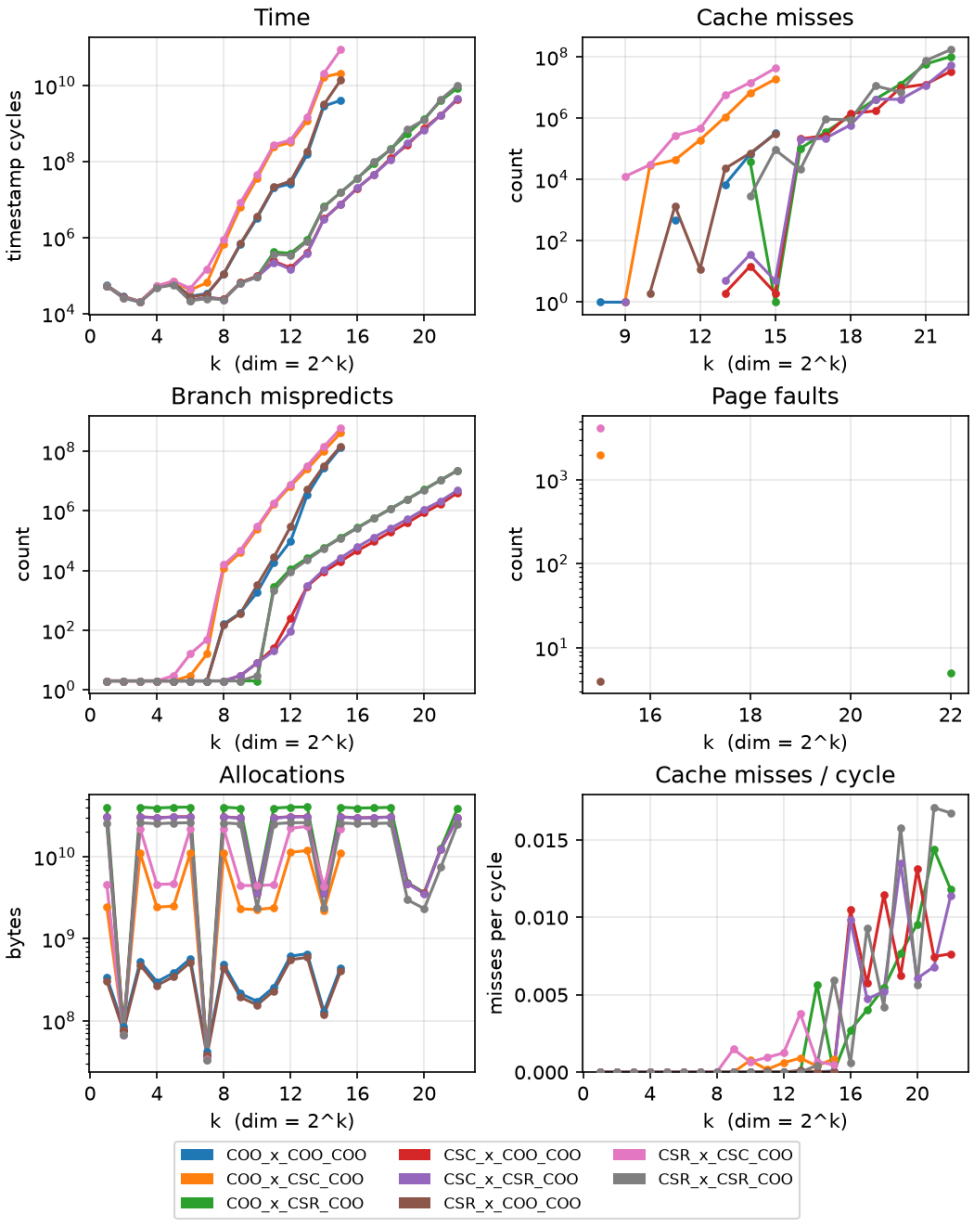


Figure B.31: Flickr Sample 2 comparison.

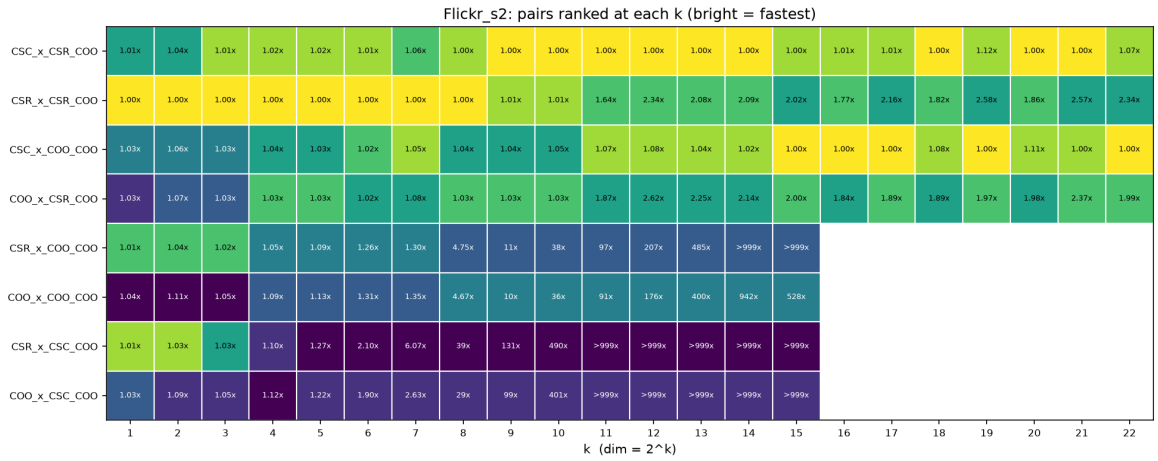


Figure B.32: Flickr Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

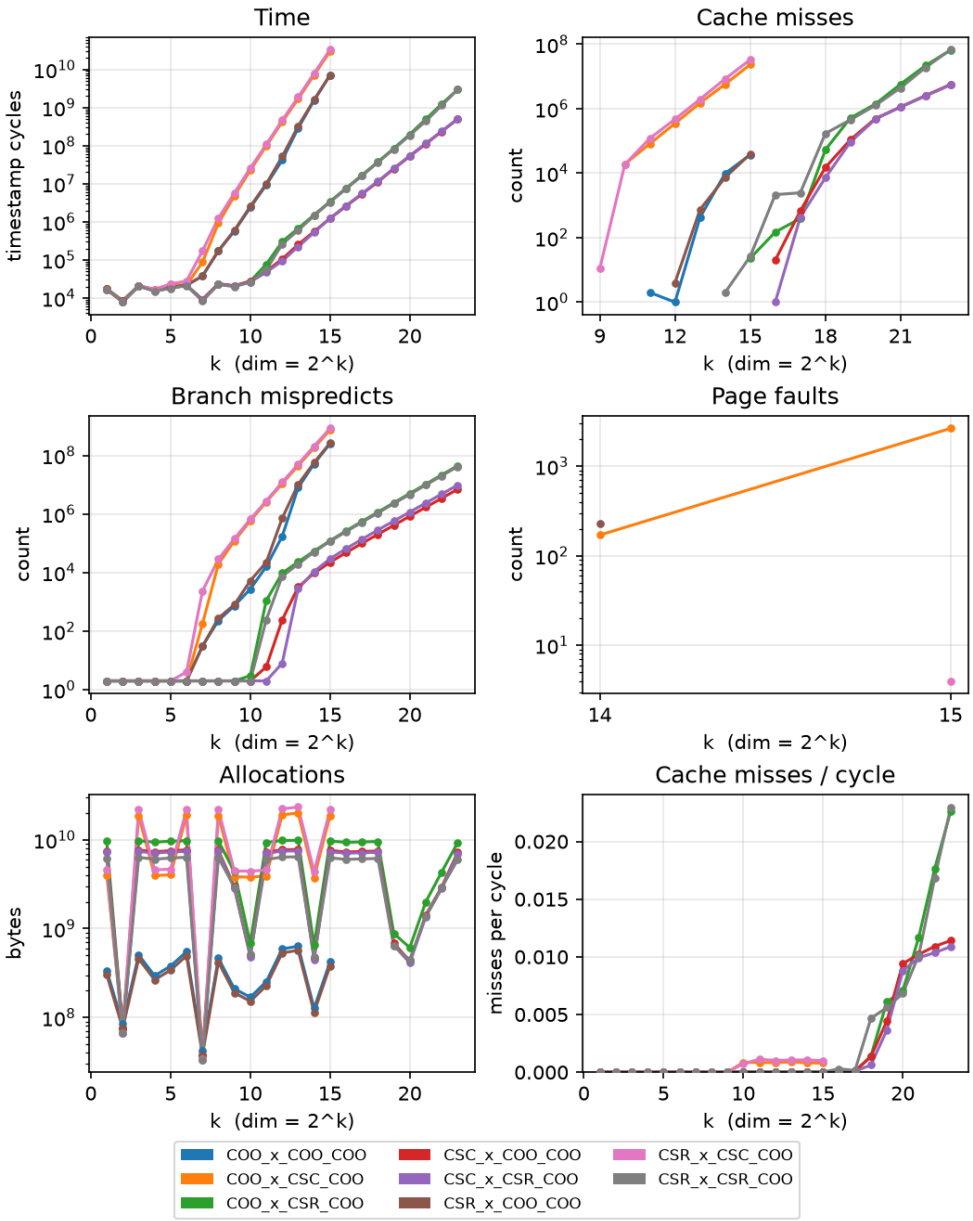


Figure B.33: Gnutella-25 Sample 2 comparison.

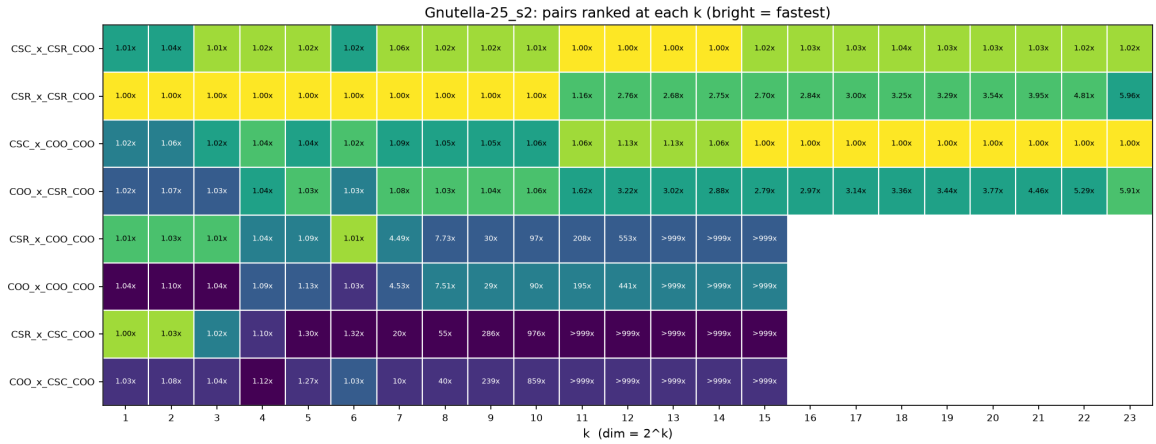


Figure B.34: Gnutella-25 Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

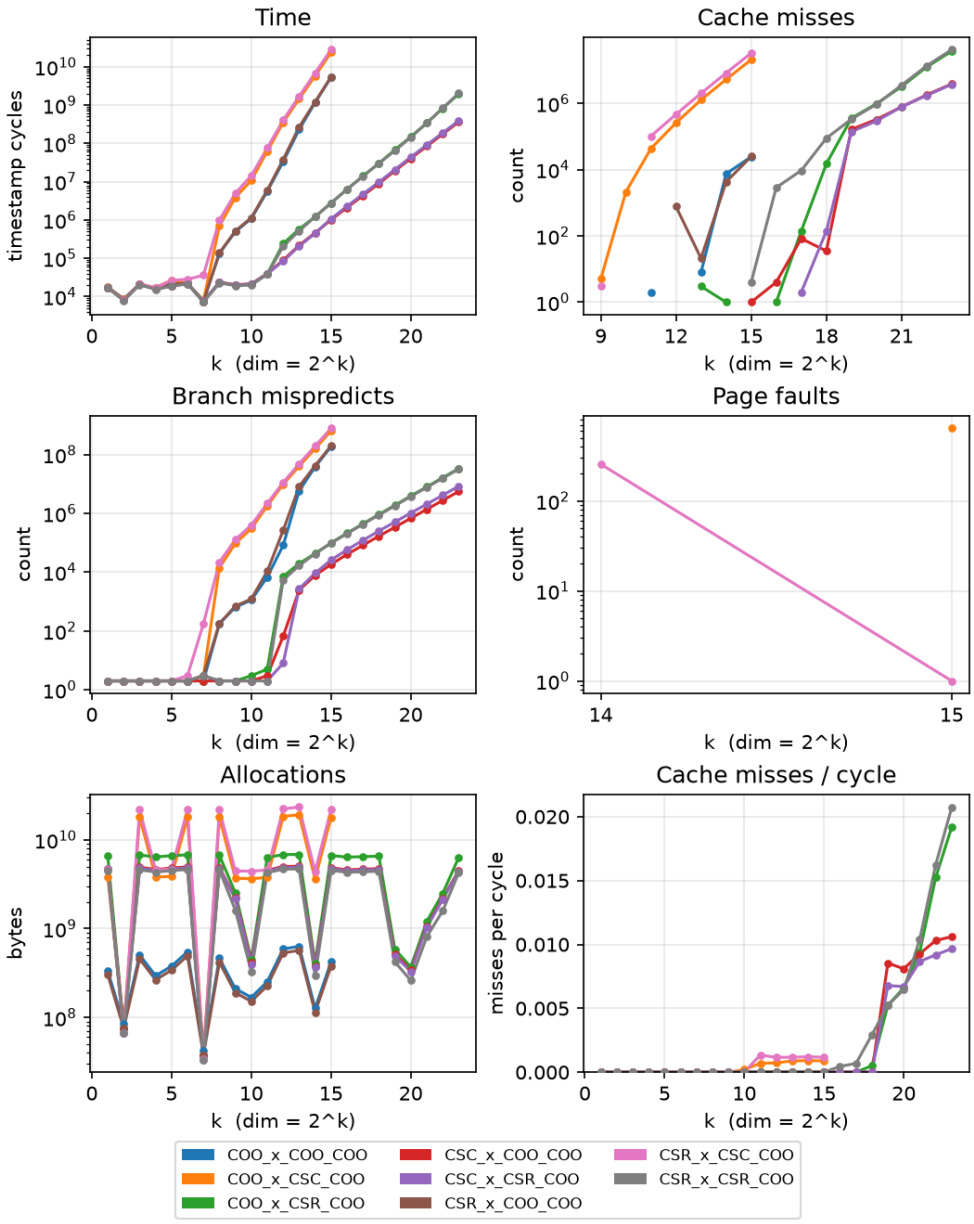


Figure B.35: Gnutella-30 Sample 2 comparison.

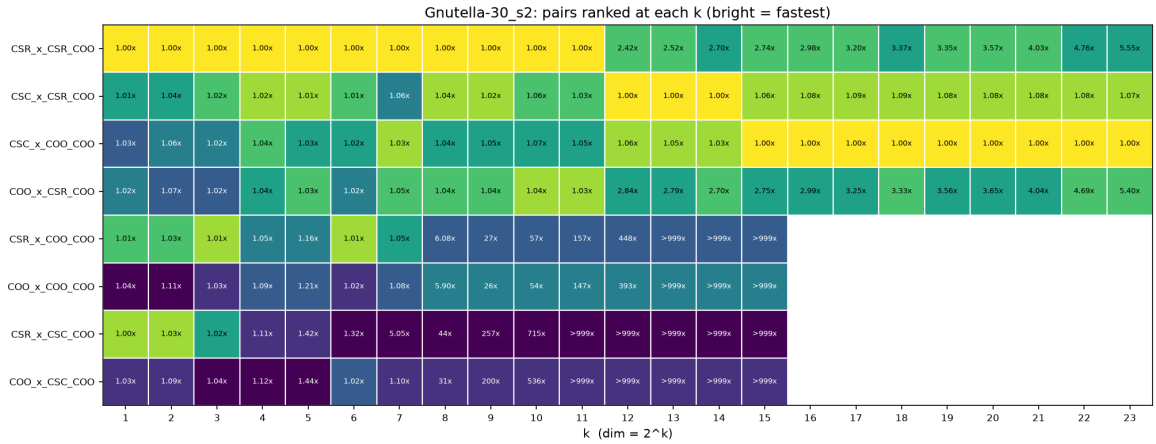


Figure B.36: Gnutella-30 Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

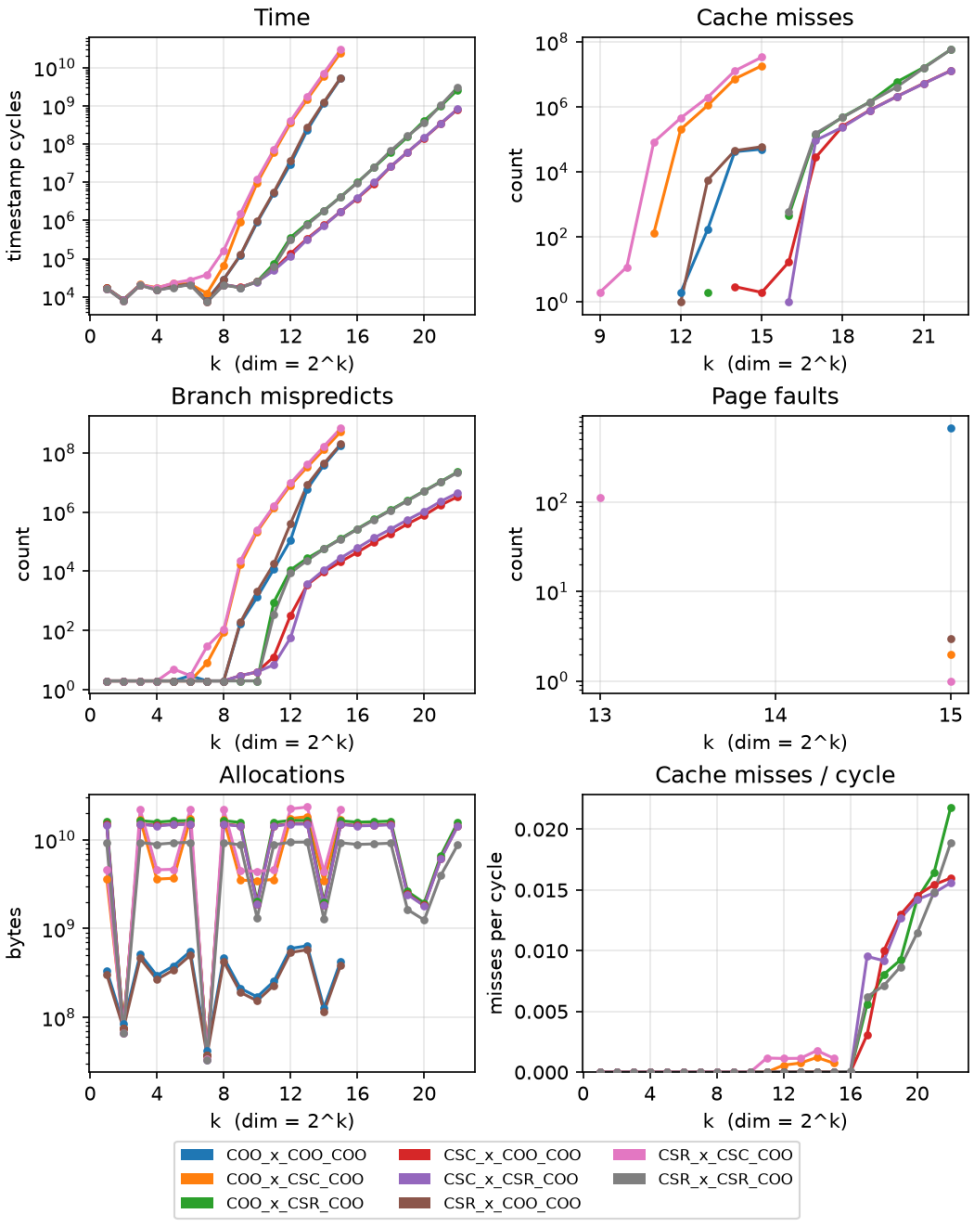


Figure B.37: Web-Notredame Sample 2 comparison.

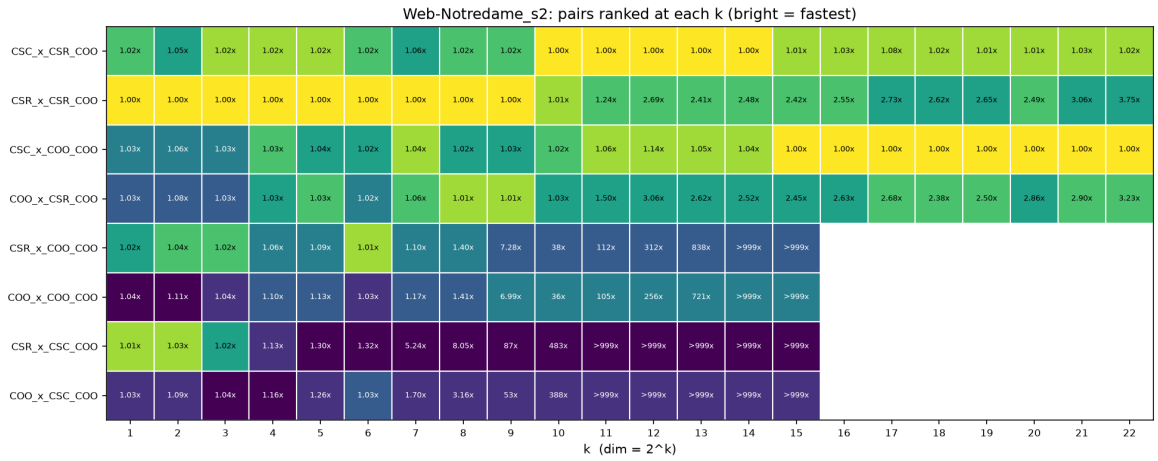


Figure B.38: Web-Notredame Sample 2: format pairs ranked at each k. Brighter cells are faster; each number is the slowdown relative to the fastest pair at that k.

Appendix C

Tiling Figures

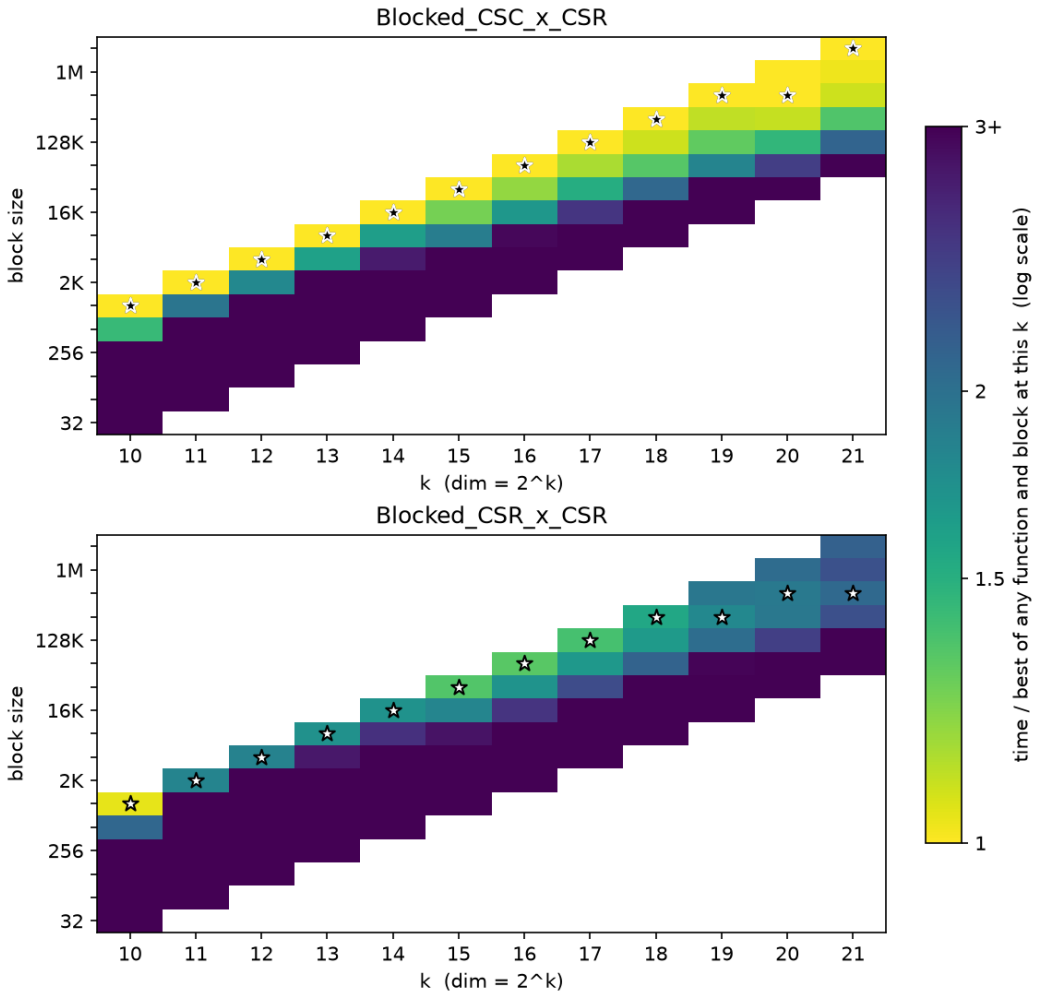


Figure C.1: AS-Newman Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

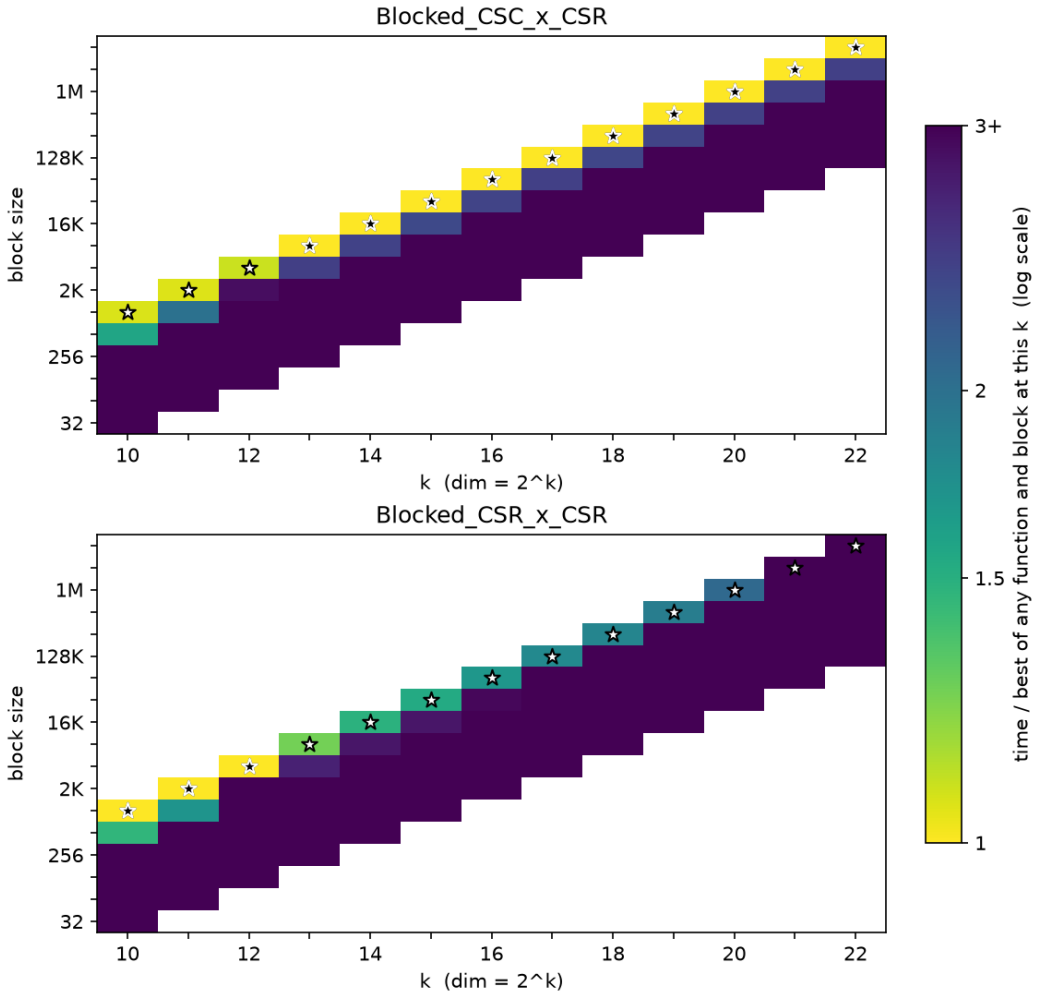


Figure C.2: ATP-GR-QC Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

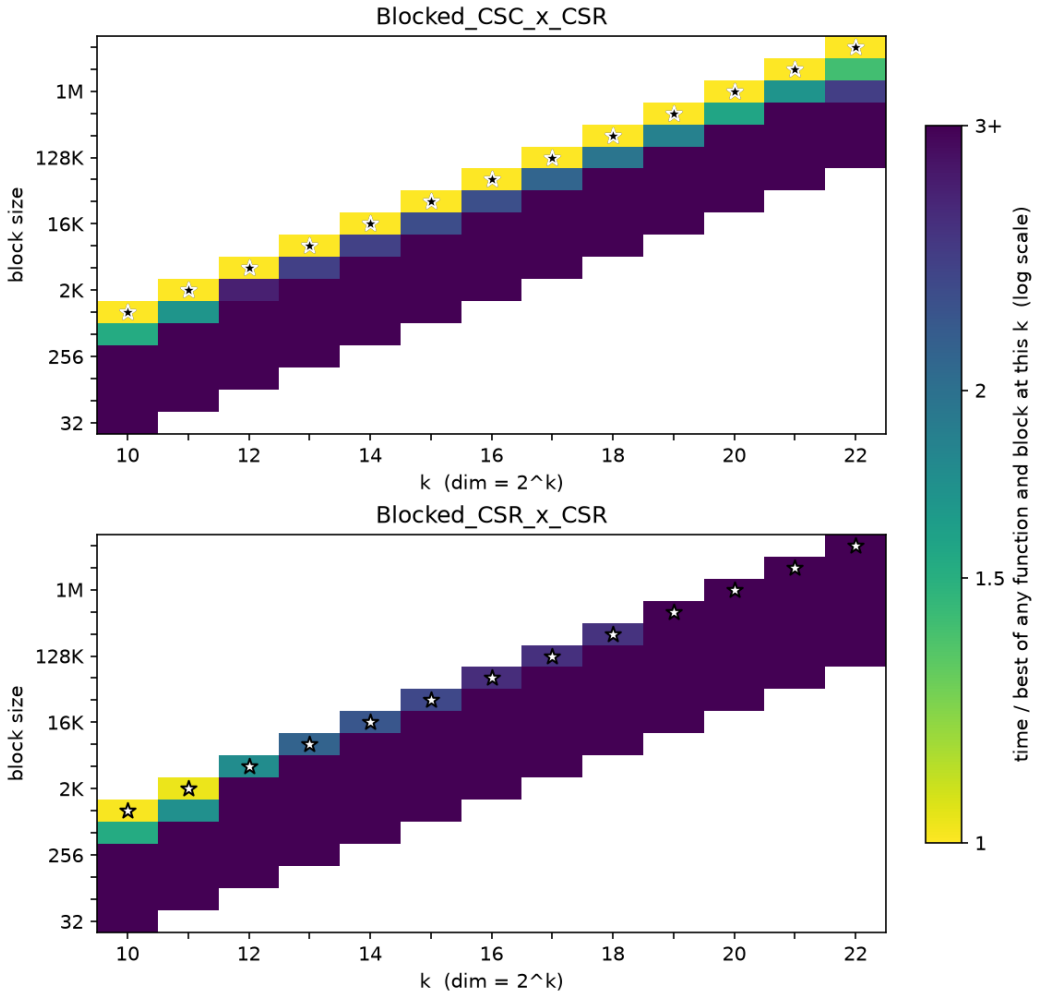


Figure C.3: Answers Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

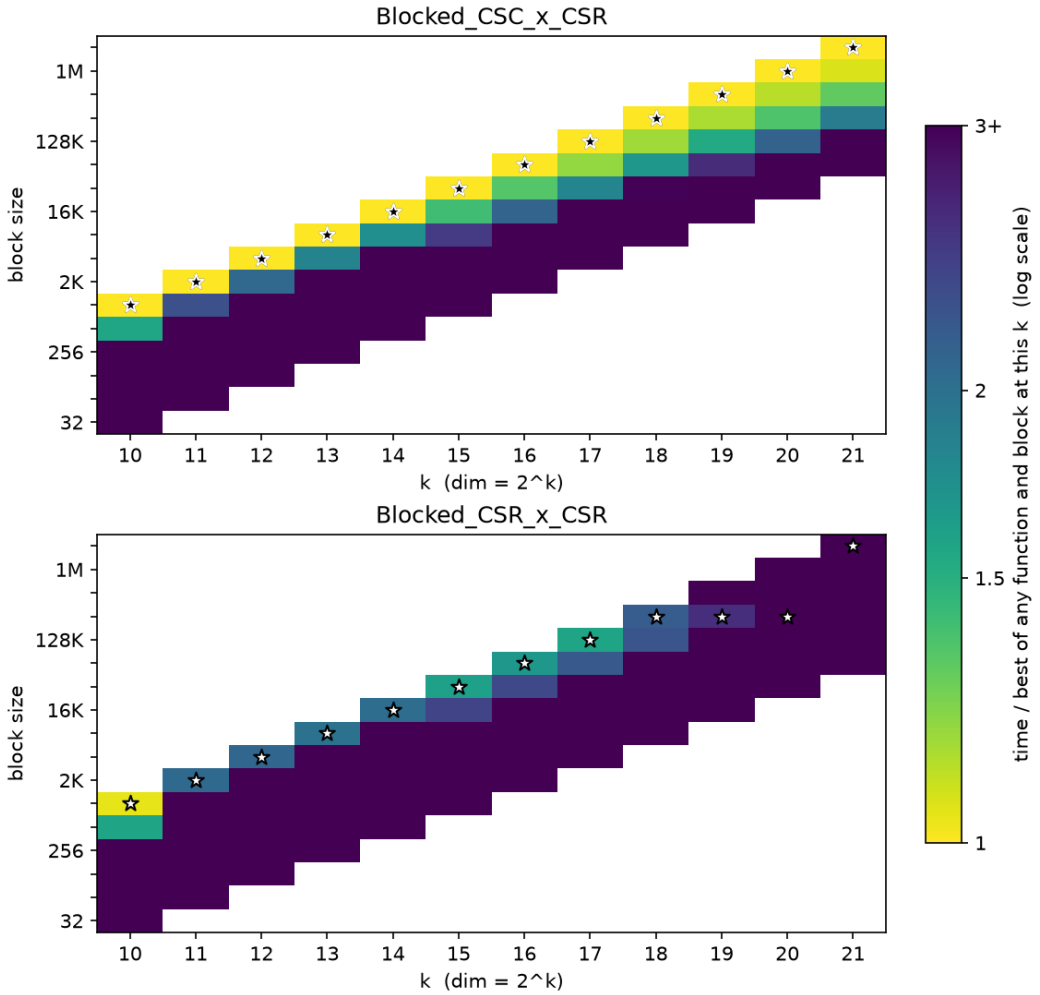


Figure C.4: Bio-Proteins Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

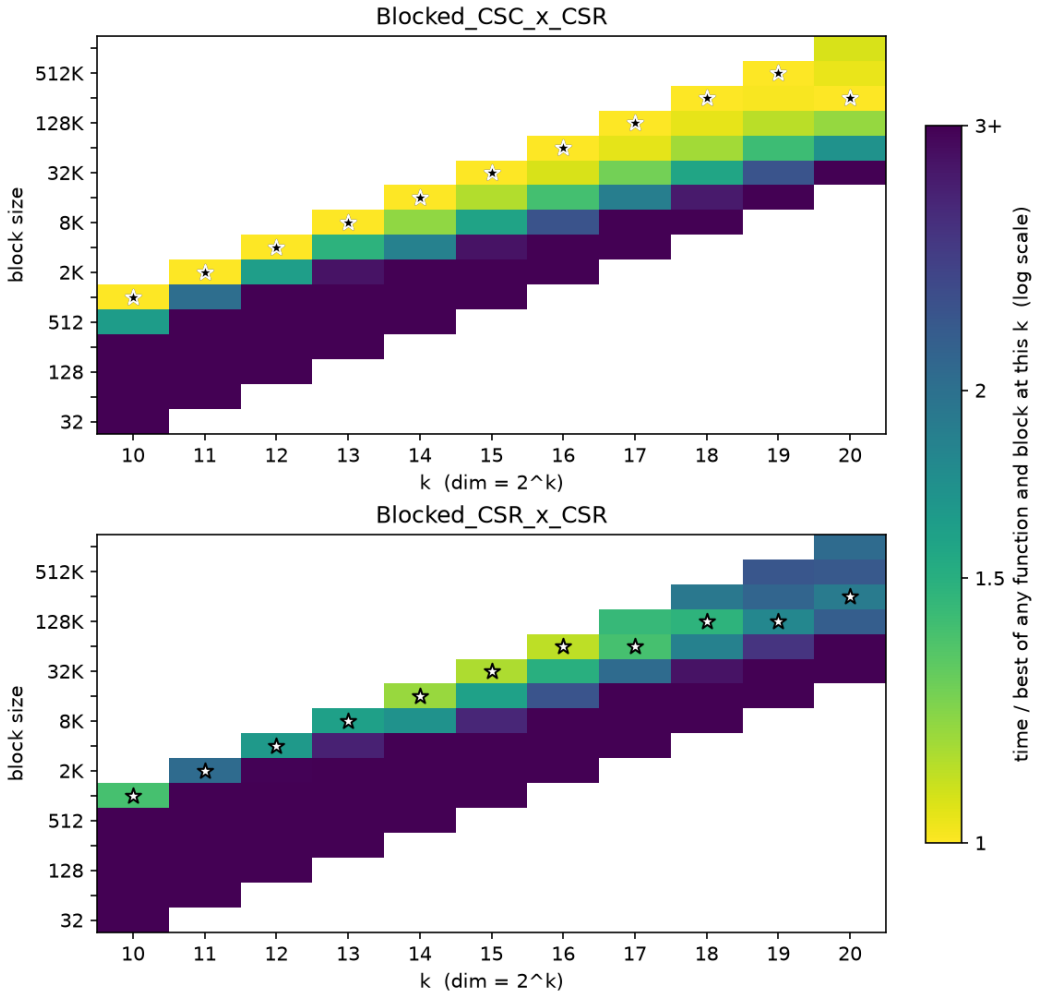


Figure C.5: Blog-Nat05-6m Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

Figure C.6: Blog-Nat06all Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

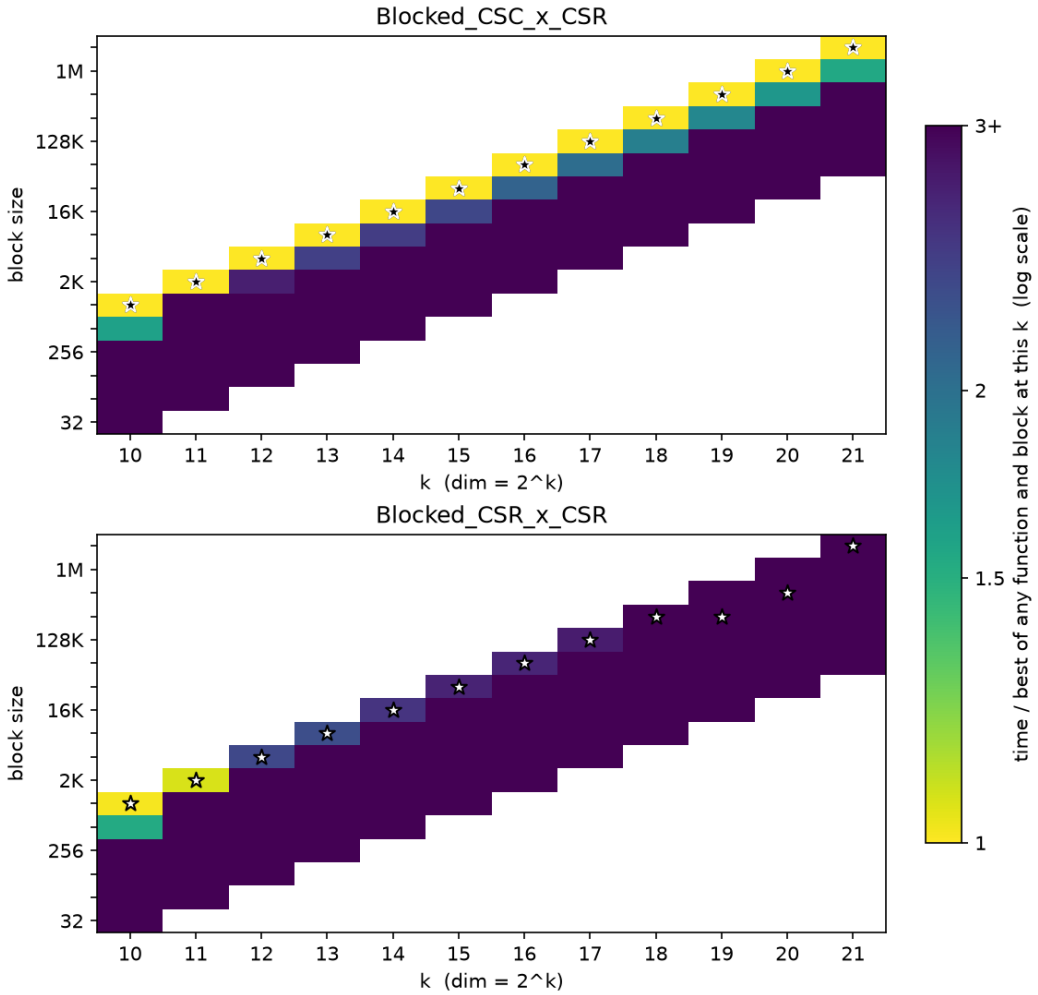


Figure C.7: CA-DBLP Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

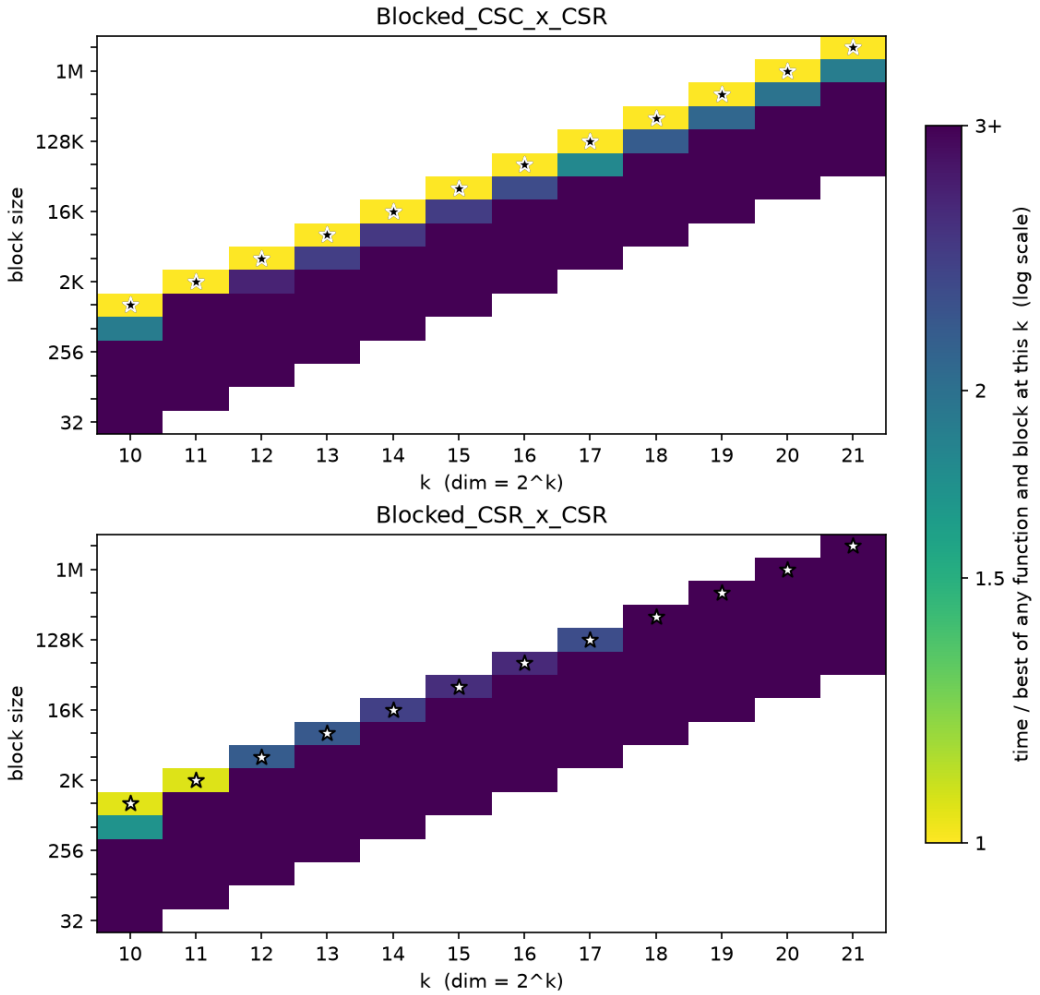


Figure C.8: CA-GR-QC Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

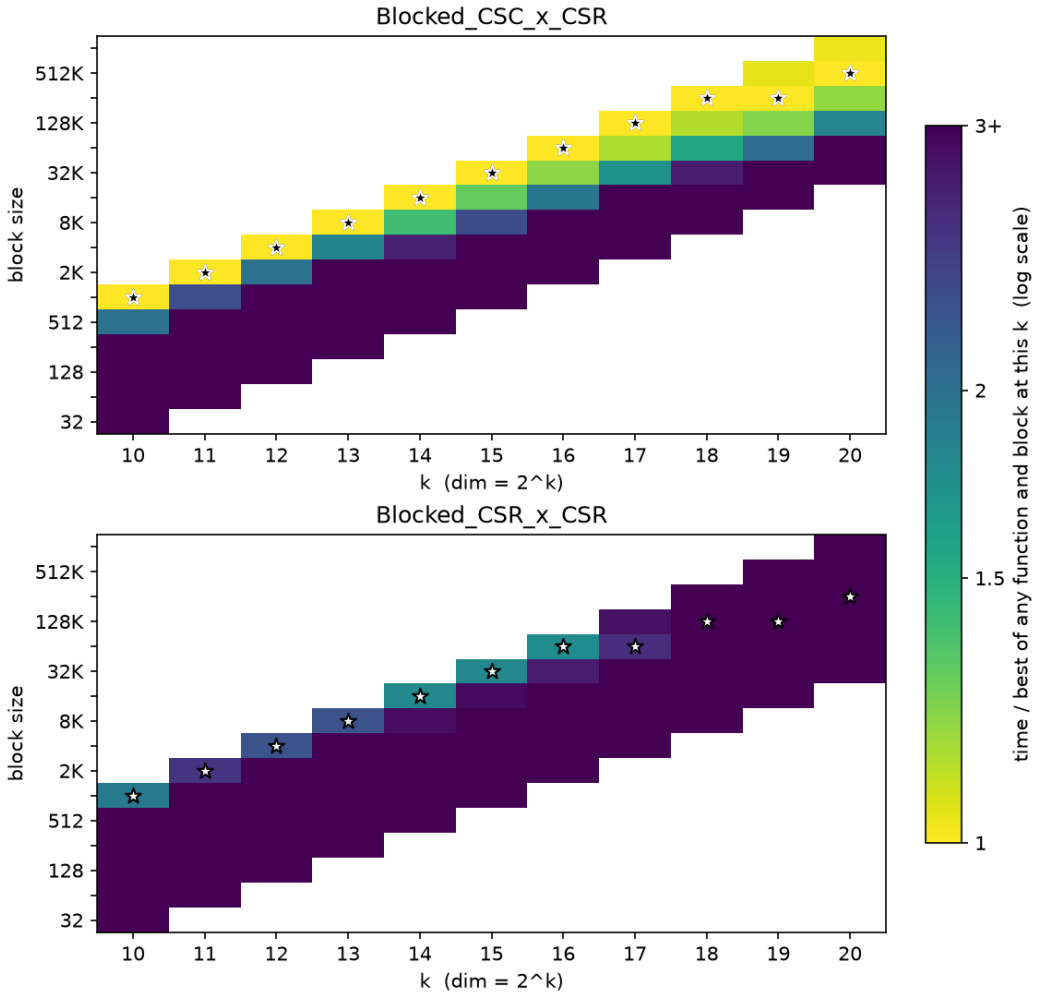


Figure C.9: CA-Hep-Ph Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

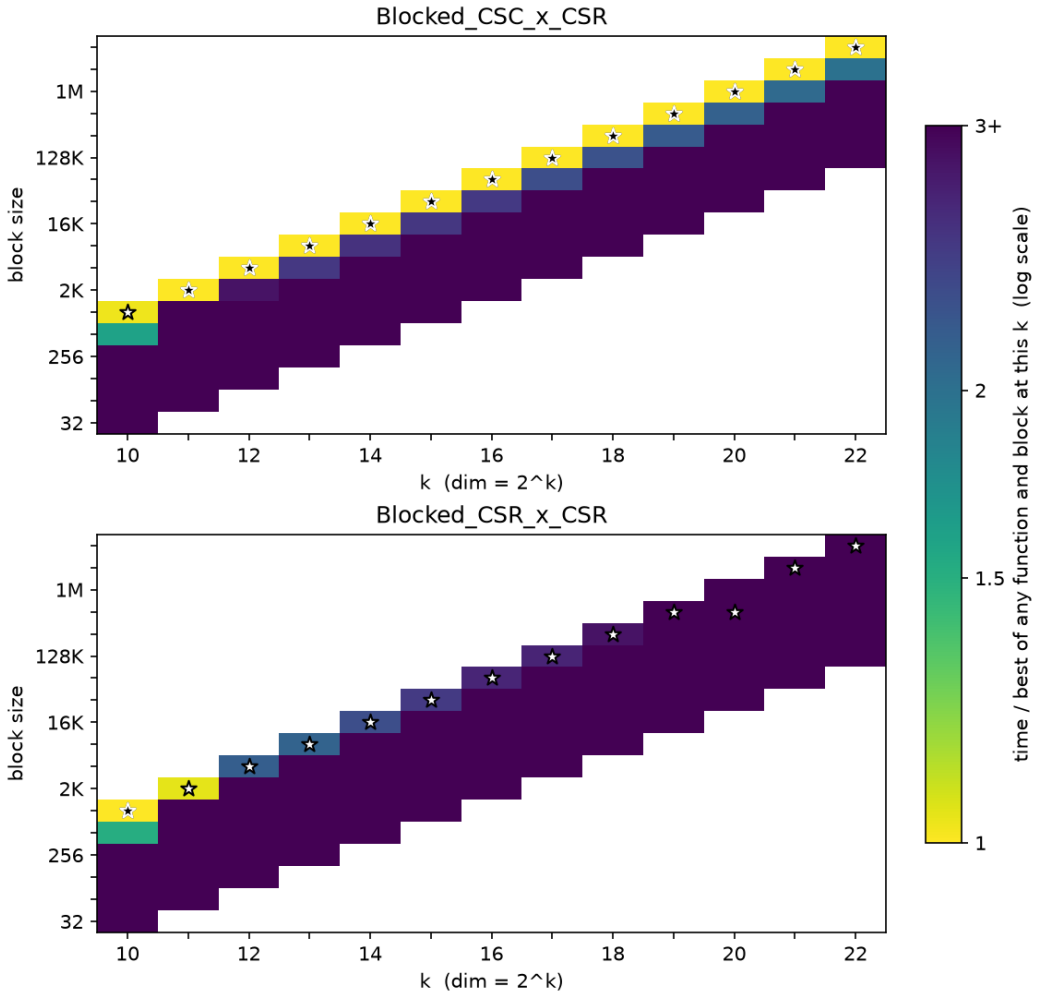


Figure C.10: CA-Hep-Th Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

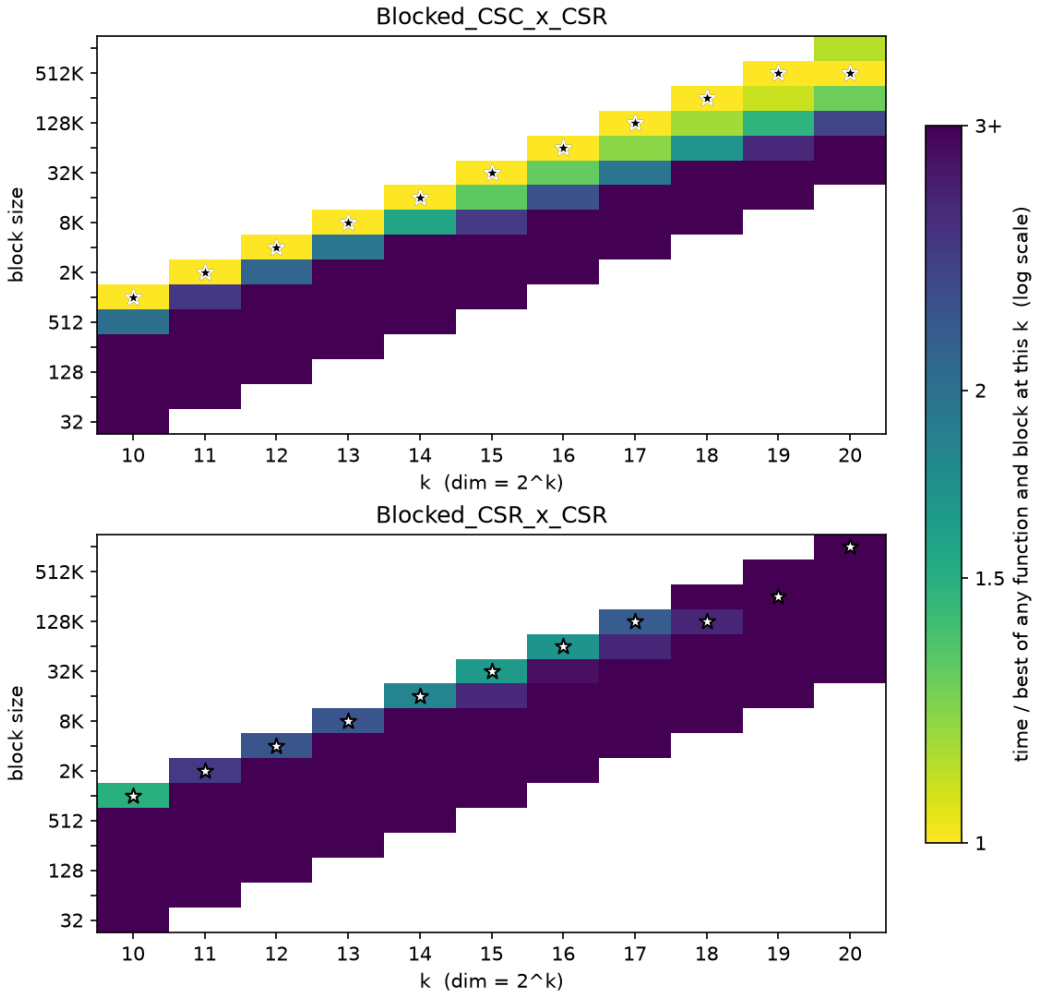


Figure C.11: Cit-Hep-Ph Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

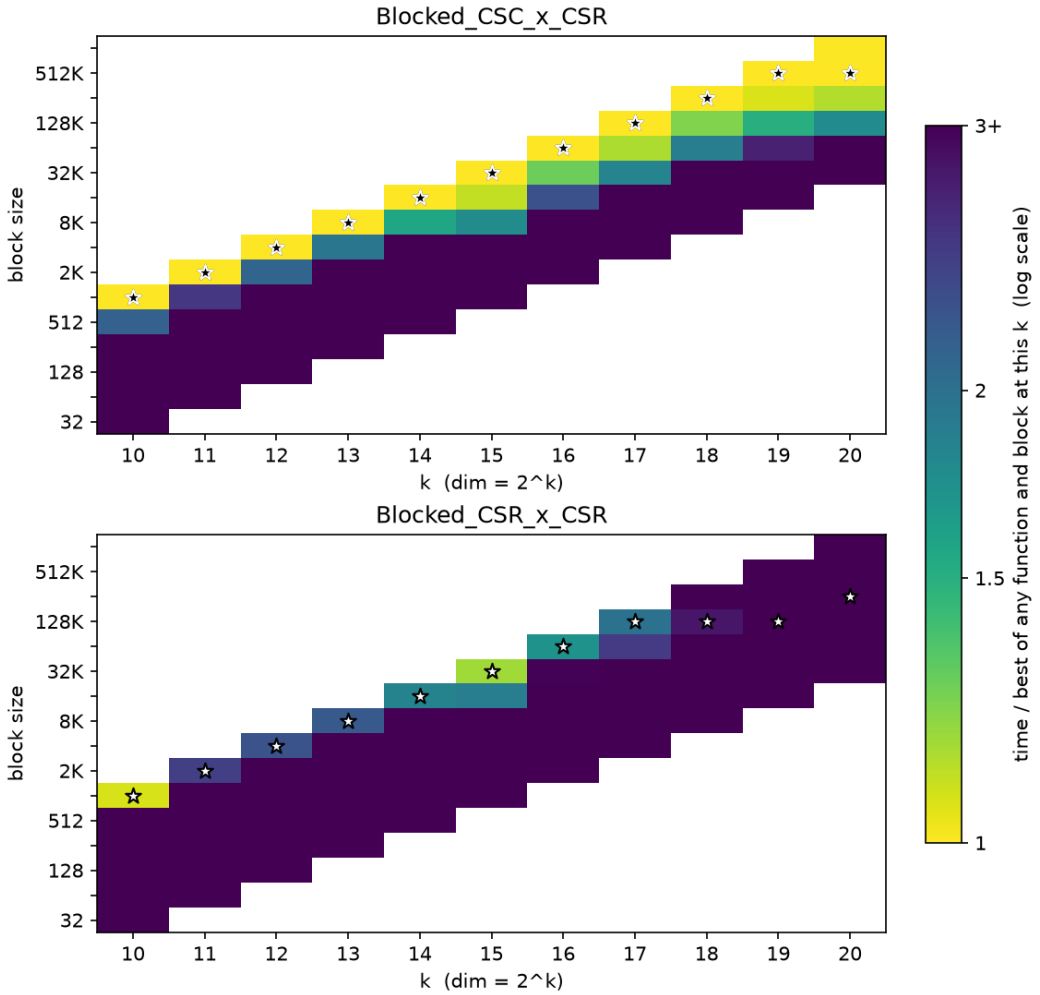


Figure C.12: Cit-Hep-Th Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

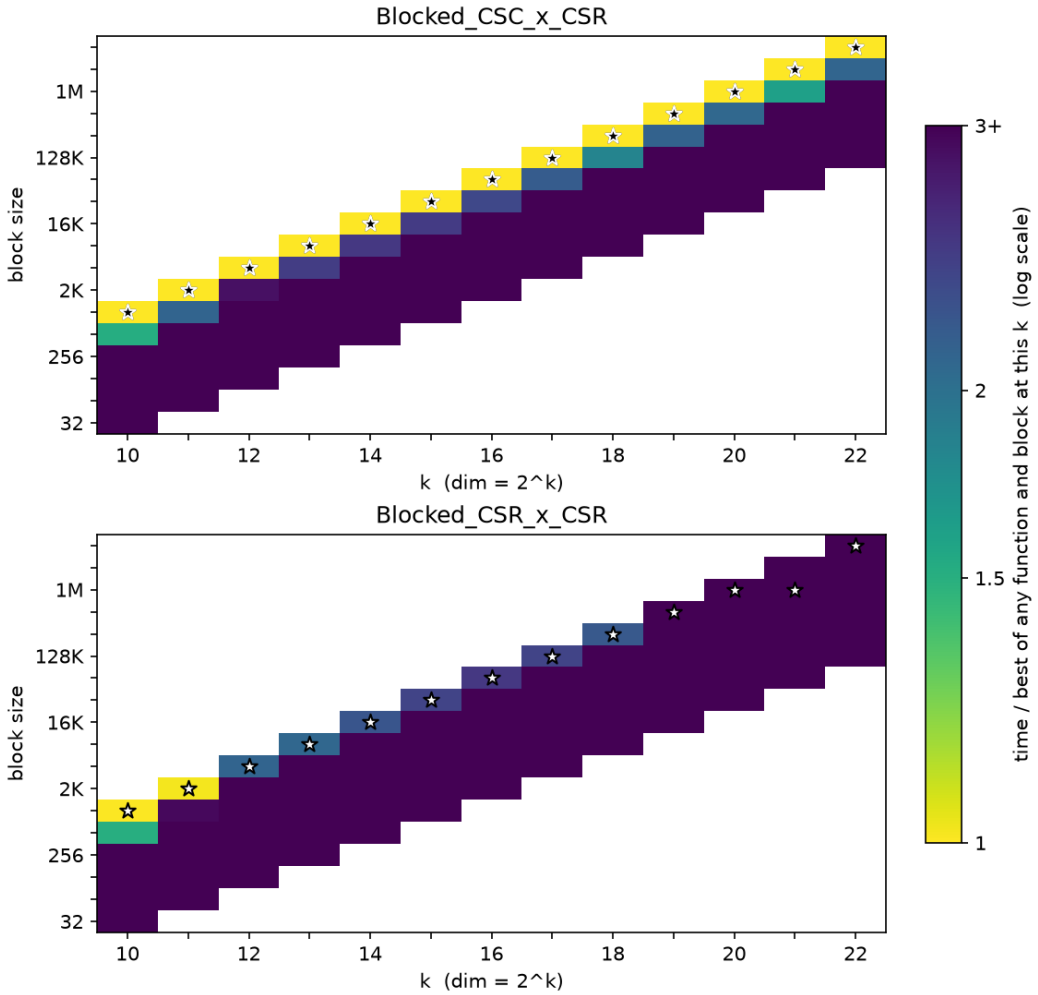


Figure C.13: Delicious Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

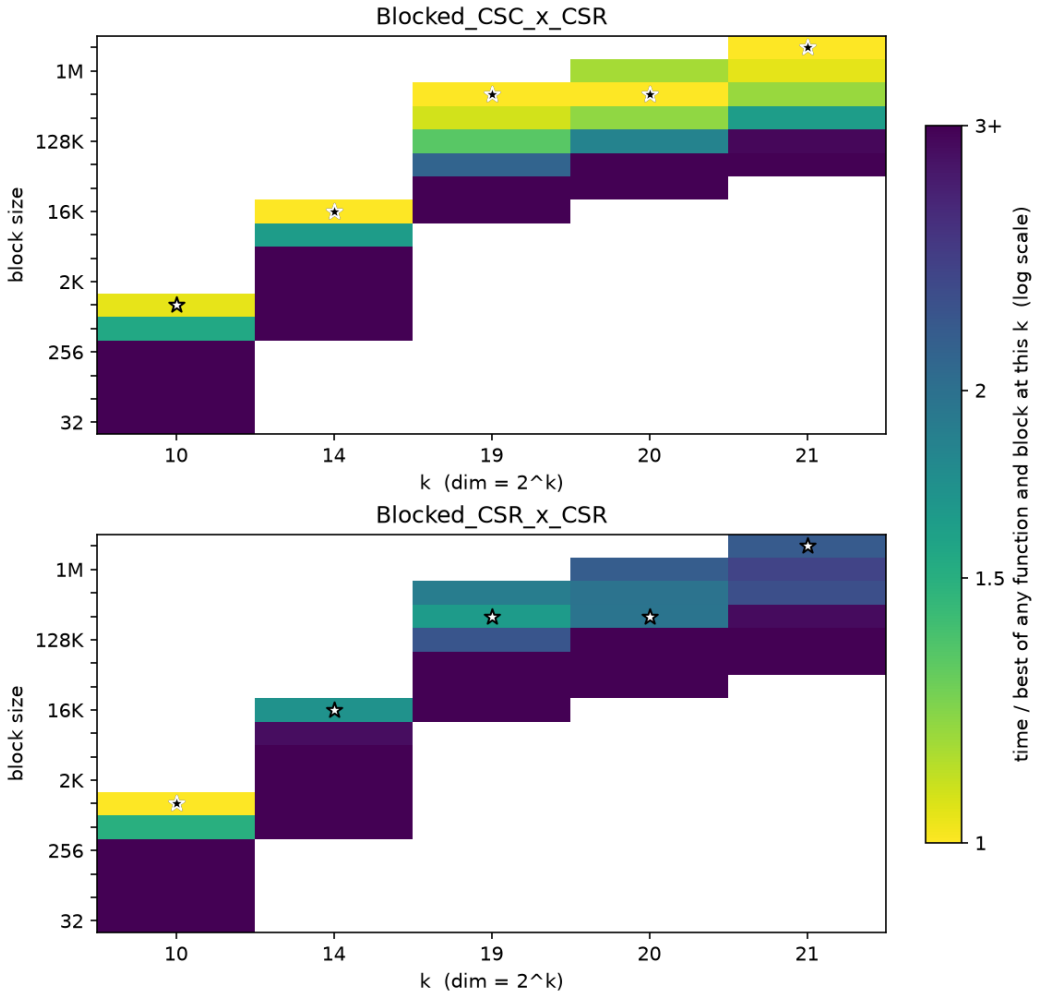


Figure C.14: Epinions Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

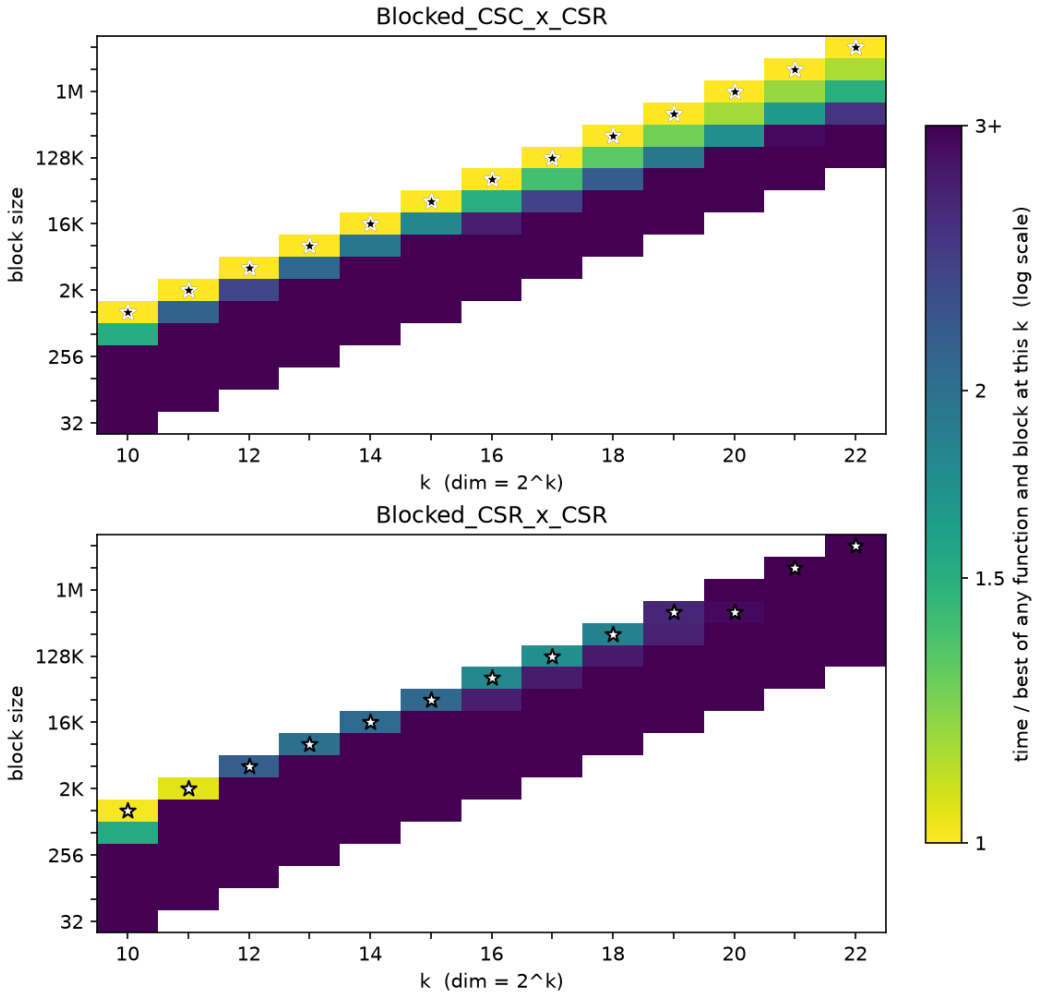


Figure C.15: Flickr Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

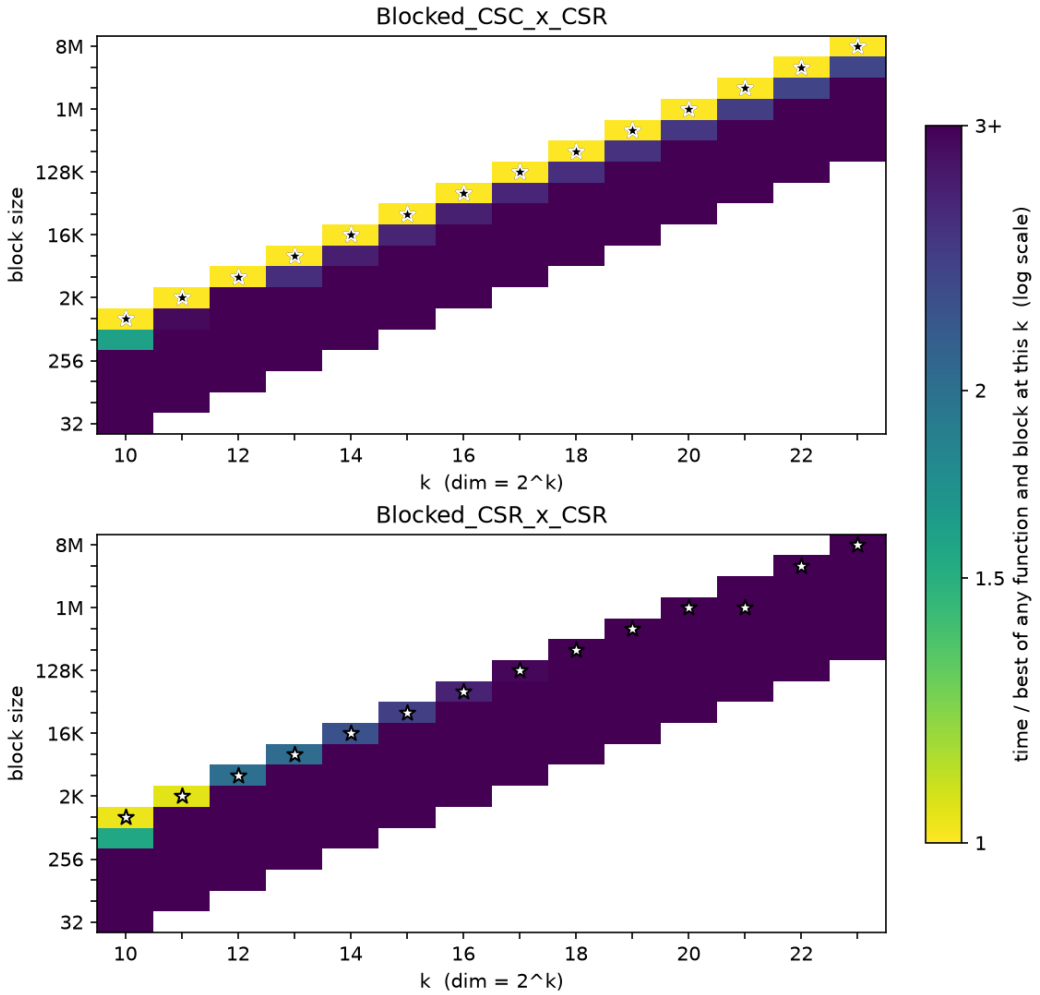


Figure C.16: Gnutella-25 Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

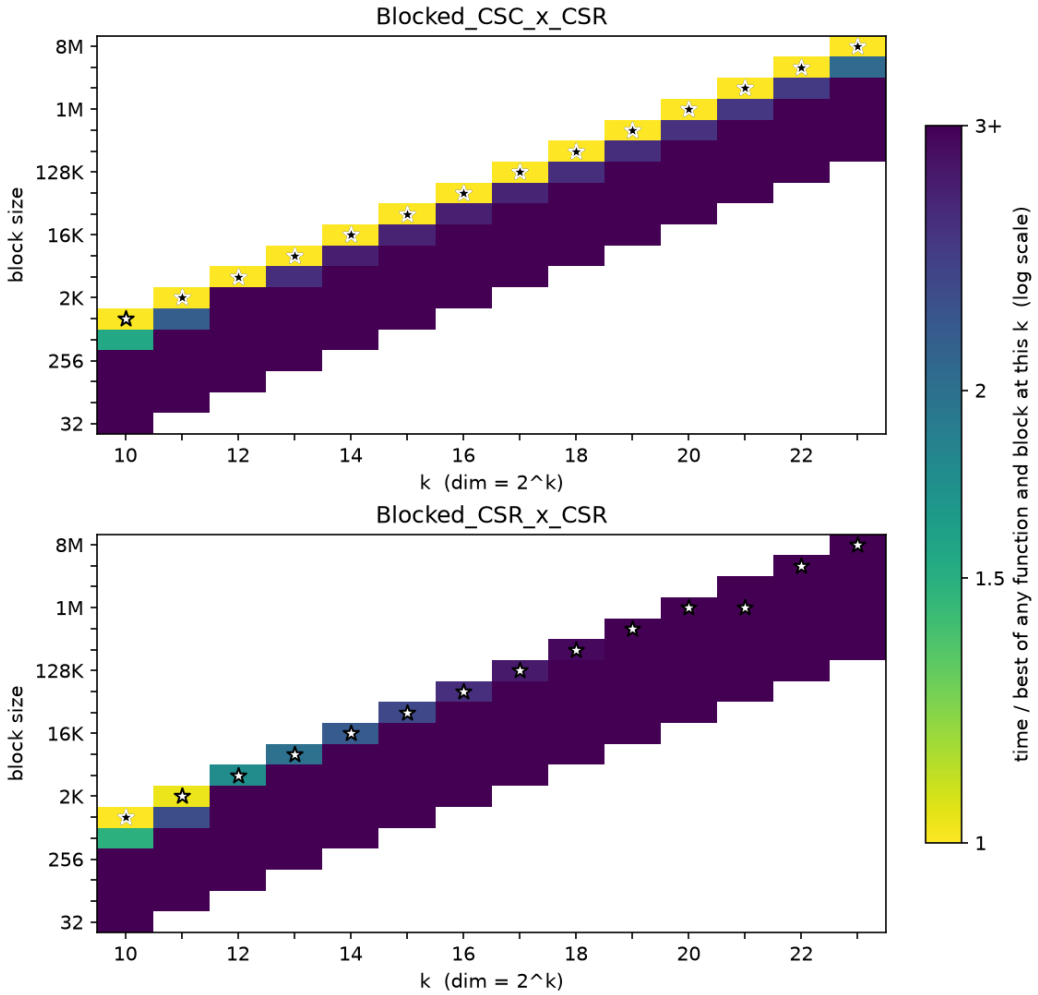


Figure C.17: Gnutella-30 Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.

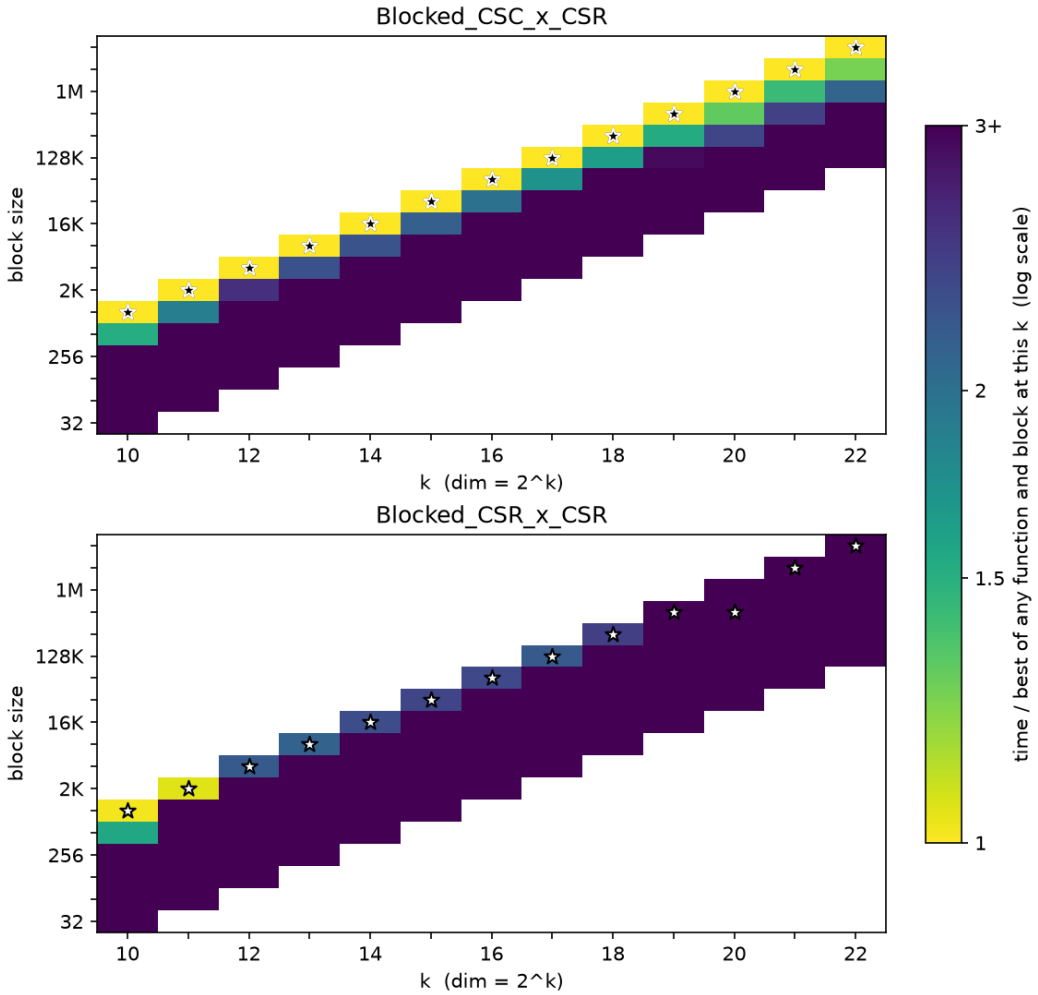


Figure C.18: Web-Notredame Sample 2 tiled block-size sweep. Colour is time relative to the best block size at each k (1 is best); stars mark the best block size.